Copernicus Global Land Operations
"Vegetation and Energy"
"CGLOPS-1"

Framework Service Contract N° 945120-IPR-2023

VALIDATION REPORT

LAND SURFACE PHENOLOGY

VERSION 2.0

Issue I1.00

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Dissemination Level		
PU	Public	X
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CO	Confidential, only for members of the consortium (including the Commission Services)	

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Document Release Sheet

Book captain:	Jorge Sánchez-Zapero (EOLAB)	Sign Jorge Sanchez-Zapero	Date 01.12.2025
Approval:	Roselyne Lacaze (HYGEOS)	Sign Roselyne Lacaze	Date 15.12.2025
Endorsement:	Nadine Gobron (JRC)	Sign	Date
Distribution:	Public		

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Change Record

Date	Page(s)	Description of Change	Release
01.12.2025	All	First issue	I1.00

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List of Acronyms

AD	Applicable Document
AG	Agriculture
AMPL	Seasonal amplitude
AOI	Area Of Interest
APU	Accuracy, Precision and Uncertainty
ATBD	Algorithm Theoretical Basis Document
B	mean Bias
BBCH	*Biologische Bundesanstalt, Bundessortenamt und CHemische Industrie*
BQ	Best-Quality
BRDF	Bidirectional Reflectance Distribution Function
CCI	Climate Change Initiative
CEOS	Committee on Earth Observation Satellites
CGLOPS	Copernicus Global Land OPerationS
CLC	Corine Land Cover
CLMS	Copernicus Land Monitoring Service
COG	Cloud-Optimized Geotiff
CUL	CULtivated
DB	Deciduous Broadleaf Forest
DBF	Deciduous Broadleaf Forest
DCF	Deciduous Coniferous Forests
DMP	Dry Matter Productivity
DN	Deciduous Needleleaf Forest
EB	Evergreen Broadleaf Forest
EBF	Evergreen Broadleaf Forest
ECF	Evergreen Coniferous Forests
ECV	Essential Climate Variable
EEA	European Environment Agency
EN	Evergreen Needleleaf Forest
EOS	End Of Season
EOSD	End Of Season Date
EOSV	Vegetation index value at EOSD
ESA	European Space Agency
EVAL	Evaluation newtork
EVI2	2-band Enhanced Vegetation Index
FAPAR	Fraction of Absorbed Photosynthetically Active Radiation
GBOV	Ground-Based Observations for Validation
GCC	Green Chromatic Coordinate

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GCOS	Global Climate Observing System
GDMP	Gross Dry Matter Productivity
GLC-100m	Global Land Cover at 100m resolution
GLSP_QC	Global Land Surface Phenology Quality Control
GPP	Gross Primary Productivity
GR	Grasslands
HER	HERbaceous
HPLM	Hybrid Piecewise Logistic Model
HR-VPP	High Resolution Vegetation Phenology and Productivity
IPCC	Intergovernmental Panel on Climate Change
JCGM	Joint Committee for Guides in Metrology
JRC	Joint Research Centre
LAI	Leaf Area Index
LANDVAL	Land Validation
LENGTH	Length of season
LCD	Land Cover Dynamics
LPV	Land Product Validation subgroup of CEOS
LSLOPE	Slope of the green-up period
LSP	Land Surface Phenology
LST	Land Surface Temperature
MAD	Median Absolute Deviation
MAR	Major Axis Regression
MAXD	Day of the maximum of season
MAXV	Maximum value of the season at MAXD
MCDxx	MODIS products based on Terra+Aqua data
MD	Median Deviation
MINV	Minimum value of the season
MODIS	Moderate Resolution Imaging Spectroradiometer
MSI	MultiSpectral Instrument
N	Number of samples
NASA	National Aeronautics and Space Agency
NBAR	Nadir BRDF-corrected TOC Adjusted Reflectance
NDVI	Normalized Difference Vegetation Index
NEON	National Ecological Observatory Network
NetCDF	Network Common Data Form
NLF	Needle-Leaf Forest
NPN	National Phenology Network
NPP	Net Primary Productivity

NRT	Near Real Time
NV	No Vegetated
OF	Other Forests
OLCI	Ocean and Land Colour Instrument
OLS	Ordinary Least Squares
PEP725	Pan-European Phenology Project
PGQ	Proportion of Good Quality
PhenoCam	Ecosystem Phenology Camera
PPI	Plant Phenology Index
PROBA-V (or PBV)	Project for On Board Autonomy – Vegetation
PQAD	Product Quality Assurance Document
PSF	Point Spread Functions
PUM	Product User Manual
QA	CLMS LSP V2 Quality Assurance layer
QAR	Quality Assessment Report
QC	Quality Control
QF	Quality Flag
R	Correlation coefficient
ReBeLS	Regularised BRDF inversion for Land Surface
RMSD	Root Mean Square Deviation
ROI	Region of Interest
RSLOPE	Slope of the green-down period
S-2	Sentinel-2
S-3	Sentinel-3
S2GM	Sentinel-2 Global Mosaic
SBA	Sparse and Bare Areas
SH	Shrublands
SHR	SHRublands
SNPP	Suomi National Polar-orbiting Partnership
SOS	Start of the season
SOSD	Start of the season date
SOSV	Vegetation index value at SOSD
SPROD	Seasonal Productivity
ST	Seasonal Trajectories
STD	Standard deviation
SWI	Soil Water Index
TIMESAT	A program for analysing time-series of satellite sensor data
TN	Tundra

TOC	Top-of-Canopy
TPROD	Total Productivity
UM	User Manual
VI	Vegetation Index
VIIRS	Visible/Infrared Imager Radiometer Suite
VNP	SNPP/VIIRS suite of products
VPP	Vegetation Phenology and Productivity
VR	Validation Report
WGS84	World Geodetic System 1984
XX	Landscape Region

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EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) produces a series of qualified bio-geophysical products on the status and evolution of the land surface. The products are used to monitor vegetation, crops, water cycle, energy budget and terrestrial cryosphere. Production and delivery of the parameters are to take place in a timely manner and are complemented by the constitution of long-term time series.

This validation report presents the quality assessment of CLMS Land Surface Phenology version 2 (LSP V2) derived from Project for On-Board Autonomy – Vegetation (PBV, 2014–2018) and Sentinel-3 Ocean and Land Colour Instrument (OLCI, 2019–2024) observations. CLMS LSP V2 retrievals based on PBV were also evaluated against OLCI-based retrievals for the 2019 overlap year, with the aim of assessing the impact of the input-sensor change. CLMS LSP V2 parameters are derived from Plant Phenology Index (PPI) retrieved from Bidirectional Reflectance Distribution Function (BRDF)-adjusted top-of-canopy (TOC) reflectance data. The products provide a total of 13 vegetation phenology and productivity (VPP) parameters, derived for up to two growing seasons. The validation methodology is described in the CLMS Product Quality Assurance Document [CGLOPS1_PQAD]. Two main validation approaches are defined: direct validation (i.e., comparison of satellite products with in situ observations) and indirect validation or product intercomparison. Several criteria of performance are evaluated, including the product consistency across sensors, the product and completeness, spatial consistency and error evaluation (accuracy, precision and uncertainty).

The direct validation of the Start Of the Season Date (SOSD) and the End Of the Season Date (EOSD) is performed with Pan-European Phenology Project (PEP725), TEMPO and National Ecological Observatory Network (NEON) datasets, which provide phenological observations. The Ecosystem Phenology Camera (PhenoCam) V3 dataset, which infers phenometrics from digital cameras, is also used for direct comparison. The product intercomparison with global satellite-based products includes comparison with phenology (SOSD and EOSD vs. MODIS V6.1, VIIRS V2 similar phenometrics) and productivity (TPROD/SPROD vs. CLMS Gross Primary Productivity (GPP) products). The intercomparison over Europe with High Resolution VPP (HR-VPP) products is also performed for start and end of season dates and for productivity layers.

CLMS LSP V2 shows strong internal spatial and temporal consistency between PBV and OLCI, with no significant discontinuities across the sensor transition. Seasonal phenology metrics (SOSD, EOSD) and productivity indicators (SPROD, TPROD) present spatially coherent global patterns, realistic latitudinal and biome-dependent variability, and high completeness for season 1. Spatial agreement with HR-VPP is generally high. CLMS LSP V2 are also spatially consistent MODIS V6.1, and VIIRS V2 over large areas, with SOSD/EOSD differences typically within ±15 days relative to MODIS and VIIRS mid-greenup and mid-greendown definitions. Productivity metrics are systematically higher than HR-VPP (~21–23%) and correlate strongly with CLMS GPP.

Direct validation with PEP725, TEMPO, NEON, and PhenoCam V3 confirms that CLMS LSP V2 reliably captures the timing of phenological transitions. SOSD is typically slightly later than observations, with biases mainly ranging from -3 to +9 days, while EOSD is slightly earlier, with biases between -18 and +5 days, resulting in marginally shorter growing seasons. Correlations range from 0.3 to 0.77, and root mean square difference (RMSD) falls within 15–34 days. These differences are partially expected due to satellite phenophase definitions compared to diverse in-situ phenophase observations. Compliance with user requirements shows percentages within threshold typically between 30% and 50%.

Inter-annual precision shows mean absolute deviation (MAD) values of 6–7 days (PBV) and 8–9 days (OLCI) for SOSD, and 7–9 days for EOSD, with no detectable impact from the 2018–2019 sensor transition. Precision is comparable to, or better than, MODIS V6.1 and VIIRS V2, ensuring robust multi-year consistency.

Overall, CLMS LSP V2 (2014–2024) is a consistent, accurate, and robust global vegetation phenology and productivity dataset, suitable for environmental monitoring.

1 BACKGROUND OF THE DOCUMENT

1.1 SCOPE AND OBJECTIVES

This document presents the results of the scientific validation of the Copernicus Land Monitoring Service (CLMS) Land Surface Phenology 300m Version 2.0 products (CLMS LSP V2), annually generated from Project for On Board Autonomy – Vegetation (PBV, 2014–2018) and Sentinel-3 Ocean and Land Colour Instrument (OLCI, 2019–2024) observations. Additionally, the year 2019 of PBV was also compared with same overlap year of OLCI in order to check the consistency between estimates with different input sensor data.

For each year, CLMS LSP V2 products were derived using, as input data, the Bidirectional Reflectance Distribution Function (BRDF) corrected Top-of-Canopy (TOC) Adjusted Reflectance (NBAR) from PBV or OLCI, covering the period from 1st of January of the given year to the end of February of the following year.

The main objective of this assessment is to evaluate the scientific quality of CLMS LSP V2 products at global scale.

1.2 CONTENT OF THE DOCUMENT

This document is structured as follows. Chapter 2 recalls the relevant product requirements. Chapters 3 and 4 describe the main characteristics of the product under evaluation and the reference datasets, respectively. Chapter 5 outlines the methodology used for the quality assessment, including the metrics and evaluation criteria. Chapter 6 presents the results of the analysis, and Chapter 7 summarizes the main conclusions of the study.

1.3 RELATED DOCUMENTS

1.3.1 Applicable documents

AD1: Part 2: Technical specifications of Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' JRC/2023/OP/0273, 19th April 2023.

Available at https://etendering.ted.europa.eu/cft/cft-display.html?cftId=13795

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1.3.2 Input documents

Document ID	Descriptor
CGLOPS1_PQAD	Product Quality Assurance Document "Vegetation & Energy".
CGLOPS1_ATBD_LSP300m-V2	Algorithm Theoretical Basis Document of Land Surface Phenology 300m Version 2
CGLOPS1_ATBD_BRDFCorrection300m-V1	Algorithm Theoretical Basis Document of the Version 1 of the correction of the Bidirectional Reflectance Distribution Function
CGLOPS1_ATBD_DMP-NPP300m-V1.1	Algorithm Theoretical Basis Document of 300m Dry Matter Productivity and Net Primary Production Version 1.1 product
CGLOPS1_ATBD_FAPAR300m-V1.1	Algorithm Theoretical Basis Document of 300m LAI, FAPAR, FCOVER Version 1.1 product, derived from Sentinel-3/OLCI data
CGLOPS1_QAR_DMP-NPP300m-V1.1	Quality Assessment Report of DMP, GDMP, NPP and GPP 300m Version 1.1

Available at: https://land.copernicus.eu/en/technical-library

1.3.3 Output documents

Document ID	Descriptor
CGLOPS1_PUM_LSP300m-V2	Product User Manual of Land Surface Phenology Version 2.0.

Available at: https://land.copernicus.eu/en/technical-library

1.3.4 External documents

Document ID	Descriptor
GCOS#245	The 2022 GCOS ECVs Requirements

https://library.wmo.int/records/item/58111-the-2022-gcos-ecvs-requirements-gcos-245?offset=1

GCOS#200 The Global Observing System for Climate: Implementation needs. World Meteorological Organization, Geneva, Switzerland (2016)

https://library.wmo.int/opac/doc_num.php?explnum_id=3417

JCGM, 2014 JCGM, 2014. International Vocabulary of Metrology–Basic and General Concepts and Associated Terms, Chemistry International -- Newsmagazine for IUPAC. Walter de Gruyter GmbH.

https://doi.org/10.1515/ci.2008.30.6.21

HR-VPP_ST&VPP_ATBD Copernicus Land Monitoring Service. High-Resolution Vegetation Phenology and Productivity (HR-VPP), Seasonal Trajectories and VPP parameters. Algorithm Theoretical Basis Document

HR-VPP_ST&VPP_UM Copernicus Land Monitoring Service. High-Resolution Vegetation Phenology and Productivity (HR-VPP), Seasonal Trajectories and VPP parameters. User Manual

HR-VPP_ST&VPP_VR Copernicus Land Monitoring Service. High-Resolution Vegetation Phenology and Productivity (HR-VPP), Seasonal Trajectories and VPP parameters. Validation Report 2021-2024

Available at: https://land.copernicus.eu/en/technical-library

MOD12Q2_User_Guide_V61 User Guide to Collection 6.1 MODIS Land Cover Dynamics (MCD12Q2) Product.

https://lpdaac.usgs.gov/documents/1417/MCD12Q2_User_Guide_V61.pdf

VNP22_User_Guide_V1 VIIRS Global Land Surface Phenology Product User Guide.Version 1.1

https://lpdaac.usgs.gov/documents/637/VNP22_User_Guide_V1.pdf

IPCC 2003 Intergovernmental Panel on Climate Change - Good Practice Guidance for Land Use, Land-Use Change and Forestry.

https://www.ipcc.ch/site/assets/uploads/2018/03/GPG_LULUCF_FULLEN.pdf

NEON.DOC.014040 NEON TOS Protocol and procedure: Plant Phenology

https://data.neonscience.org/documents/10179/2139401/NEON.DOC.014040vD/6a4f81a7-20b5-ba40-ad63-7541ffe959d9?version=1.0

2 REQUIREMENTS

According to the applicable document [AD1], the technical requirements relevant for the LSP are described below.

2.1 SPECIFIC TECHNICAL DETAILS AND REQUIREMENTS

Table 1: CLMS relevant technical requirements for LSP.

PRODUCT SPECIFICATION	
Geometric Properties:	
Spatial resolution	300 m
Geolocation precision	Better than 0.5 pixels
Coordinate position	Centre of the pixel
Geodetic datum	WGS84
Geographic projection	Regular latitude/longitude grid
Geographic coverage	Global
LSP – LAND SURFACE PHENOLOGY	
LSP indicates the timing, date of occurrence, and corresponding value of the vegetation parameter, of key phases of the vegetation development cycle; Phenological metrics related to the start, end of growing season, length of the season and maximum development, seasonal total and partly integrals (as productivity parameters), as well as growth and senescence rates, intra-annual coefficient of variation, etc.	
Time span	Yearly
Timeliness	after one month of last year observation
Uncertainty (2-sigma)	Dates related metrics: Threshold: 10 days, Goal: 5 days
Stability (per decade, i.e. 10 years)	Threshold: 10% Goal: 5%

2.2 FURTHER REQUIREMENTS

2.2.1 Output product composition

Products may contain various information layers and ancillary information – the base reference for product packages are the operational products as on 01.03.2023.

2.2.2 Data Structure

Data coding[1] shall be compatible with the Global Land products as on 01.03.2023 and/or follow the INSPIRE specifications, where applicable.
Ancillary information shall be as currently used and include at least the following:

- The number of measurements per pixel used to generate any synthesis product
- The per-pixel date of the individual measurement or the start-end dates of the period covered
- Quality indicators, with explicit per-pixel identification of the cause of anomalous parameter result.

The product naming and filename conventions that are used in the Copernicus Global Land component production as on 01.03.2023 shall be followed. This may be adapted for complete product collections upon agreement with the contracting authority during the Framework.

2.2.3 Data format

To ensure interoperability with the current Global Land component (operational product data formats and archive data formats) and other Copernicus services, all datasets will be available in NETCDF. Additional format such as Cloud Optimized Geotiff (COG) or ZAR format can be proposed for production or could be requested by the contracting authority during the framework contract.

2.2.4 Uncertainties and Validation

Uncertainties indicated in the product specifications above follow the threshold and goal proposed by GCOS#245.

Uncertainties estimates should account for the error propagation uncertainty coming from input data though the retrieval algorithms as during the contract period, ESA plans to imbed uncertainties in the Sentinel-3 ground segment products and, for Sentinel-2, there is an offline tool to determine Evel 1 uncertainties; these can be used for propagation in the production chain.

Validation of the products shall conform to at least the CEOS LPV standards. Wherever appropriate the bio-geophysical variables shall be validated and compared to CEOS CAL/VAL data sets and/or Ground-Based Observations for Validation (GBOV) of Copernicus Global Land component when biophysical parameters are available.

2.2.5 Input data

Copernicus sentinel data are available from https://dataspace.copernicus.eu

Bio-geophysical variable products should be based on common base reflectance data:

[1] Data coding is the provision for the number of bits, range of values, usage of reserved values, content of status map, conversion to physical values formulas.

- Sentinel 3: Derived Top-of-Canopy reflectance may be brokered or produced as it is the case on 01.03.2023 under the CGLOPS contracts;
- Sentinel 2: the Global Land Sentinel-2 Global Mosaic (S2GM)[2] component provides temporal mosaic of surface spectral bands that can be brokered and/or directly used.

Up to and including 2019, products archives of the Global Land component have been based on SPOT VGT, Proba-V, ENVISAT, MODIS, TOPEX/Poseidon, Jason-1, Jason-2, Jason-3, datasets, which are available through the https://dataspace.copernicus.eu.

LST and SWI can be based on geostationary and other satellite data.

Ancillary satellite data that is purchased through Copernicus and put at disposal of the Services is available through the Data Warehouse and will become available on the Copernicus Dataspace Ecosystem.

Ancillary data sets, other than satellite imagery described above, that might be required shall be the responsibility of the contractor.

2.2.6 Product delivery

Products shall be delivered to the Copernicus Land Component dissemination.

The Copernicus Data Space Ecosystem infrastructure will be used to provide access to the final map products.

[2] https://land.copernicus.eu/en/products/global-image-mosaic

3 CLMS LAND SURFACE PHENOLOGY (LSP) V2

The calculation of annual CLMS LSP V2 products [CGLOPS1_ATBD_LSP300m-V2] is based on the following four steps:

(1) **Data preparation**: generating NBAR using Regularised BRDF Inversion for Land Surface (ReBeLS) retrieval algorithm [CGLOPS1_ATBD_BRDFCorrection300m-V1]. The BRDF algorithm normalizes the TOC reflectance to the nadir viewing at 12:00 local noon.

(2) **Generation of the Plant Phenology Index (PPI)**, which is a physically based vegetation index developed from a simplified solution to the radiative transfer equation (Jin and Eklundh, 2014). The input data for the LSP estimation are global 5-day PPI time series calculated from PROBA-V and Sentinel-3/OLCI NBAR in the red band (Oa08 band for OLCI) and the near infrared band (Oa17 band for OLCI).

(3) To smooth and gap-fill data, **PPI seasonal trajectories** are generated, which are based on cubic spline functions implemented in a program for analyzing time-series of satellite sensor data (TIMESAT - Jönsson and Eklundh, 2004), version 4.2. These functions adapt smoothly to seasonal and inter-annual vegetation index variations. Cubic spline functions have high fidelity and flexibility to represent satellite measurements of plant phenology (Bolton et al., 2020; Cai et al., 2017).

(4) From the seasonal trajectories (data not stored as output in the CLMS product), **13 phenology and productivity parameters for up to two growing seasons per year are extracted** (Figure 1).

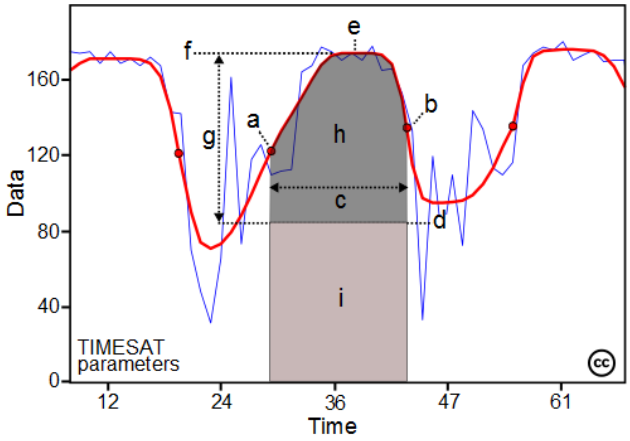


Figure 1. Examples of TIMESAT-derived phenology and productivity parameters: (a) start of season, (b) end of season, (c) length of season, (d) minimum value, (e) time of maximum, (f) maximum value, (g) amplitude, (h) seasonal productivity, (h+i) total productivity. Source: https://web.nateko.lu.se/timesat/timesat.asp.

The computation of start and end of seasons is based on defined threshold points on the smoothed seasonal trajectory. TIMESAT determines the timing of start of season (SOSD) and end of season (EOSD) by applying a threshold value in relation to the seasonal amplitude (25% for SOSD and 15% for EOSD). Once start and end dates have been defined, the other parameters are extracted.

CLMS LSP V2 product provides information on vegetation phenology and productivity up to two growing seasons. Each season includes 13 parameters (Table 2) and is accompanied by a Quality Assurance (QA) flag layer to inform about data reliability [CGLOPS1_PUM_LSP300m-V2]. QA is 8-bit encoded, where two bits are assigned for each aspect: SOS (bits 0-1), peak (bits 2-3), EOS (bits 4-5) and overall (bits 6-7) quality. The details of bit values and descriptions are listed in Table 3.

Table 2: Vegetation Phenology and Productivity parameters generated in CLMS LSP V2 products.

No.	Name	Data type	Scale, offset	Data range	Fill value	Description	Unit
1	SOSD	INT16	1, 0	[-365, 365]	-9999	Day of start-of-season	Day-Of-Year
2	EOSD	INT16	1, 0	[0, 730]	-9999	Day of end-of-season	
3	MAXD	INT16	1, 0	[0, 366]	-9999	Day of maximum-of-season	
4	SOSV	INT16	0.001, 0	[0, 5000]	-9999	Vegetation index value at SOSD	PPI unit $m^2.m^{-2}$
5	EOSV	INT16	0.001, 0	[0, 5000]	-9999	Vegetation index value at EOSD	
6	MINV	INT16	0.001, 0	[0, 5000]	-9999	Average vegetation index value of minima on left and right sides of each season	
7	MAXV	INT16	0.001, 0	[0, 5000]	-9999	Vegetation index value at MAXD	
8	AMPL	INT16	0.001, 0	[0, 5000]	-9999	Season amplitude (MAXV – MINV)	
9	LENGTH	INT16	1, 0	[0, 730]	-9999	Length of Season (number of days between start and end)	day
10	LSLOPE	INT16	0.001, 0	[0, 1000]	-9999	Slope of the green-up period	$m^2.m^{-2}.day^{-1}$
11	RSLOPE	INT16	0.001, 0	[0, 1000]	-9999	Slope of the green-down period (absolute value of decreasing rate)	
12	TPROD	INT16	1, 0	[0, 2000]	-9999	Total productivity. Growing season integral computed as the sum of all daily values between SOSD and EOSD.	$m^2.m^{-2}.day$
13	SPROD	INT16	1, 0	[0, 2000]	-9999	Seasonal productivity. Growing season integral computed as sum of all daily values minus their base level value.	
Aux	QA	UINT8	-	-	255	Quality flag	-

Table 3. CLMS LSP V2 QA flags and descriptions of phenology quality.

QA byte	Quality	Description
SOS/Peak/EOS quality (two-bit values)		
0	Best	More than 2 valid observations and average best PPI quality
1	Good	More than 2 valid observations and average good PPI quality
2	Fair	More than 2 valid observations and average fair PPI quality
3	Poor	Average poor PPI data quality, or fewer than 3 observations, or more than > 24 observations*
Overall quality (two-bit values)		
0	Best	No poor quality in three phenology phases
1	Good	Maximum one phase with poor quality
2	Fair	Maximum two phases with poor quality
3	Poor	All three phases with poor quality**
Summary of QA (byte values)		
QA <127* **	Good	Overall good quality, no more than one poor results in three phases
QA >127* **	Poor	Overall poor quality, more than one poor results in three phases
QA=255	Failure	No input, or no phenology estimation, or all poor estimates

Notes:
* Includes three situations: 1) fewer than 3 valid observations, 2) phenology phase shorter than 10 days, and 3) phenology phase longer than 4 months.
** Overlaps QA=255 for situations of no inputs, or no phenology estimation. In any case, the LSP retrieval is considered a failure.
*** There is no case QA=127

4 REFERENCE DATASETS

4.1 SATELLITE PRODUCTS

This section provides an overview of the retrieval algorithms of the satellite-based products used in this exercise. The main characteristics of the satellite products are summarized per two families: phenology (Table 4) and productivity (Table 5). Summary of retrieval algorithms are provided hereafter.

Table 4: Characteristics of the remote sensing phenology products used in this exercise.

Product	Satellite / Sensor	Resolution / Period	Coverage	Input:	Fitting / Smoothing methods	Phenology Transition	Reference
CLMS LSP V2*	PROBA/V S-3/ OLCI	300m / 2014-2024	Global	BRDF-normalized PPI	Cubic spline and double-logistic fitting	25% and 15% of amplitude for green-up and green-down	[CGLOPS1_ATBD_LSP 300m-V2]
MODIS LCD V6.1	Terra / MODIS	500m / 2001-current	Global	NBAR EVI2	Cubic spline fitting	15%, 50%, and 90% of amplitude for green-up and green-down	(Zhang et al., 2006) [MOD12Q2_User_Guide_V61]
VIIRS LCD V2	-SNPP / VIIRS	500m / 2013-current**	Global	NBAR EVI2	Stavisky-Golay filter, mean/median filter, logistic fitting	Third derivative (extreme curvature change rate)	(Zhang et al., 2018b) [VNP22_User_Guide_V1]
HR-VPP	S-2 / MSI	10m / 2017-current	Europe	PPI	Cubic spline and double-logistic fitting	25% and 15% of amplitude for green-up and green-down	[HR-VPP_ST&VPP_ATBD]

*dataset under evaluation
** year 2024 was not available at the moment of the validation report

Table 5: Characteristics of the remote sensing productivity products used in this exercise.

Product	Satellite / Sensor	Resolution / Period	Coverage	Input	Method	Reference ATBD
CLMS LSP V2*	PROBA/V S-3/ OLCI	300m / 2014-2024	Global	BRDF-normalized PPI	PPI integral between SOSD and EOSD	[CGLOPS1_ATBD_LSP300m-V2]
CLMS GPP	PROBA/V S-3/ OLCI	300m / 2014-present	Global	FAPAR	(Monteith, 1972)	[CGLOPS1_ATBD_DMP-NPP300m-V1.1].
HR-VPP	S-2 / MSI	10m / 2017-current	Europe	PPI	PPI integral between SOSD and EOSD	[HR-VPP_ST&VPP_ATBD]

*dataset under evaluation

4.1.1 Reference phenology global products

4.1.1.1 NASA MODIS Land Cover Dynamics Version 6.1 (MODIS LCD V6.1)

The Terra and Aqua combined Moderate Resolution Imaging Spectroradiometer (MODIS) Land Cover Dynamics (MCD12Q2) Version 6.1 data product[3] provides global maps of phenometrics at 500 m with an annual time step since 2001 (Zhang et al., 2006) [MOD12Q2_User_Guide_V61]. The MCD12Q2 Land Cover Dynamics Version 6.1 data product (**MODIS LCD V6.1** hereafter) is derived from time series of the 2-band Enhanced Vegetation Index (EVI2) calculated from MODIS Nadir BRDF-NBAR. Vegetation phenology metrics at 500-meter spatial resolution are identified for up to two detected growing cycles per year. For pixels with more than two valid vegetation cycles, the data represent the two cycles with the largest NBAR-EVI2 amplitudes. For areas where the NBAR-EVI2 values are missing due to cloud cover or other reasons, the data gaps are filled with good quality NBAR-EVI2 values from the year directly preceding or following the product year.

The layers provided by the product correspond to the total number of vegetation cycles detected for the product year (up to two cycles/year), the onset of greenness, green-up midpoint, maturity, peak greenness, senescence, green-down midpoint, dormancy, EVI2 minimum, EVI2 amplitude, integrated EVI2 over a vegetation cycle, as well as overall and phenology metric-specific quality information. MODIS LCD V6.1 provides different phenological dates using 15% (*Greenup* and *Dormancy* for start and end of season), 50% (*MidGreenup* and *MidGreendown*), and 90% (*Maturity* and *Senescence*) EVI2 amplitude.

Comparison of vegetation phenology retrieved from MODIS with in situ measurements revealed realistic estimates over Harvard Forest (Zhang et al., 2006). More generally, the spatial distribution of phenological metrics estimated from MODIS data were qualitatively realistic, and exhibits strong correspondence with temperature patterns in mid- and high-latitude climates, with rainfall seasonality in seasonally dry climates, and with cropping patterns in agricultural areas (Zhang et al., 2006).

4.1.1.2 NASA VIIRS Land Cover Dynamics Version 2 (VIIRS LCD V2)

The Suomi National Polar-Orbiting Partnership (SNPP) Visible/Infrared Imager Radiometer Suite (VIIRS) LCD data product[4] generates phenological metrics of individual vegetation growing cycles on the vegetated land surface at a spatial resolution of 500m across the globe (Zhang et al., 2018b)

[3] https://www.earthdata.nasa.gov/data/catalog/lpcloud-mcd12q2-061 (last access on 18/11/2025)

[4] https://www.earthdata.nasa.gov/data/catalog/lpcloud-vnp22q2-002 (last access on 18/11/2025). Note: year 2024 was not available for this data access.

[VNP22_User_Guide_V1]. The VNP22Q2 V2 dataset (**VIIRS LCD V2** hereafter) is derived from time series of EVI2 calculated from VIIRS Nadir BRDF-NBAR. Vegetation phenology metrics at 500-meter spatial resolution are identified for up to two detected growing cycles per year. VIIRS LCD V2 products are generated with same algorithm as previous V1, but introduces some improvements or changes:

- Improved calibration algorithm and better coefficients for entire SNPP mission.
- Improved geolocation accuracy and updates to fix outliers around maneuver periods and other events.
- Corrections to the aerosol quantity flag (low, average, high) mainly over brighter surfaces in the mid to high latitudes such as desert and tropical vegetation areas. This has an impact on the retrieval of other downstream data products such as VIIRS Vegetation Indices and BRDF/Albedo.
- Improved cloud mask input product for corrections along coastlines and artifacts from use of coarse resolution climatology data.
- Replaced the land/water mask input product with MODIS heritage seven class land/water mask.

The main changes respect to the predecessor based on MODIS (MODIS LCD V6.1) are: first, VIIRS LCD V2 reconstructs the EVI2 time series using a physically-based hybrid piecewise logistic model (HPLM), while MODIS LCD V6.1 fits the EVI2 time series using a penalized cubic smoothing spline. Second, VIIRS V2 identifies phenological transition dates using the curvature change rate derived from the HPLM without any predefined thresholds involved, whereas MODIS V6.1 detects phenological dates using thresholds of EVI2 amplitude. Third, VIIRS V2 records the phenological metrics occurring within a given calendar year or product year. However, MODIS V6.1 records the metrics for the vegetation cycles for which the peak date falls within the calendar year of the product year.

The product contains six phenological transition dates: onset of greenness increase, date of mid-greenup, onset of greenness maximum, onset of greenness decrease, date of mid-senescence and onset of greenness minimum phases. The product also includes the growing season length. The greenness related metrics consist of EVI2 onset of greenness increase, EVI2 onset of greenness maximum, EVI2 growing season, rate of greenness increase and rate of greenness decrease. The confidence of phenology detection is provided as greenness agreement growing season, Proportion of Good Quality (PGQ) growing season, PGQ onset greenness increase, PGQ onset greenness maximum, PGQ onset greenness decrease, and PGQ onset greenness minimum. The final layer (Global Land Surface Phenology Quality Control, GLSP_QC) specifies the overall quality of the product. A low-resolution browse image showing green-up is also available when viewing each VIIRS LCD V2 granule.

Comparison of VIIRS LCD resulting phenological metrics over United States with species-specific field phenological observations, Landsat phenology retrievals and the MODIS phenology detections showed that the VIIRS phenometrics are of high quality and in a good agreement with the other independent satellite and field observations (Zhang et al., 2018b). Globally, the phenological dates derived from VIIRS EVI2 were compared with MODIS detections (Zhang et al., 2018a), which showed a good agreement with an average absolute difference less than a week except for the senescence onset. Phenological dates derived from VIIRS EVI2 and PhenoCam were generally comparable for the vegetation green-up phase, but differed considerably for the senescence phase (Zhang et al., 2018a). The continuity between VIIRS V2 and MODIS V6.1 phenometrics was assessed and they agreed well and had little systematic bias (Román et al., 2024).

4.1.2 Reference global productivity products

4.1.2.1 CLMS Gross Primary Productivity Collection 300m V1.1 (CLMS GPP)

The CLMS GPP is provided in the CLMS Dry Matter Productivity/Gross Dry Matter Productivity (DMP/GDMP) & Net Primary Productivity/Gross Primary Productivity (NPP/GPP) V1.1 product suite [CGLOPS1_ATBD_DMP-NPP300m-V1.1]. The products are delivered, over the globe, every 10 days with a maximum of 3 days in Near Real Time (NRT) followed by consolidation in the course after 6 dekads.

GPP is derived from the GDMP product, which represents the total dry matter fixed by vegetation through photosynthesis. GDMP is generated using a Monteith-based light-use-efficiency model (Monteith, 1972) that integrates CLMS FAPAR 300m V1.1 10-daily products [CGLOPS1_ATBD_FAPAR300m-V1.1] and land cover information based on European Space Agency (ESA) Climate Change Initiative (CCI) Land Cover Map epoch 2010. GPP is obtained by converting GDMP from dry-matter units to carbon units using a standard 0.5 carbon-fraction factor from Intergovernmental Panel on Climate Change (IPCC) guidelines [IPCC 2003].

$$\text{GPP } [\text{gC/m}^2\text{/day}] = \text{GDMP } [\text{kgDM/ha/day}] \times 0.5 \times 0.1$$

QAR [CGLOPS1_QAR_DMP-NPP300m-V1.1] for July 2018 – April 2019 period revealed that, in the overall global comparisons with MODIS, CLMS GDMP OLCI product exceeds MODIS for most biomes. A validation with Fluxes Network (FLUXNET) data spread over 66 sites shows that there is a fairly good overall relation between satellite-based products and the FLUXNET in situ based GDMP, but the uncertainty is high and precision of the relation is low. The results obtained in this report are also valid for NPP and GPP, as these are a simple rescaling from kgDM/ha/day to gC/m²/day, taking into account the Carbon Use Efficiency, which is taken as a constant term (0.5 according to [IPCC 2003]).

4.1.3 CLMS High-Resolution Vegetation Phenology and Productivity (HR-VPP)

The HR-VPP product suite, produced within the Pan-European component of the CLMS, are derived from the Sentinel-2 constellation (Sentinel-2A and Sentinel-2B) at high spatial resolution (10m x 10m) with a revisit time of 5 days [HR-VPP_ST&VPP_UM]. They are generated over the entire administrative boundaries of the 39 European Environment Agency (EEA) countries (EEA39: 32-member countries, the United Kingdom and 6 cooperating countries in the Western Balkans) from January 1, 2017, onwards.

The HR-VPP product suite contains 3 product groups:

- The **raw Vegetation Indices** (VI). This group consists of products generated in NRT providing, for every pixel, the status of the vegetation vigor. The status is provided in four ways: Leaf Area Index (LAI), FAPAR, Normalized Difference Vegetation Index (NDVI) and PPI;
- The **Seasonal Trajectories** (ST) products are provided yearly after the end of the vegetation growing season. These are derived as a regular time-series of every 10 days by fitting a smoothing and gap filling function to the raw PPI, generated in the product group VIs. TIMESAT version 4 software package (Jönsson and Eklundh, 2004) is used to produce STs from PPI retrievals;
- The **Vegetation Phenology and Productivity** (VPP) parameters products bundle are derived from the STs of the PPI index, on a yearly basis, after the end of the growing season. VPP metrics are provided for up to two growing seasons, being e.g. start of the season, end of season, seasonal productivity, etc. These metrics are defined in concordance with CLMS LSP V2 products, following similar algorithm [HR-VPP_ST&VPP_ATBD] (Tian et al., 2021).

Different versions of HR-VPP were generated depending on the production period [HR-VPP_ST&VPP_UM]: V102-V104 (2017–2020), which corrected several issues identified in the initial V101 run; and V105 (2021–2024), which incorporates an additional outlier-detection step to remove minor spurious second seasons. The latest version, V105, is used in this study.

Validation of HR-VPP phenology metrics for 2021-2024 period [HR-VPP_ST&VPP_VR] showed good agreement with existing medium-resolution products (MODIS LCD V6.1 and VIIRS LCD V1) over Europe. The accuracy of phenology metrics (SOSD, EOSD) as compared with ground PEP725 observations showed a slight positive bias of 4 days for SOSD (i.e., late start of season) and negative bias of -6 days for EOSD (i.e., early end of season).

4.2 IN-SITU REFERENCE PRODUCTS

4.2.1 Pan-European Phenology Project (PEP725)

The Pan-European Phenology Project (PEP725, http://www.pep725.eu/) (Templ et al., 2018) has the objective to establish an open access database of plant phenology for science, education and research. The database presently holds more than 16 million records, about 97 growing stages and 338 plant species (including cultivars). However, this dataset has some limitations for remote sensing validation purposes as is not evenly distributed around the world and the geolocation accuracy of the observations is often poor (2 – 4 km). For that reason, a strategy using Corine Land Cover (CLC) was performed similar to that used in Tian et al., 2021. CLC 2018 (v2020_20u1) provides inventory of land cover in 44 different classes, with a Minimum Mapping Unit of 35 ha for areal phenomena and a minimum width of 100 m for linear phenomena in Europe. Several species were grouped into two vegetation groups (Table 6).

Table 6: Grouping strategy for linking CLMS LSP V2 with PEP725 phenology observations.

Vegetation group	PEP725 genus	CLC classes
Deciduous broad-leaved trees	Acer, Aesculus, Alnus, Betula, Fagus, Fraxinus, Populus, Quercus, Robinia, Salix, Sambucus, Sorbus, Syringa and Tilia.	Broad-leaved forest, Transitional woodland-shrub, Mixed forest, Green urban areas.
Evergreen coniferous trees	Larix, Picea and Pinus.	Coniferous forest.

For each site, a 4 km x 4 km window around the station is selected, then the CLC is resampled to the CLMS LSP V2 300m spatial resolution and the percentage of each CLC class is computed. Then, the pixels corresponding to the main vegetation group (i.e., deciduous broad-leaved trees or evergreen coniferous trees) are identified. The median CLMS LSP V2 SOSD and EOSD values for each vegetation group is obtained and only used in the comparison if the vegetation group represents more than 10% of the area. Figure 2 shows an example of aggregated vegetation group for PEP725 site ID=110. In the example, only DBF pixels are used in the comparison, as represent 35.6% in the 4 km x 4 km area. The phases considered in the comparison are the same to that used in Tian et al., 2021 and are displayed in Table 7.

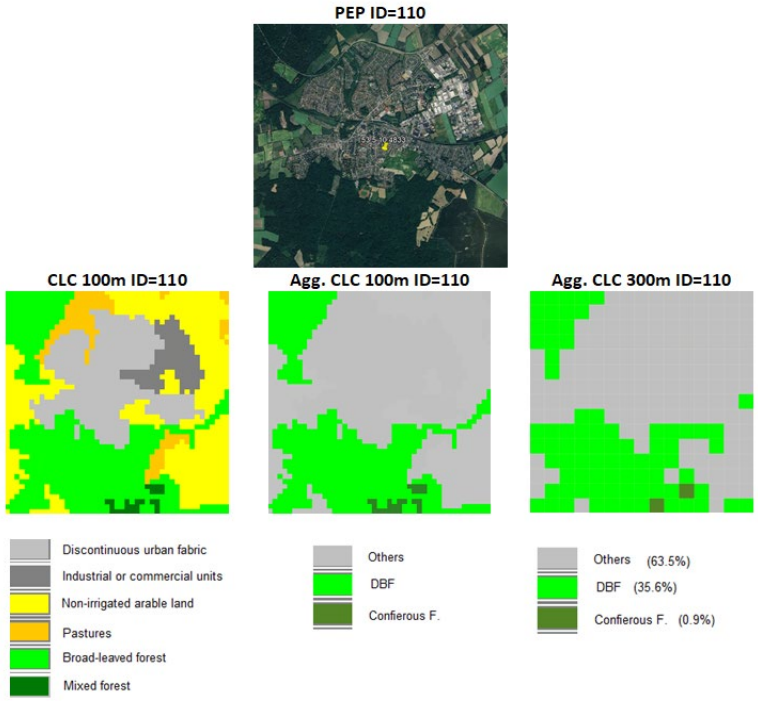


Figure 2: Vegetation group strategy. Top: Google Earth view of 4 km x 4 km around PEP site ID=110. Bottom: CLC around the site at 100m of spatial resolution (left), aggregated CLC around the site at 100 m resolution (center) and aggregated CLC around the site resampled at 300 m resolution (right).

Table 7: Phases used for linking CLMS LSP V2 SOSD and EOSD with PEP725 phenology observations.

PEP725 Phase_id	Description	Short name	Metrics
10	Gramineae: Fist true leaf emerged from coleoptile, Dicotyledons and Monocotyledons: Cotyledons completely unfolded, Perennial plants: First leaves separated	First leaves separated	SOSD
11	First true leaf, leaf pair or whorl unfolded, Perennial plants: First leaved unfolded	First leaves unfolded	SOSD
223	Leaf unfolding (≥50%)	Leaf unfolding >=50%	SOSD
95	50% of leaves fallen	50% of leaves fallen	EOSD
205	Autumnal coloration ≥50%	Autumnal colouration >=50%	EOSD

A total of 2244 out of 7355 available sites in the current PEP725 dataset[5], providing data in 2014–2024 period, are used in the validation after performing the vegetation group selection with availability of selected phenophases.

4.2.2 TEMPO

TEMPO (https://tempo.pheno.fr/) (Maury et al., 2023) is the French national network of observatories whose objective is to answer how climate change influences in the seasonal rhythms of living organism and productivity. TEMPO counts with 10 observatories and more than 95 partners (research units, technical and public institutions, etc.).

As in the comparison with PEP725, several species were grouped into two vegetation groups (Table 8). Phenological stages are recorded following *the Biologische Bundesanstalt, Bundessortenamt und CHemische Industrie* (BBCH) reference system, which describes annual crops and fruit trees phenological stages. Similar growth stages of each plant species are given the same BBCH code. The phenological stages used as potential start of season are "Germination / sprouting / bud development" and "Leaf development (main shoot)" and the stage used as potential end of season is "Beginning of dormancy, die off of leaves, stem" (see Table 9).

Table 8: Grouping strategy for linking CLMS LSP V2 SOSD and EOSD with TEMPO phenology observations.

Vegetation group	TEMPO genus	CLC classes
Deciduous broad-leaved trees	Aesculus, Alnus, Betula, Fagus, Fraxinus, Populus, Quercus, Robinia, Salix, Sambucus, Sorbus and Syringa.	Broad-leaved forest, Transitional woodland-shrub, Mixed forest, Green urban areas.
Evergreen coniferous trees	Larix, Picea and Pinus.	Coniferous forest.

[5] http://www.pep725.eu (data access 20/10/2025)

Table 9: Phases used for linking CLMS LSP V2 SOSD and EOSD with TEMPO phenology observations.

TEMPO phenological event code	Description	Metrics
0	Germination / sprouting / bud development	SOSD
1	Leaf development (main shoot)	SOSD
9	Beginning of dormancy, die off of leaves, stem, …	EOSD

For each site, genus and phenophase, the median values of the TEMPO samples are compared to the satellite products at native resolution. The Copernicus Global Land Cover 100m (GLC-100m) (Buchhorn et al., 2020) has been used to discard sites classified as urban or water bodies. The current dataset (V2[6]) counts with more than 170 species and 12737 sites in the 2014–2024 period. A total of 2102 sites are available after performing the vegetation group selection with availability of selected phenophases.

4.2.3 National Science Foundation's National Ecological Observatory Network (NEON)

The National Science Foundation's National Ecological Observatory Network (NEON, https://www.neonscience.org/) is a continental-scale observation facility designed to collect long-term open access ecological data to better understand ecosystem changes. NEON provides free and open data that characterize plants, animals, soil, nutrients, freshwater and the atmosphere.

The NEON Plant Phenology observations dataset contains quality-controlled native sampling resolution data from in-situ observations of plant leaf development and reproductive phenophases [NEON.DOC.014040]. Phenophase status and intensities follow those of the USA National Phenology Network (USA-NPN). The status and intensity data are reported per phenophase, individual or patch for every sampled day.

The phenophases used as potential start of season are "breaking leaf/needle buds" and "emerging leaves/needles," and the phenophases used as end of season are "colored leaves/needles" and "falling leaves/needles". Each species of NEON is classified according to their growth form. The available growth forms were aggregated according to Table 10. For each plot, a recount of how many individual species belong to an aggregated growth form has been made. The final phenophase used to compare with satellite products is the median phenophase of the individual species, weighted by the proportion of aggregated growth form they represent at the plot.

[6] https://tempo.pheno.fr/eng/data/data-portal (data access 20/10/2025)

Table 10: Growth Forms aggregation for NEON data.

Aggregated growth forms	Growth forms
DBF (Deciduous Broadleaf Forests)	Deciduous broadleaf Drought deciduous broadleaf
DCF (Deciduous Coniferous Forests)	Deciduous conifer
EBF (Evergreen Broadleaf Forests)	Evergreen broadleaf Evergreen broadleaf –no leaf buds Evergreen broadleaf – no leaves Evergreen broadleaf – no needles Semi-evergreen broadleaf
ECF (Evergreen Coniferous Forests)	Evergreen conifer Pine
HER (Herbaceous)	Cactus Graminoid Evergreen forb Forb Semi-evergreen forb

The whole dataset (DP1.10055.001[7]) counts with more than 600 species and has 47 sites and typically 2 different plots per site. A total of 37 plots, available in 2014–2024 period, provide data for the growth forms and phenophases selected for validation.

4.2.4 Ecosystem Phenology Camera (PhenoCam V3)

The Ecosystem Phenology Camera network (PhenoCam, https://phenocam.nau.edu/webcam/) provides automated near-surface remote sensing of canopy phenology. For that, high-resolution digital cameras were distributed mainly throughout north-eastern United States and adjacent Canada, and currently extended to other regions around the globe. Images of the cameras are uploaded to PhenoCam server every half hour, extracting quantitative colour information from each picture. Canopy greenness indices provide information about the amount of foliage present and its colour. The image analysis of these cameras provides an objective means by which canopy phenology can be monitored and quantified.

The PhenoCam standard index is the green chromatic coordinate (GCC), which is the relative brightness of the green channel of cameras normalized against the overall brightness of the red, green and blue channels. From GCC time series, PhenoCam also provides an estimation of the green-up

[7] NEON. Plant phenology observations (DP1.10055.001), provisional data. Dataset accessed from https://data.neonscience.org/data-products/DP1.10055.001 (data access 20/10/2025)

and green-down transition dates, which correspond with the start and end of the growing season. To estimate the transition dates, first the maximum and minimum values of greenness are estimated. Then, the 10%, 25% and 50% of the greenness curve range is computed as transition dates.

PhenoCam V3 (pre-release[8]) counts with a total of 1383 Region of Interest (ROI) masks and data from 2000 to 2024. Each PhenoCam site can have different ROI's with different vegetation types in the same coordinate. In order to choose the vegetation type ROI more similar to the satellite retrievals, a strategy of selecting ROIs was developed, choosing the ROIs with vegetation type similar to the Copernicus GLC-100m aggregated classes (i.e., PhenoCam forest vegetation types was selected if GLC-100m indicates majority of forest and PhenoCam non-forest vegetation types if GLC-100m indicates majority of non-forest, as shown in Table 11). Finally, a total of 752 ROIs are used.

Table 11: Strategy for choosing PhenoCam ROIs vegetation types according to GLC-100m.

	GLC-100m aggregated classes	PhenoCam Vegetation Types
Forests	EBF (Evergreen Broadleaf Forests) DBF (Deciduous Broadleaf Forests) NLF (Needleleaf Forests) OF (Other Forests)	EB (Evergreen Broadleaf Forests) DB (Deciduous Broadleaf Forests) EN (Evergreen Needleleaf Forests) DN (Deciduous Needleleaf Forests)
Non-forests	CUL (Cultivated) HER (Herbaceous) SHR (Shrublands) SBA (Sparse and Bare Areas)	AG (Agriculture) GR (Grasslands) TN (Tundra) SH (Shrublands) XX (Landscape Region) NV (No vegetated)

[8] https://phenocam.nau.edu/phenocam_explorer_prerelease/ (data access 20/10/2025)

5 VALIDATION METHODOLOGY

The Quality Assessment follows the procedures described in the PQAD [CGLOPS1_PQAD]. The procedures are mainly based on the Committee on Earth Observation Satellites (CEOS) Land Product Validation subgroup (LPV) validation protocol for biophysical variables (Fernandes et al., 2014) and on the state-of-the-art for validation of satellite-based phenology products (Moon et al., 2021; Bolton et al., 2020; Zhang et al., 2018b; Peng et al., 2017; Melaas et al., 2016; Rodriguez-Galiano et al., 2015;). The proposed methodology relies on direct validation and product intercomparison approaches.

- The **direct validation,** comparing satellite products to ground observations of the corresponding biophysical variables. Direct validation is usually limited by a small number of reference sites. PEP725 (see section 4.2.1), TEMPO (section 4.2.2) and NEON (section 4.2.3) datasets provide phenological observations and are used to validate the start and end of the growing season. In addition, PhenoCam V3 (section 4.2.4) infers phenometrics from digital cameras, allowing direct comparison with near-ground-based estimates.
- The **indirect validation** includes intercomparison with similar satellite-based products. The indirect validation allows extending the evaluation over large areas or time periods, but only relative discrepancies between existing datasets can be assessed. Reference global products providing phenology metrics (MODIS V6.1 and VIIRS V2, see section 4.1.1) and productivity parameters (CLMS GPP, see section 4.1.2) are used in this study, as well as the CLMS HR-VPP products (sections 4.1.3) covering Europe.

5.1 VALIDATION CRITERIA AND METRICS

The following criteria are analysed: product completeness, spatial consistency and error evaluation, which involves the quantification of Accuracy, Precision and Uncertainty (APU). In addition, the conformity test against LSP uncertainty requirements (Chapter 2) is also performed.

- **Product completeness**

Completeness corresponds to the absence of spatial and temporal gaps in the data. Missing data are mainly due to cloud or snow contamination, poor atmospheric conditions or technical problems during the acquisition of the images and is generally considered by users as a severe limitation of a given product. It is therefore mandatory to document the completeness of the product. Yearly global maps and percentages of valid and missing data are provided.

- **Spatial consistency**

Spatial consistency refers to the realism and repeatability of the spatial distribution of retrievals over the globe. A first qualitative check can be performed through a systematic visual analysis of global maps and maps over selected regions of interest, allowing the detection of unusual patterns or artefacts (e.g., missing values, stripes, unrealistic values) based on the expert knowledge of the scientist. The assessment of spatial consistency can be achieved by comparing the spatial distribution of the product under study with that of a reference validated product, as well as by analyzing the spatial distribution of differences.

Additionally, the analysis of phenological trends as a function of latitude provides a complementary means to evaluate spatial consistency. Spatial variability is assessed by computing the standard deviation (STD) and median absolute deviation (MAD) over homogeneous areas, serving as indicators of spatial precision.

- **Temporal consistency**

The temporal behavior of the product is qualitatively assessed by analysing the evolution of global mean values (excluding croplands to avoid artificial anomalies) across the full time series.

- **Error evaluation**

Accuracy is the degree of the *closeness of the agreement between the result of a measurement and a true value of the measurand* [JCGM, 2014]. Commonly, accuracy represents systematic errors and often is computed as the statistical mean bias (B), i.e., the difference between the short-term average measured value of a variable and the true value. The short-term average is the average of a sufficient number of successive measurements of the variable under identical conditions, such that the random error is negligible relative to the systematic error. The latter can be introduced by instrument biases or through the choice of remote sensing retrieval schemes [GCOS#200].

Precision or repeatability is the *closeness of the agreement between the results of successive measurements of the same measurand carried out under the same conditions of measurement* [JCGM, 2014]. Commonly, precision represents the dispersion of product retrievals around their expected value and can be estimated by the standard deviation of the difference (STD) between retrieved satellite product and the corresponding reference estimates.

Uncertainty is a *parameter, associated with the result of a measurement that characterizes the dispersion of the values that could reasonably be attributed to the measurand* [JCGM, 2014]. Uncertainty includes systematic and random errors and can be estimated by the Root Mean Square Deviation (RMSD).

In addition to these metrics, other statistics are useful to evaluate the goodness of fit between two datasets including linear model fits. For this purpose, Major Axis Regression (MAR) is computed instead Ordinary Least Squares (OLS) because is specifically formulated to handle error in both of the x and y variables (Harper, 2014). Other metrics are used, such as number of samples (N),

indicative of the power of the validation, or the correlation coefficient (R, estimated as Pearson coefficient), which indicates descriptive power of the linear accuracy test.

It should be noted that strong and/or multiple outliers affect the classical metrics described above (i.e., B and STD): in such cases using the median deviation (MD) instead of the mean bias to estimate systematic error and the median absolute deviation (MAD) as a measure of precision is preferable.

APU quantities are evaluated by several metrics (Table 12) reporting the goodness of fit between the products and the corresponding reference dataset. Scatter-plots and validation metrics (Table 12) versus references are produced. This analysis is complemented per main biome type. In addition, the inter-annual precision is evaluated as the computation of MAD between consecutive years. Temporal evolution of inter-annual precision and B/RMSD compared with references (i.e., a proxy of stability) is analysed.

Finally, the conformity test is provided, which corresponds to the computation of the percentage of pixels within the product user requirements. Two uncertainty predefined levels (goal and threshold, see Table 1) are defined based on CLMS technical requirements [AD1].

Table 12: Validation metrics. 'x' and 'y' refer to reference and evaluated products.

Statistics	Comment
N	Number of samples. Indicative of the power of the validation
B	Mean Bias. Difference between average values of y and x (y-x). Indicative of accuracy and offset. Bias (%) is the relative mean bias between the average of x and y.
MD	Median deviation between y and x. Best practice reporting the accuracy. MD (%) is the relative MD between the average of x and y.
STD	Standard deviation of the pair differences. Indicates precision. STD (%) is the relative STD between the average of x and y.
MAD	Median absolute deviation between x and y. Best practice reporting the precision. MAD (%) is the relative MAD between the average of x and y.
RMSD	Root Mean Square Deviation. RMSD is the square root of the average of squared errors between x and y. Best practice reporting uncertainty. RMSD (%) is the relative RMSD between the average of x and y.
MAR	Slope and offset of the Major Axis Regression linear fit. Indicates some possible bias
R	Correlation coefficient. Indicates descriptive power of the linear accuracy test. Pearson coefficient is used.

5.2 SAMPLING

In addition to global-scale analyses, several samplings were selected to capture both specific and broad conditions. Some samples focus on homogeneous areas or representative biomes to assess local spatial variability and precision, while others cover larger European or global extents to evaluate overall consistency across diverse environments. This combined approach ensures assessment of both local detail and large-scale patterns, as described below for the spatial consistency and error evaluation analysis.

- **Spatial consistency**
  - The methodology for visual analysis includes the visualization of the global maps of the product content at reduced resolution (i.e., 1/48), and the zooms over 6 subcontinental areas of 10º latitude x 15º longitude and 6 Areas of Interest (AOIs, see Table 13) of 20 km x 20 km over Europe at full resolution (see Figure 3).
  - The spatial consistency with reference products includes:
    - Global maps and histograms of differences with global reference satellite products.
    - Comparison of maps and histograms with HR-VPP over the 6 AOIs over Europe.
  - Additionally, the phenological trend as a function of latitude is analysed. A latitudinal transect is defined as the geodesic line from Norway (70.315ºN, 20.119ºE) to South Africa (34.872ºS, 20.119ºE) (see Figure 3) with a step of 2 km between samples.
  - The spatial variability (i.e., STD and MAD, indicative of precision) over 6 selected homogeneous ROIs of 5 km x 5 km (Table 14). One ROI for each main biome type (EBF, DBF, NLF, crop, herbaceous, shrublands) is selected for the analysis.

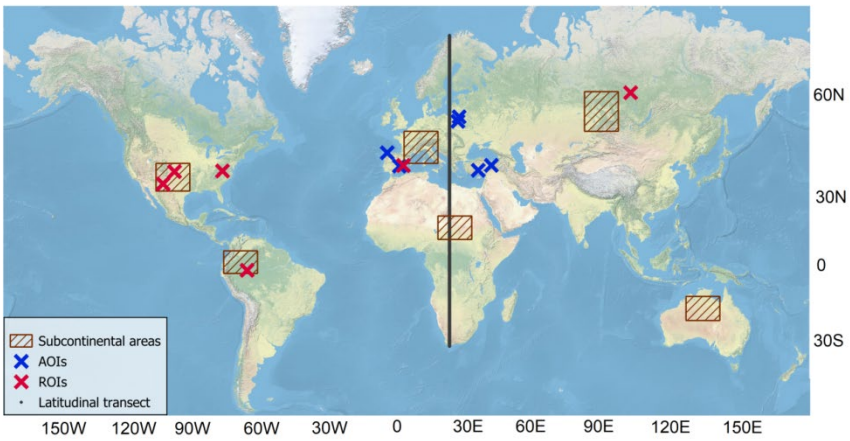
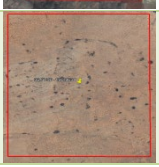


Figure 3: Location of the subcontinental areas (brown boxes), AOIs (blue crosses), ROIs (red crosses) and latitudinal transect (grey vertical line) used for spatial consistency.

Table 13: Summary of central coordinates of selected AOIs (20 km x 20 km) for spatial consistency.

No.	Name	Biome	Central coordinates	
			Latitude	Longitude
1	Barrax	CUL	39.05437° N	-2.10068° E
2	LUCAS 2777	DBF	43.497616° N	-7.482012° E
3	Bialowieza National Park	DBF	52.749876° N	23.867391° E
4	LUCAS 1208	NLF	53.942444° N	24.406548° E
5	EOLAB 53	CUL	37.619363° N	32.86563° E
6	Spar 113	HER	39.326087° N	38.681288° E

Table 14: Relevant information and central coordinates of selected ROIs (5 km x 5 km).

ROI #	Name	Country	Biome	Central coordinates		Google Earth View
				Latitude	Longitude	
1	*Evare*	Brazil	EBF	-3.6131° N	-69.6280° E	
2	*Mountain Lake*	USA (Virginia)	DBF	37.3783° N	-80.5248° E	
3	*Oskoba*	Russia	NLF	59.7619° N	100.4345° E	
4	*Albufera*	Spain	Crop (rice)	39.2734° N	-0.3188° E	
5	*Cimarron National Grassland*	USA (Kansas)	Herbaceous	37.1369° N	-101.8065° E	
6	*Jornada*	USA (New Mexico)	Shrublands	32.6756	-106.8631° E	

- **Error evaluation**
  - The comparison with in-situ measurements depends on the availability of the data during the period under study (2014–2024). Figure 4 shows the location of the 5108 validation plots (2244 PEP725, 2102 TEMPO, 37 NEON and 725 Phenocam V3) used in this exercise, mostly distributed across Europe and North America.

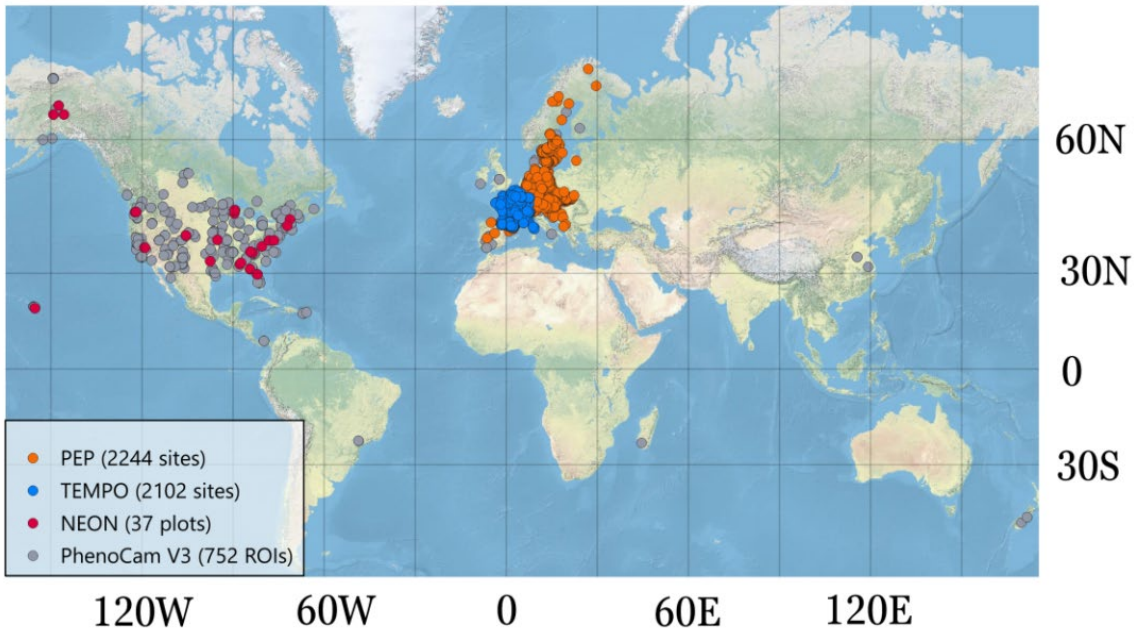


Figure 4: Distribution of PEP725, TEMPO, NEON and PhenoCam locations used in direct validation for 2020–2024 period.

  - For product intercomparison, two different samplings are used depending on the reference product:
    - For global products, the LAND VALidation version 2 (LANDVAL V2) network of sites (Figure 5, left side) is used (Martínez-Sánchez et al., 2024). It provides a global sampling representative of global conditions in term of land cover, continental area and ecoregion. LANDVAL V2 counts with a total of 2000 sites and is based on Copernicus GLC-100m classification (Buchhorn et al., 2020) and World Wild LiFe Terrestrial Ecoregion of the World classification (Olson et al., 2001). The following quality criteria to select the coordinates along the globe are: i) same land cover and ecoregion (>90% homogeneity) over an area of 3 km x 3 km; ii) distance between samples at least of 2 degrees on the hypotenuse in Plate Carrée projection; and iii) slope within the 3 km x 3 km area lower than 100 m. LANDVAL V2 provides almost identical distribution than the global real distribution of ecosystems, improving

the global representative of LANDVAL V1 used in several validation studies for vegetation LAI/FAPAR or surface albedo products (Fuster et al., 2020; Sánchez-Zapero et al., 2023, 2020).

- For the comparison over Europe with HR-VPP, the EVALuation network (EVAL) composed by 3800 evaluation sites is used (Camacho et al., 2024). This sampling, representative of the European region, was defined for the validation of HR-VPP products [HR-VPP_ST&VPP_VR]. Sites were defined to be homogeneous in land cover over areas of, at least, 700 m^2. The sites are well distributed across the European region, as shown in Figure 5 (right side).

These analyses are achieved per aggregated land cover class based on the 8 generic classes derived from the GLC-100m classification (Buchhorn et al., 2020): Evergreen Broadleaf Forest (EBF), Deciduous Broadleaf Forest (DBF), Needle-Leaf Forest (NLF), Other Forests (OF), Cultivated (CUL), Herbaceous (HER), Shrublands (SHR) and Sparse and Bare areas (SBA).

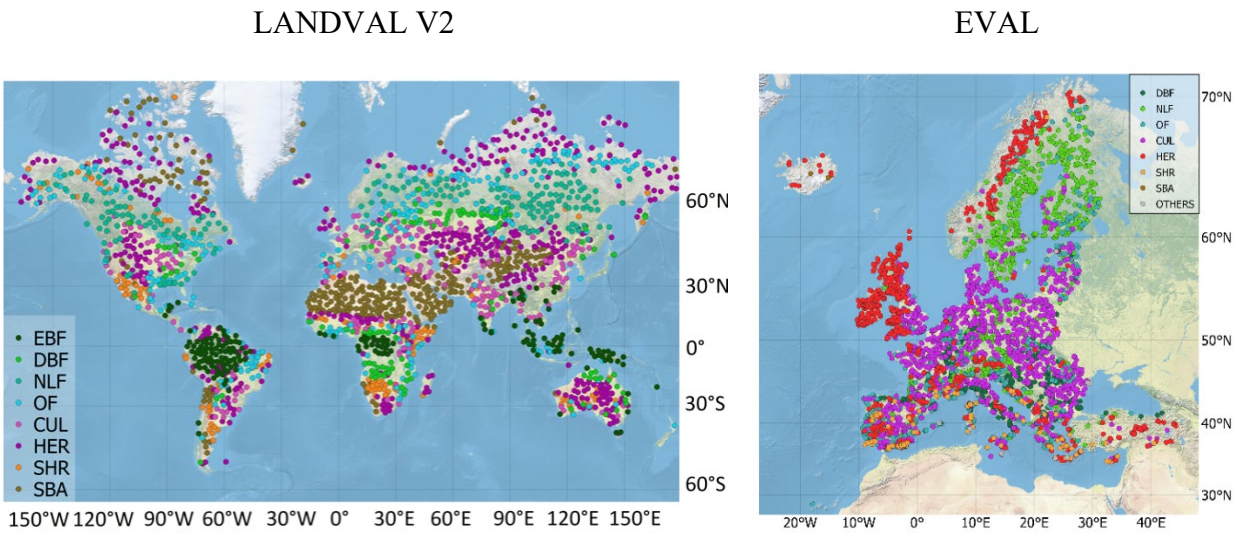


Figure 5: Sampling strategy for product intercomparison with reference global (LANDVAL V2, 2000 sites, left side) and European (EVAL, 3800 sites, right side) products.

5.3 SPATIO-TEMPORAL COLLOCATION

The quality flag information of each reference satellite product was used to retain only the *best-quality* (BQ) pixels for the analysis (Table 15). All assessments were therefore performed using BQ retrievals, with a single exception: the spatial consistency with VIIRS LCD V2, due to the very

restrictive quality filtering of this product (see completeness results in section 6.3) its quality flag was not applied in this specific case in order to increase spatial coverage.

Table 15: Quality Flag information used to select BQ pixels.

Product	Quality Flag
CLMS LSP v2	QA < 127 (overall good quality)
MODS LCD V6.1	QA_Overall ≤ 1 (best/good)
VIIRS LCD V2	GLSP_QC bits 0-2='000' (processed, good quality)
CLMS OLCI GPP	Not used, as heritage from CLMS FAPAR 300m products [CGLOPS1_ATBD_FAPAR300m-V1.1].
HR-VPP	QF ≥ 7 (high and medium quality data)

Satellite products must be compared on common spatial and temporal supports to enable a quantitative comparison between products.

- Regarding the spatial support:
  - A spatial support of 1 km^2 is used. This corresponds to 3x3 pixels in case of CLMS LSP V2 products, 2x2 pixels in case of MODIS and VIIRS based products and 100x100 pixels in case of Sentinel-2 HR-VPP products. All reference products were first resampled to the *Plate Carrée* projection, matching the CLMS LSP V2 grid.
  - For the specific case of the spatial consistency analysis with high-resolution (HR-VPP, 10 m) products, the CLMS LSP V2 300m grid was used as spatial support. The HR-VPP datasets were aggregated to the CLMS LSP V2 spatial resolution and projection using an effective Point Spread Functions (PSF) approach, which improves the evaluation performance compared to simple averaging (Mira et al., 2015). The PSF was approximated through the convolution of a Gaussian function representing the optical sensor response (Nestola et al., 2017). For each AOI, an iterative optimization was applied to determine the pixel extent and the Full Width at Half Maximum in both directions that maximize the correlation between the aggregated HR-VPP and the CLMS LSP V2 300 m products.
- Regarding the temporal support:
  - In the case of CLMS LSP V2 SOSD and EOSD error evaluation (quantitative evaluation) with equivalent products, only retrievals provided within the current year of the product are used, excluding SOSD/EOSD dates that correspond with previous or following adjacent years. For the spatial consistency, dates of adjacent years are used with the aim of improving the spatial coverage of the maps.

- For the comparison with MODIS LCD V6.1, MODIS *Greenup*, *MidGreenup* and *Maturity* dates are compared with CLMS SOSD. MODIS *Senescence*, *MidGreendown* and *Dormancy* dates are compared with CLMS EOSD.
- For the comparison with VIIRS LCD V2, CLMS SOSD is compared with VIIRS *onset of greenness increase*, *mid-greenup* and *maximum* dates whilst CLMS EOSD is compared with VIIRS *onset of greenness decrease*, *mid-senescence* and *minimum* dates.
- HR-VPP provides the same SOSD and EOSD phenometrics as CLMS LSP V2, with same thresholds definitions.

o In the case of CLMS LSP V2 TPROD and SPROD productivity layers:

- For the comparison with CLMS GPP, the sum of CLMS GPP 10-daily values between CLMS LSP V2 SOSD and EOSD is computed, as CLMS LSP V2 TPROD is defined as the sum of all PPI daily values between SOSD and EOSD. The same approach is applied for SPROD, which is defined as TPROD minus the base-level value
- For the comparison with HR-VPP, equivalent TPROD and SPROD layers are used, which have the same definitions.

For **direct validation** purposes, the spatio-temporal collocation procedure is detailed separately for each in-situ network (see details in section 4.2), since specific approaches were applied in each case.

5.4 SUMMARY OF THE VALIDATION PROCEDURE

First, SOSD/EOSD and TPROD/EOSD CLMS LSP V2 layers are compared using different PBV and OLCI input data for the overlap year (2019) in order to assess the impact of the sensor transition. The summary of the methods for PBV vs. OLCI consistency is provided in Table 16.

Table 16: Summary of the PBV vs. OLCI consistency of CLMS LSP V2.

Quality Criteria	Evaluated CLMS LSP V2 layer	Reference Product	Spatial /Temporal Coverage
Spatial Consistency	SOSD, EOSD, SPROD, TPROD	PBV vs. OLCI	Global / 2019
	Visual inspection of global maps.		
Temporal consistency	SOSD, EOSD	PBV vs. OLCI	LANDVAL V2 / 2014–2024
	Temporal evolution of mean values (excluding crops).		
Error evaluation	SOSD, EOSD, SPROD, TPROD	PBV vs. OLCI	LANDVAL V2 / 2019
	Scatter-plots PBV vs. OLCI for the overlap year.		

Secondly, CLMS LSP V2 are evaluated with external reference reference datasets. The validation primarily focuses on the two key phenological metrics (SOSD and EOSD), for which ground observations are available and user requirements are defined. In addition, the two productivity layers (SPROD and TPROD) are evaluated against comparable satellite-based reference products. For the remaining CLMS OLCI LSP V2 layers, completeness and product content are also examined. A summary of the quality assessment procedures is provided in Table 17.

Table 17: Summary of the CLMS LSP V2 validation procedure.

Quality Criteria	Evaluated CLMS LSP V2 layer	Reference Product	Spatial / Temporal Coverage
Completeness	All	MODIS LCD V6.1 VIIRS LCD V2	Global / 2014–2024
	Global maps of valid and missing data. Percentage of valid and missing data per year.		
Spatial Consistency	All	-	Global and subcontinental regions / 2014–2024
	Visual inspection of product content maps.		
	SOSD, EOSD, SPROD, TPROD	HR-VPP	AOIs / 2023
	Visual inspection of maps and histograms of values.		
	SOSD, EOSD	MODIS LCD V6.1 VIIRS LCD V2	Global & Latitudinal transect / 2014–2024
	Global maps of differences & phenological trend as a function of the latitude.		
	SOSD, EOSD, SPROD, TPROD	-	ROIs / 2017–2020
	Visual inspection of maps and spatial variability (precision)		
Error evaluation (direct validation)	SOSD, EOSD	PEP725, TEMPO, NEON, Phenocam	PEP725, TEMPO, NEON, Phenocam / 2014–2024
	Scatter-plots, validation metrics and conformity test. Temporal evolution of B/RMSD per year.		
Error evaluation (product intercomparison)	SOSD, EOSD, SPROD, TPROD	HR-VPP	EVAL /2021–2023
	Scatter-plots and validation metrics. Conformity test.		
	SOSD, EOSD	MODIS LCD V6.1 VIIRS LCD V2	LANDVAL V2 / 2014–2024, 2014–2023
	Scatter-plots and validation metrics. Conformity test. Analysis per biome type.		
	SPROD, TPROD	CLMS GPP	LANDVAL V2 / 2014–2024
	Scatter-plots and correlation. Analysis per biome type.		
	SOSD, EOSD	MODIS LCD V6.1 VIIRS LCD V2	LANDVAL V2 (excluding crops) / 2014–2024
	Inter-annual precision: temporal evolution of MAD between consecutive years.		

6 RESULTS

6.1 CONSISTENCY BETWEEN ESTIMATES FROM PBV AND OLCI

Figure 6 shows scatter-plots of CLMS LSP V2 (SOSD/EOSD and SPROD/TPROD layers) comparing PBV and OLCI for the overlap year (2019) to assess the impact of the sensor transition. The mean values of SOSD and EOSD across 2014–2024 period are displayed in Figure 7. Finally, the global maps of SOSD/EOSD and SPROD/TPROD for 2019 are shown in Figure 8 and Figure 9.

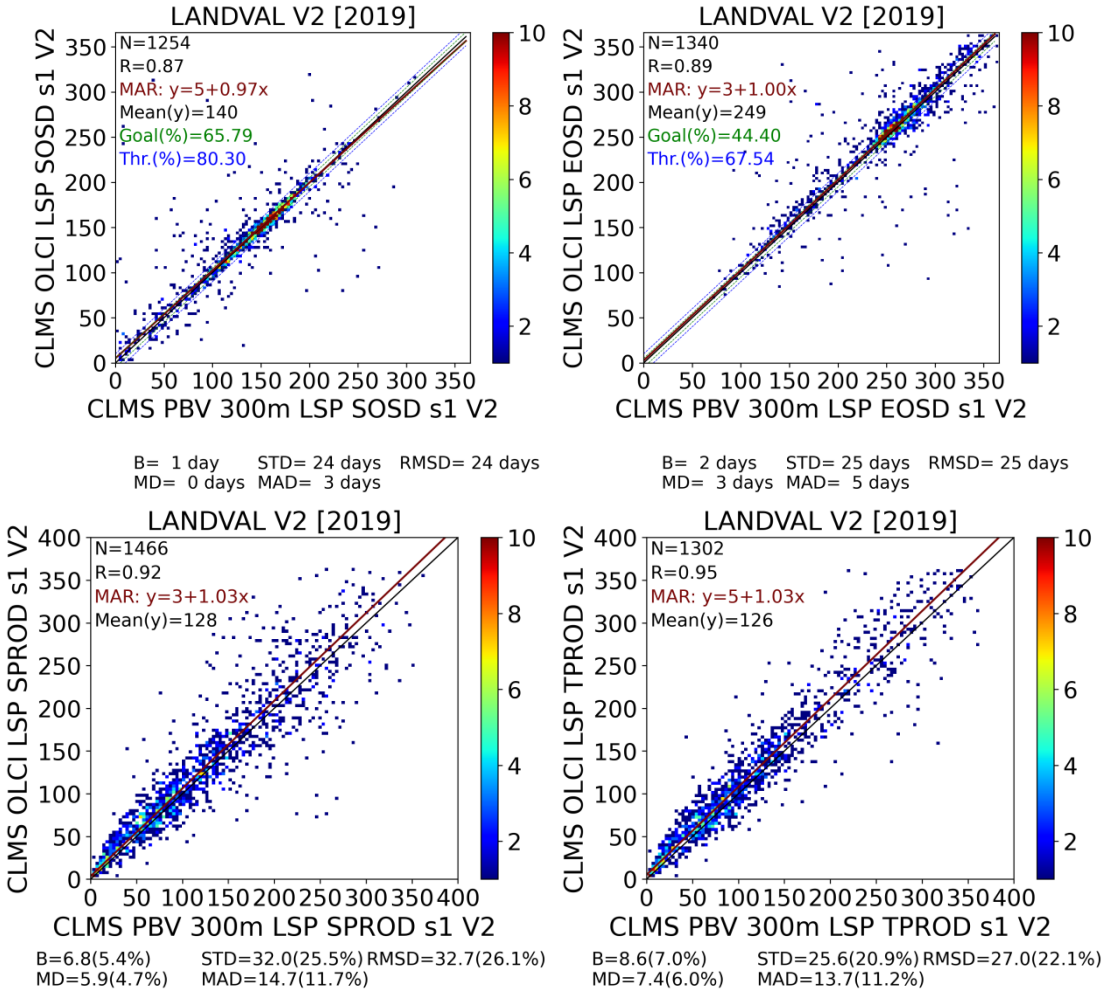


Figure 6: Scatter-plots of CLMS LSP V2 SOSD/EOSD (top) and SPROD/TPROD (bottom) comparing PBV (X-axis) and OLCI (Y-axis) for the year 2019. Results are computed over LANDVAL V2 sites. For SOSD/EOSD, the green and blue lines around the 1:1 line indicate the goal and threshold requirements, respectively.

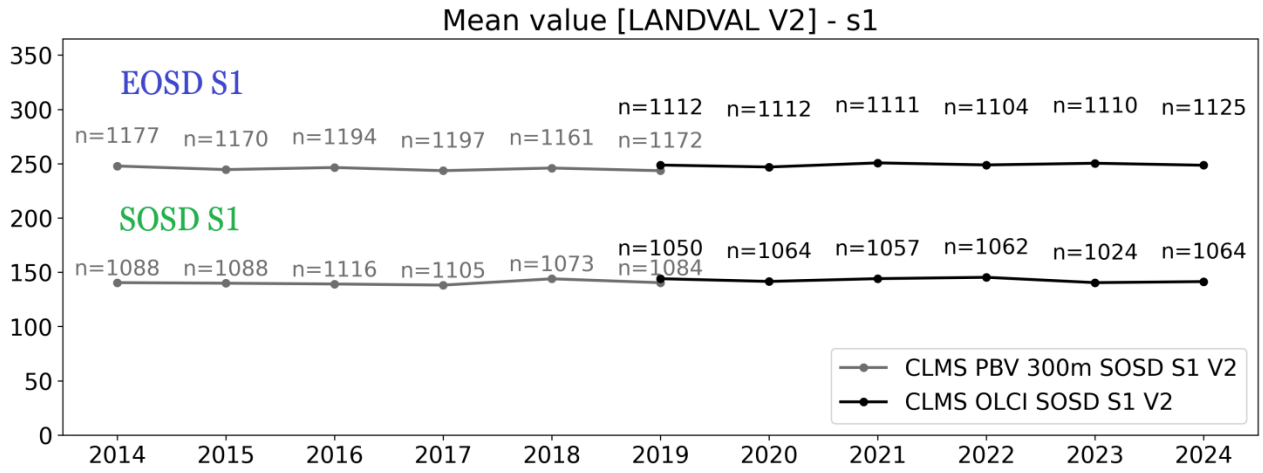


Figure 7: Mean value of SOSD and EOSD layers of CLMS LSP V2 for PBV (2014–2019, season 1) and OLCI (2019–2024). Computation over LANDVAL V2 excluding crops.

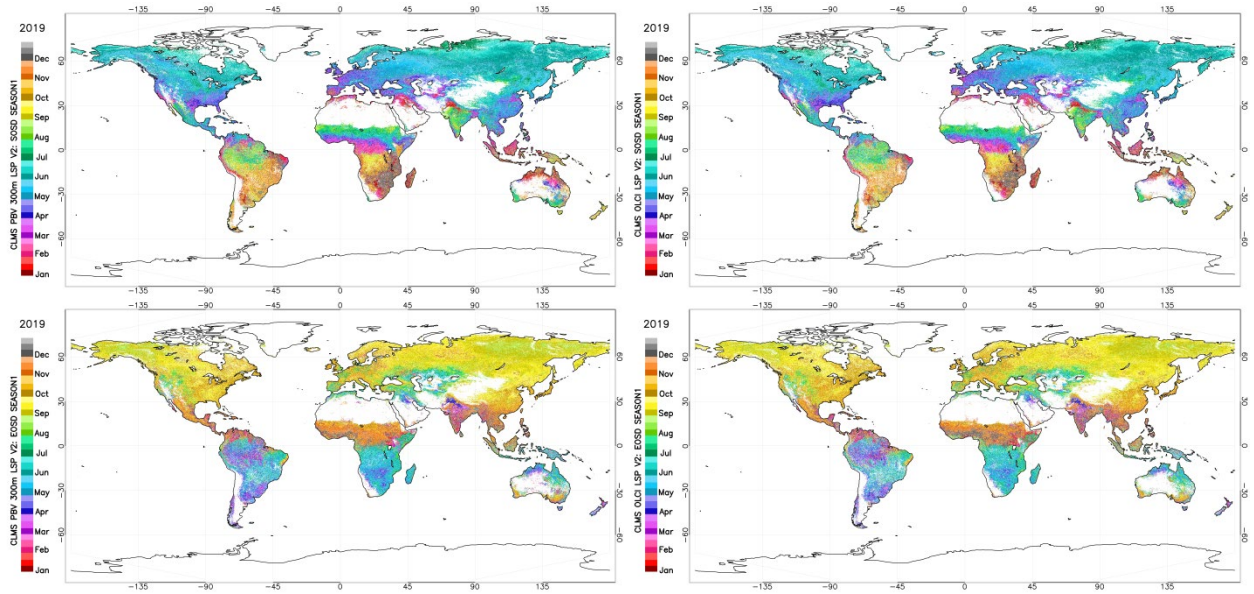


Figure 8: Global maps of CLMS PBV (left) and OLCI (right) LSP V2 SOSD (top) and EOSD (bottom) layers for 2019 (season 1).

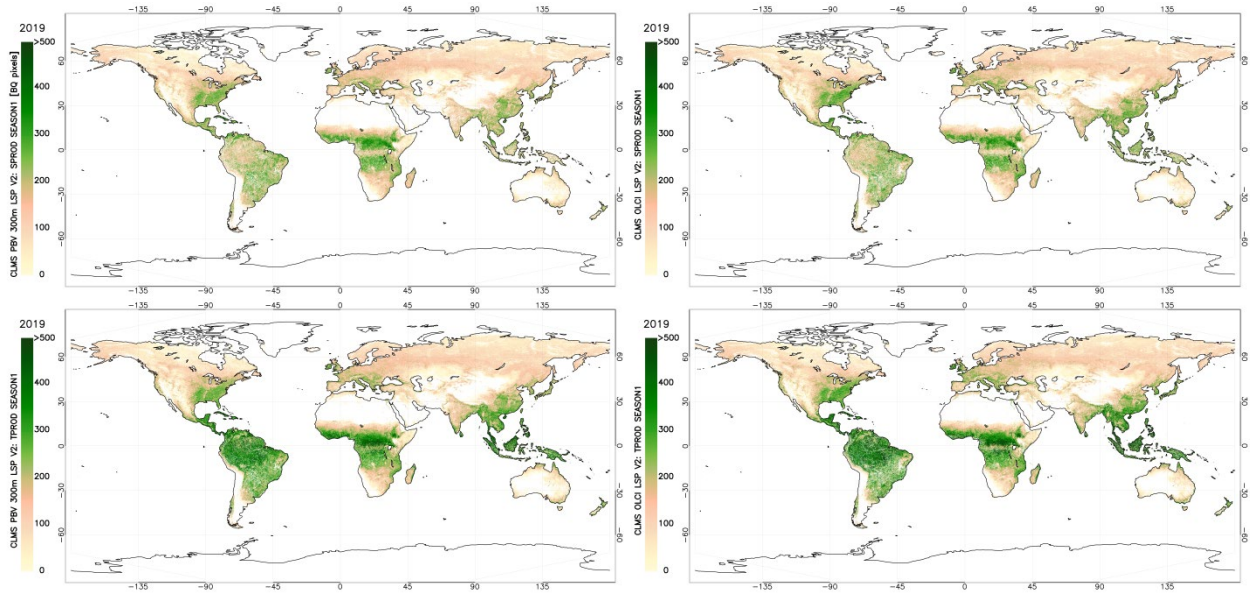


Figure 9: Global maps of CLMS PBV (left) and OLCI (right) LSP V2 SPROD (top) and TROD (bottom) layers for 2019 (season 1).

Main findings of the consistency between OLCI and PBV are:

- Strong agreement is observed during the transition from PBV to OLCI, with correlations of 0.87 and 0.89 for SOSD and EOSD, and 0.92 and 0.95 for SPROD and TPROD.
- SOSD shows almost no bias (B = 1 day, MD = 0 days), while EOSD shows a slight positive bias (and MD) of 2 days (OLCI slightly later than PBV), resulting in overall slightly longer seasons of about 1-2 days for OLCI. These small positive biases are also reflected in the temporal evolution of mean values (Figure 7), confirming that the change in sensor input has only a minimal impact on the CLMS LSP V2 time series.
- OLCI SPROD and TPROD values are slightly higher than PBV by ~7–9 $m^2 \cdot m^{-2} \cdot day$, consistent with the slightly longer season observed.
- The global maps of SOSD/EOSD and SPROD/TPROD (Figure 8) show nearly identical distributions between PBV and OLCI, demonstrating strong spatial agreement between the sensors. It should be noted that PBV marginally retrieves some values over desert areas (e.g., Sahara) where OLCI provides no data.

6.2 PRODUCT COMPLETENESS

This section evaluates the product completeness of the global CLMS LSP V2, MODIS V6.1, and VIIRS V2 phenology products. Maps of valid (blue) and missing (yellow) data are displayed in Figure 10 and Figure 11 for season 1 and season 2 of the year 2020, respectively (shown as an example). Note that, on the right-hand side, quality flags are applied to remove suboptimal-quality pixels (Table 15). At the bottom of Figure 10 and Figure 11, the temporal evolution of missing data across the 2014–2024 period is summarized.

Main results of the product completeness analysis are:

- When considering all pixels from season 1:
  - CLMS LSP V2 shows a similar completeness to VIIRS LCD V2, with approximately 75–80% of valid data. CLMS improves the completeness of the VIIRS product over tropical areas. However, VIIRS exhibits better completeness over some sparsely vegetated regions (e.g., western and southern America and central Asia).
  - CLMS LSP V2 improves completeness of MODIS LCD V6.1, which shows the lowest completeness, with around 60% of valid data.
- When applying the quality flag for season 1:
  - Approximately 5% of the global data are filtered out in the case of CLMS LSP V2.
  - The VIIRS V2 product applies a more restrictive criterion for *best-quality* retrievals, decreasing from about 80% to 30% of valid data when only best-quality pixels are considered (i.e., roughly more than 60% of data are excluded).
  - The MODIS V6.1 quality flag typically filters out less than 2% of the global data.
- CLMS LSP V2 shows a slightly higher proportion of *best-quality* data (about 2%) during the PROBA-V period compared to the OLCI period.
- All products show very few valid cases for season 2, with only around 2–3% of global pixels for CLMS LSP V2, about 2% for MODIS V6.1, and 5% for VIIRS V2 (only 1% after filtering).

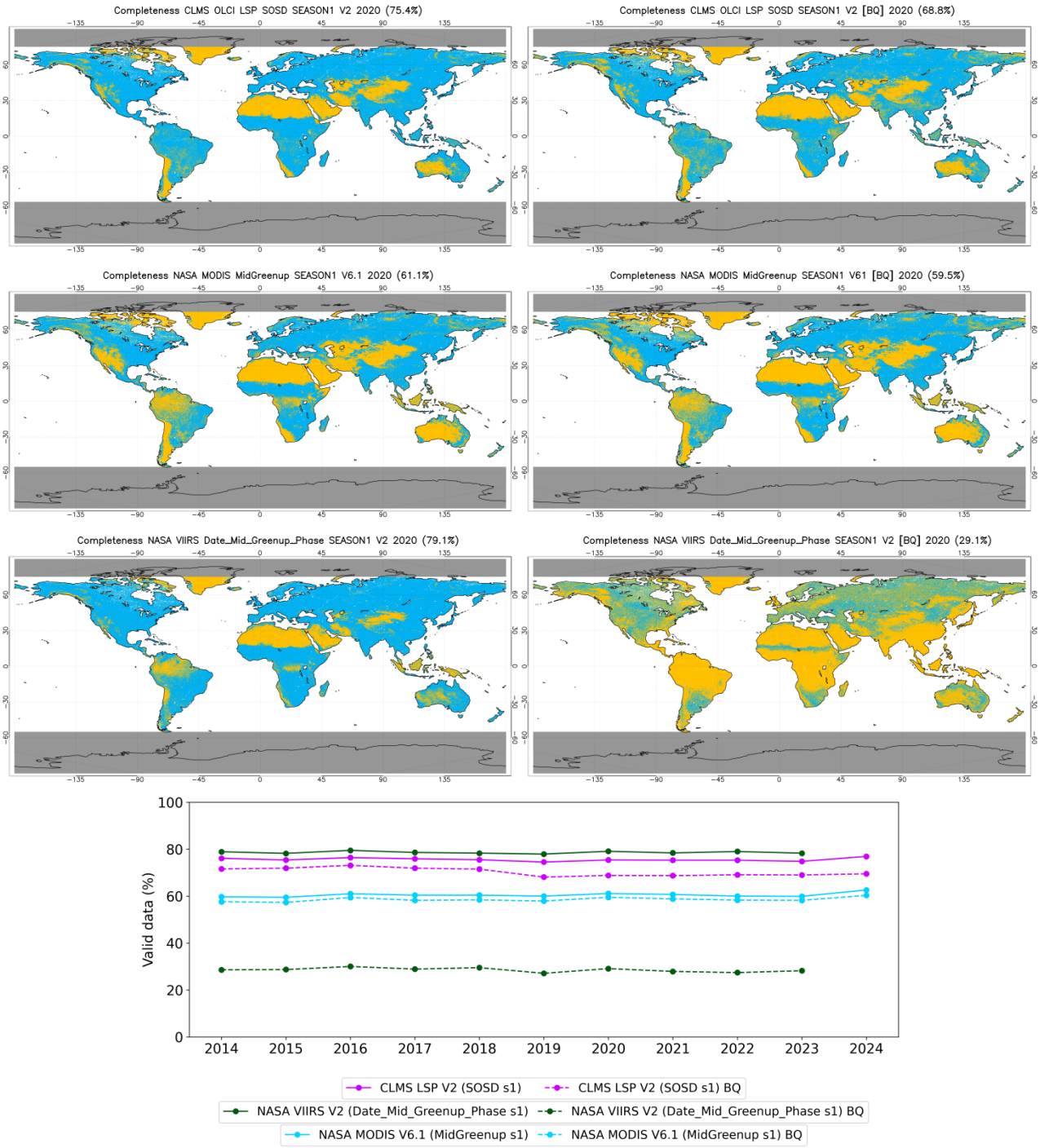


Figure 10: Global maps of valid (blue) and non-valid (yellow) observations of start-of-season layers for season 1 (2020) from CLMS LSP V2 (1st row), MODIS LCD V6.1 (2nd row) and VIIRS LCD V2 (3rd row). Left column shows all pixels; right column shows only *best-quality* pixels based on quality flags. Grey stripes at high northern and southern latitudes indicate areas without CLMS coverage. Percentages of valid data for all and best-quality pixels are summarized at the bottom for the 2014–2024 period.

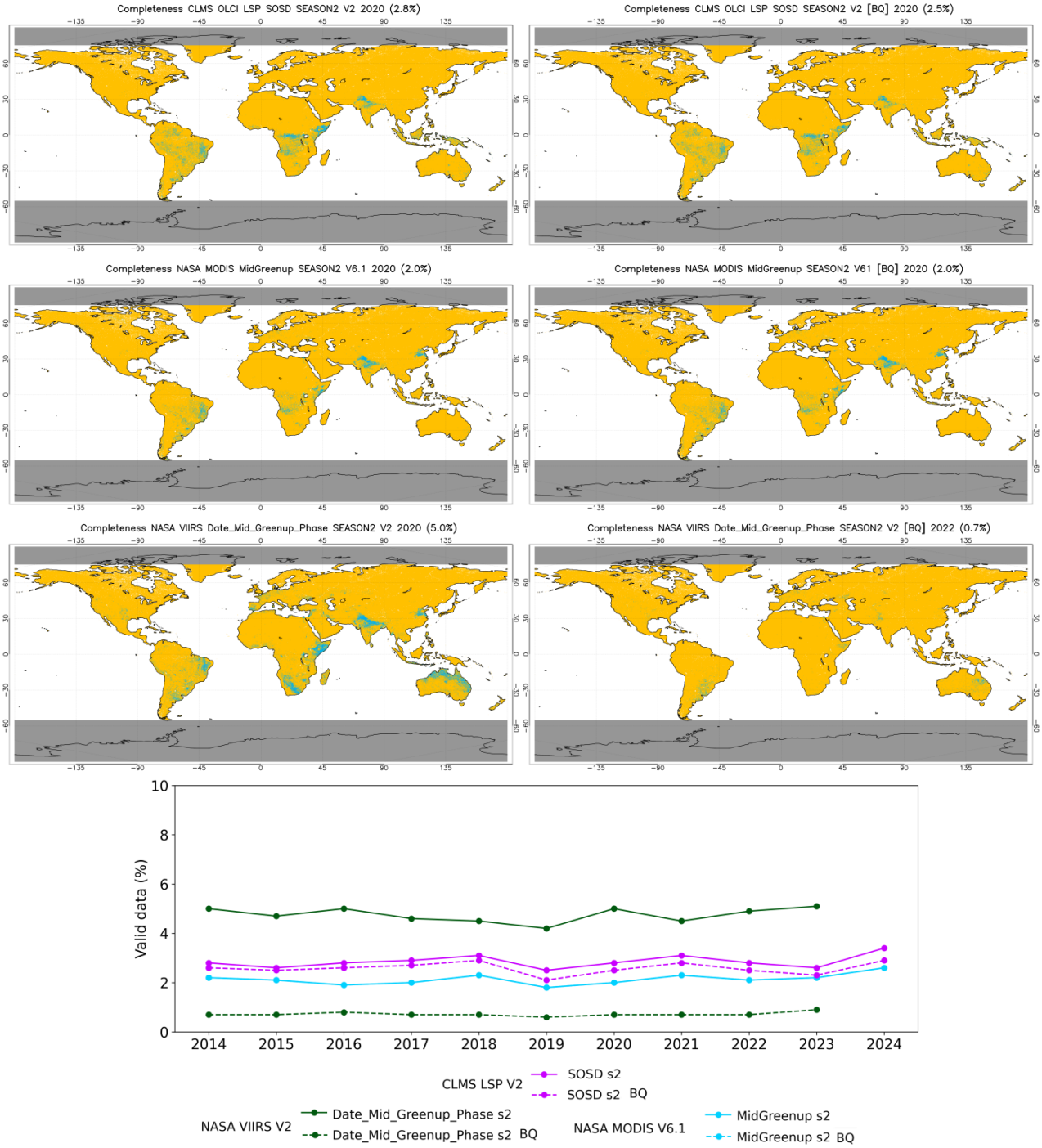


Figure 11: Same as in Figure 10 but for season 2.

6.3 SPATIAL CONSISTENCY

6.3.1 Visual inspection of product content maps

Global maps of the 13 CLMS LSP V2 parameters and the QA layer are analyzed in this section. For brevity, only the years 2018 (PBV) and 2020 (OLCI) are shown, although similar patterns are observed for all other years.

SOSD, EOSD and MAXD layers represent the day of year (DOY) corresponding to the start, end and maximum of the growing season, respectively. LENGTH indicates the duration of the season, defined as the number of days between start and end dates. These layers are displayed for PBV (2018) and OLCI (2020) in Figure 12 and Figure 13, respectively, showing both season 1 (left side) and 2 (right).

The main results of the visual inspection of the CLMS LSP V2 global maps for SOSD, EOSD, MAXD and LENGTH (Figure 12 and Figure 13) are as follows:

- Overall, PBV (Figure 12) and OLCI (Figure 13) exhibit consistent spatial patterns over the globe. It should be noted that some marginal pixels over desert regions (e.g., the Arabian and Sahara deserts) display seasonal cycles despite the likely absence of actual vegetation activity.
- Season 2 occurrences are limited and mainly observed in tropical regions.

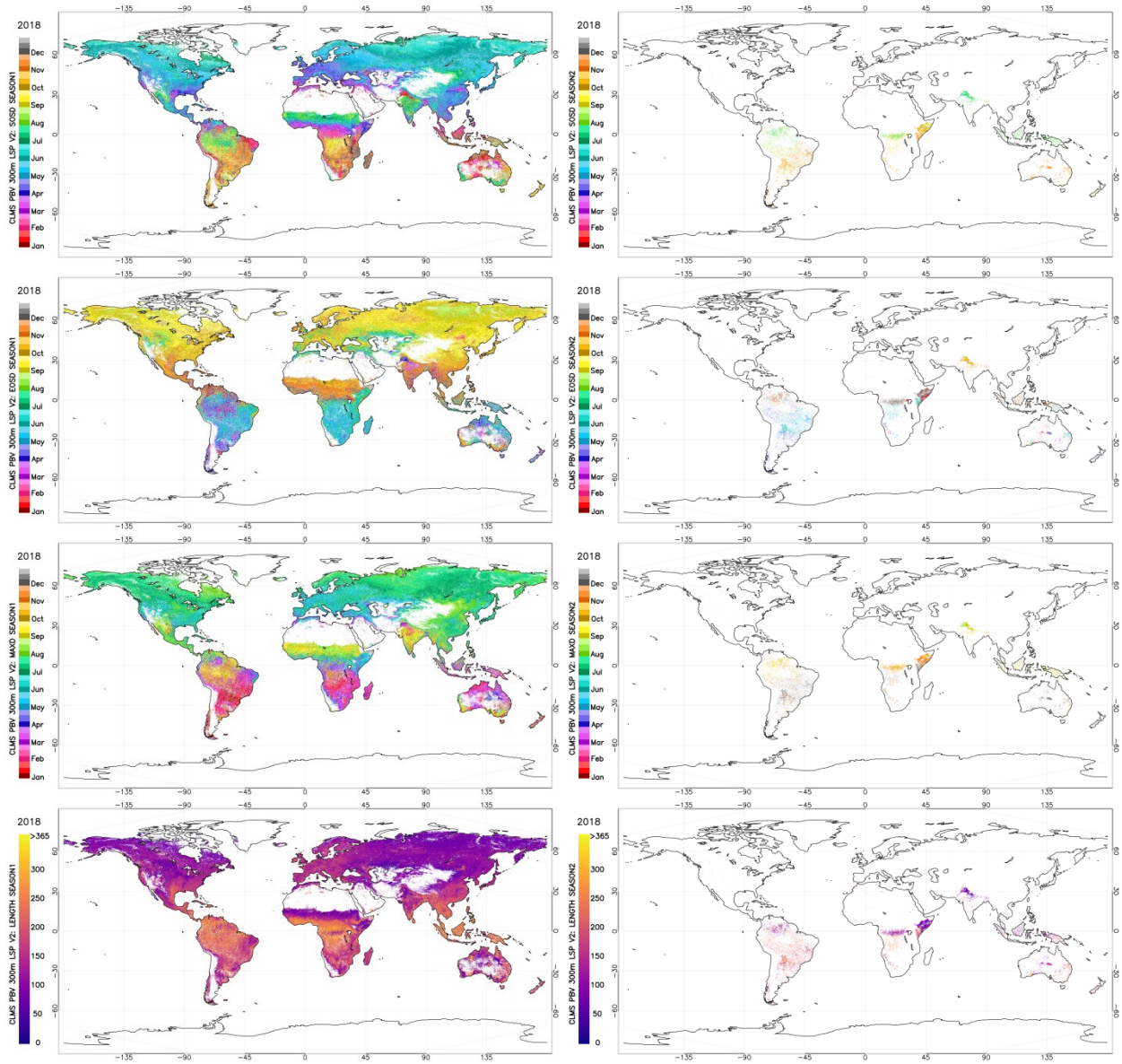


Figure 12: Global maps of CLMS PBV LSP V2 layers for 2018: SOSD, EOSD, MAXD, and LENGTH (from top to bottom). Season 1 and Season 2 are shown on the left and right panels, respectively.

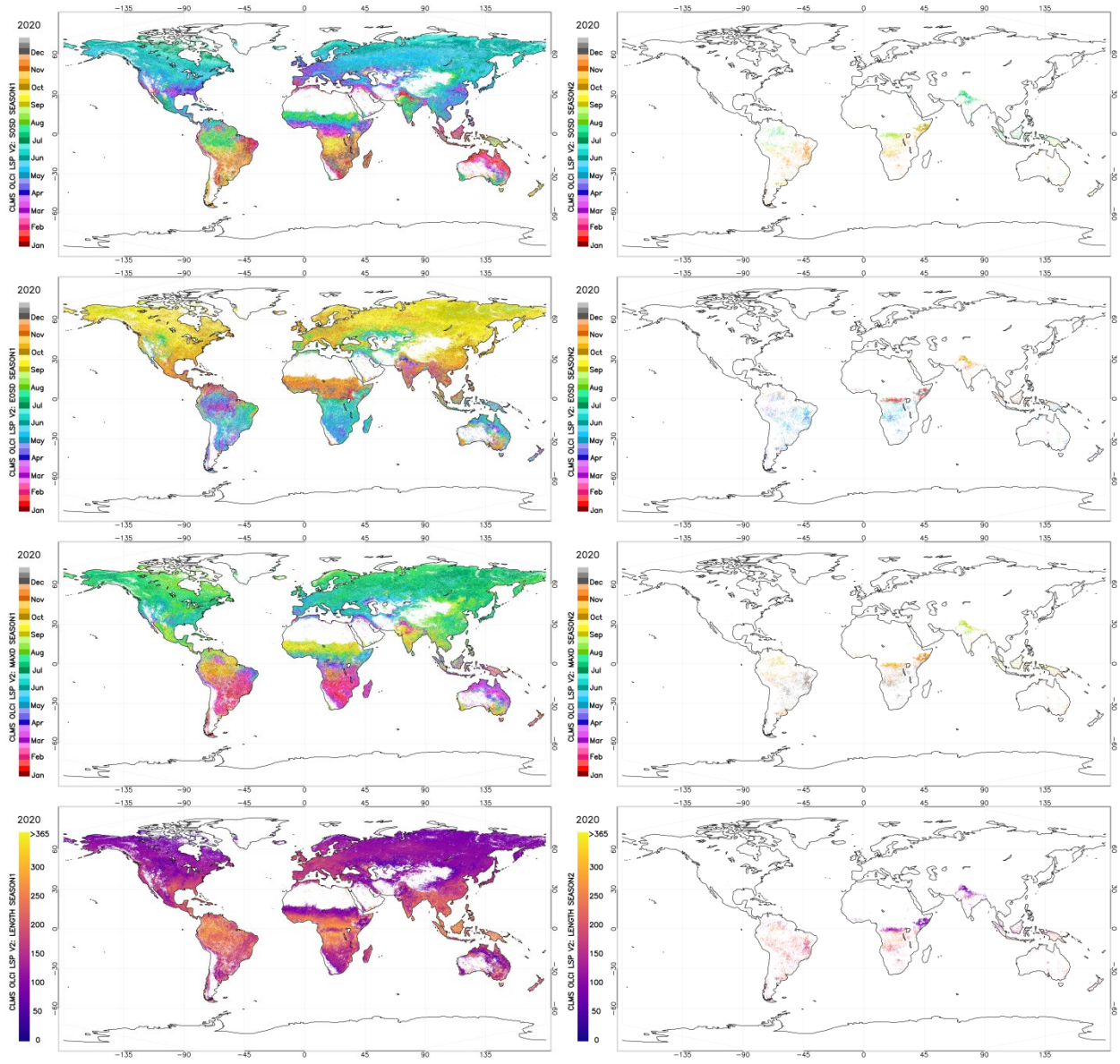


Figure 13: Global maps of CLMS OLCI LSP V2 layers for 2020: SOSD, EOSD, MAXD, and LENGTH (from top to bottom). Season 1 and Season 2 are shown on the left and right panels, respectively.

Several CLMS LSP V2 layers are defined at the vegetation index level (i.e., PPI units [m^2m^{-2}]) for different phases of the phenological cycle. SOSV and EOSV correspond to the PPI value at the start (SOSD) and end (EOSD) of the season, respectively, whilst MINV and MAXV represent the average PPI of the seasonal minima (left and right sides) and the PPI at the maximum of the season (MAXD). The seasonal amplitude (i.e., MAXV – MINV) is provided in the AMPL layer. Global maps of these

layers are shown in Figure 14 and Figure 15 for PBV (2018) and in for OLCI (2020). Only season 1 is presented, for sake of brevity.

The main findings from the visual inspection of the global maps of CLMS LSP V2 SOSV, EOSV, MINV, MAXV and AMPL layers (Figure 14 and Figure 15) are:

- Consistent patterns are observed globally, with SOSV and EOSV showing similar PPI values, and MINV and MAXV displaying the expected low and high values, respectively. SOSV values are slightly higher than EOSV, consistent with the product definition (SOSD and EOSD are defined at 25% and 15% of the seasonal amplitude, respectively). Higher values are typically found over densely vegetated areas such as forests, while lower values occur over sparsely vegetated regions.
- AMPL, defined as MAXV – MINV, shows values similar to MAXV but slightly lower, as expected.

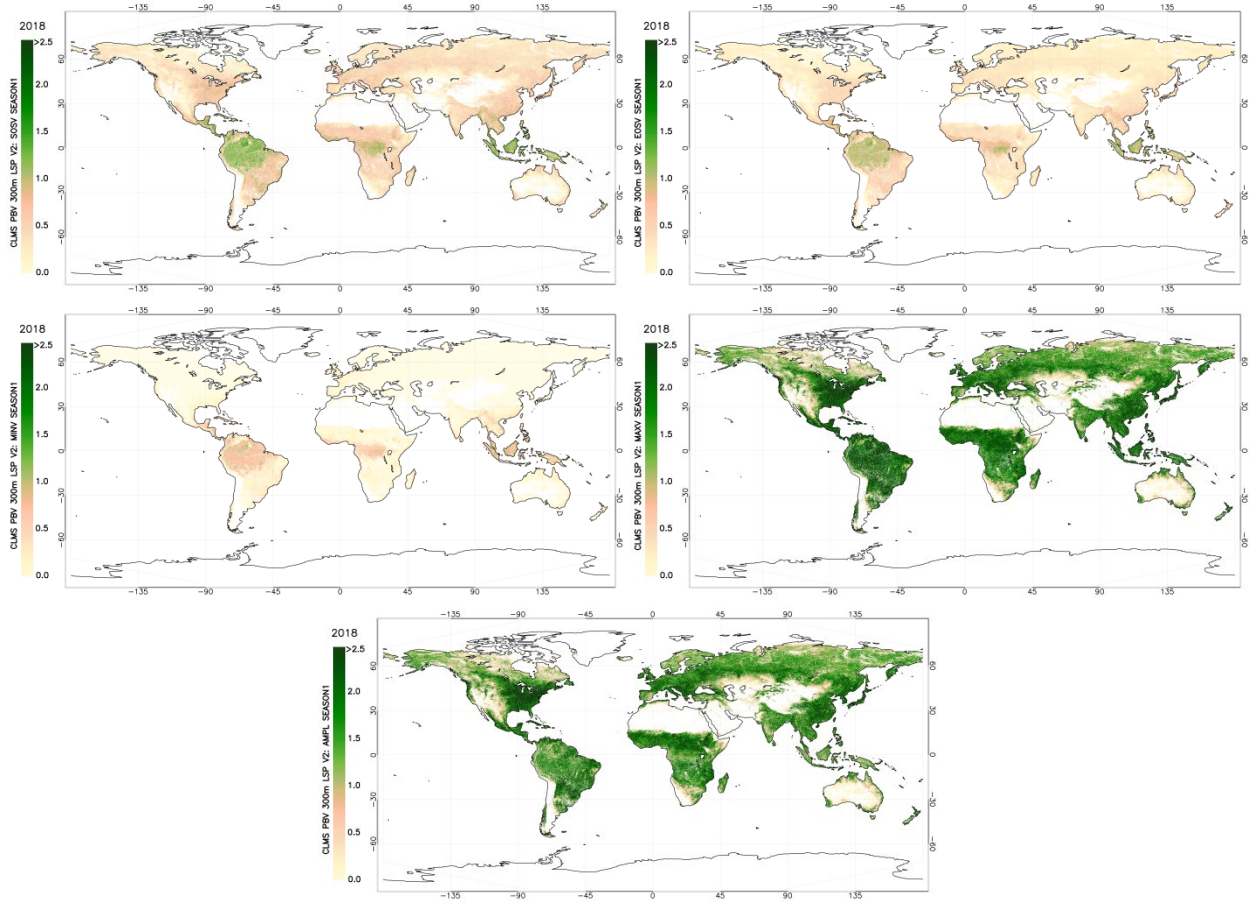


Figure 14: Global maps of CLMS PBV LSP V2 layers for 2018, season 1: SOSV, EOSV, MINV, MAXV, and AMPL (from left to right and top to bottom). Units: $m^2 \cdot m^{-2}$.

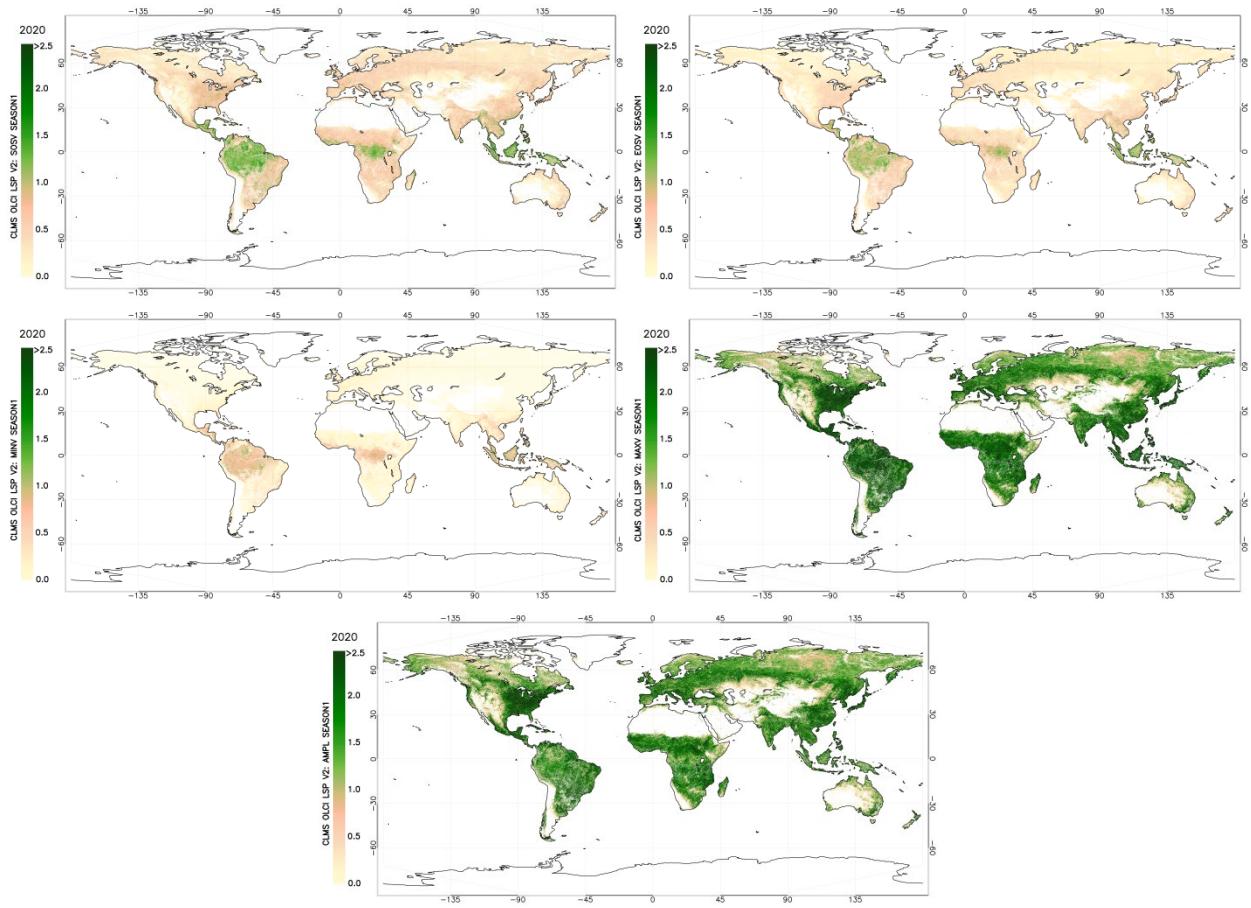


Figure 15: Global maps of CLMS OLCI LSP V2 layers for 2020, season 1: SOSV, EOSV, MINV, MAXV, and AMPL (from left to right and top to bottom). Units: $m^2 \cdot m^{-2}$.

The slope of increase and decrease of the vegetation index during the green-up and green-down periods [$m^2 \cdot m^{-2} \cdot day^{-1}$] are provided in the LSLOPE and RSLOPE layers, respectively. Global maps of both layers are presented in Figure 16 for PBV (2018) and in Figure 17 for OLCI (2020).

The visual inspection of LSLOPE and RSLOPE layers (Figure 16 and Figure 17) shows:

- The slope is generally slightly higher during the green-up (LSLOPE) period than during the green-down (RSLOPE) period, as expected.
- RSLOPE values are provided as positive values, representing the absolute magnitude of the slope during the senescence phase (but the sign of the slope should be negative).
- Larger slopes are observed over cropland and herbaceous regions, while lower slopes are found in tropical and coniferous forests, consistent with expected vegetation dynamics.

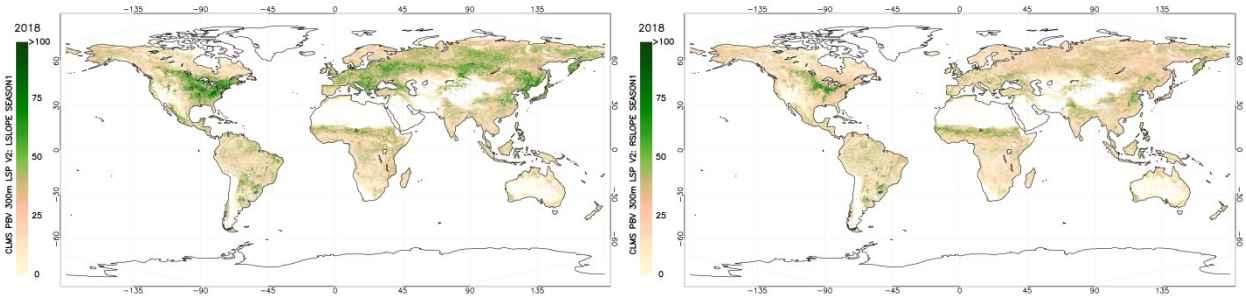


Figure 16: Global maps of CLMS PBV LSP V2 LSLOPE (left) and RSLOPE (right) for 2018, season 1. Unit: $m^2 \cdot m^{-2} \cdot day^{-1}$.

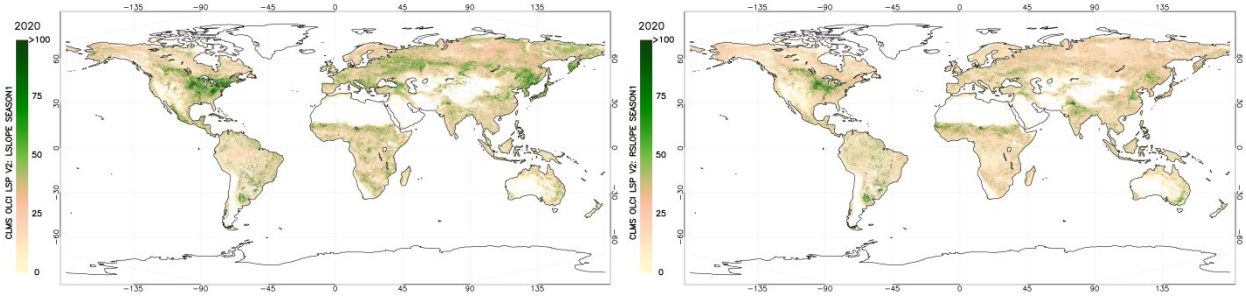


Figure 17: Global maps of CLMS OLCI LSP V2 LSLOPE (left) and RSLOPE (right) for 2020, season 1. Unit: $m^2 \cdot m^{-2} \cdot day^{-1}$.

Total and seasonal productivity [$m^2 \cdot m^{-2} \cdot day$] are provided in TPROD and SPROD layers. TPROD represents the growing season integral, computed as the sum of all daily values between SOSD and EOSD. SPROD is defined as the seasonal productivity above the base level, calculated as TPROD minus the baseline value. Global maps of TPROD and SPROD are presented in Figure 18 for PBV (2018) and in Figure 19 for OLCI (2020).

Global maps of TPROD and SPROD (Figure 18 and Figure 19) show:

- Consistent global patterns, with higher productivity over densely vegetated regions (e.g., tropical forests) and in areas with large seasonal amplitudes (see the strong spatial correspondence between TPROD/SPROD and the AMPL layer in Figure 14 and Figure 15).

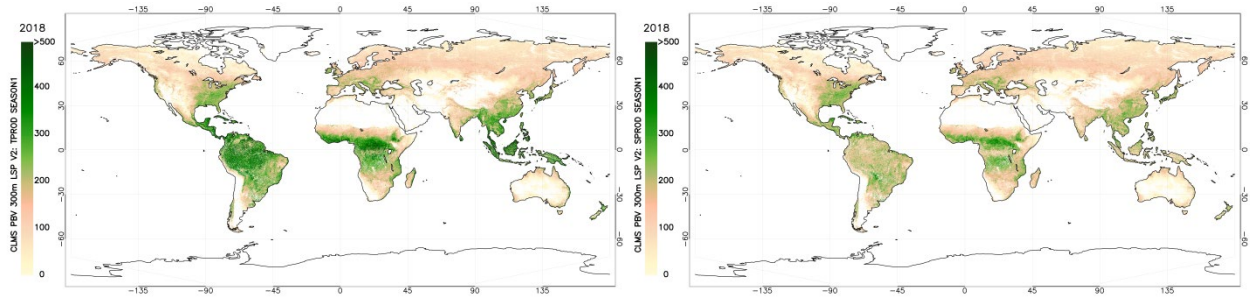


Figure 18: Global maps of CLMS PBV LSP V2 TPROD (left) and SPROD (right) for 2018, season 1. Unit: $m^2 \cdot m^{-2} \cdot day$.

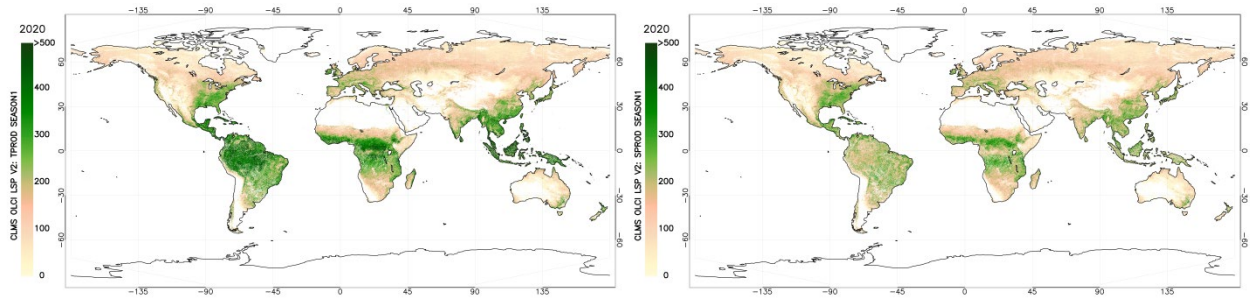


Figure 19: Global maps of CLMS OLCI LSP V2 TPROD (left) and SPROD (right) for 2020, season 1. Unit: $m^2 \cdot m^{-2} \cdot day$.

Finally, the QA layer is assigned based on both the average of PPI data quality and the total number of valid observations in each of phenological phase: green-up (SOS), green peak (peak), and green-down (EOS). Additionally, the overall QA is determined based on the comprehensive evaluation of the quality across all three phases (see Table 3).

Figure 20 and Figure 21 show the global maps of SOS, peak, EOS and overall quality provided in QA layer for PROBA-V (2018) and OLCI (2020), respectively.

Main findings from the maps of QA layer are:

- The maps of the quality of each phase (SOS, Peak and EOS quality) show that *best* or *good* typically account for more than 94% of global cases for SOS and EOS, and more than 90% for Peak. *Poor* quality generally represents less than 10% of global cases and is mainly observed over areas affected by cloud contamination.
- For the overall quality, *best* or *good* quality (i.e., maximum one phase with *poor* quality) represents around 96–97% of the cases, with no instances of poor overall quality (i.e., all three phases with poor quality). This condition is used to select *best-quality* pixels (see Table 15).

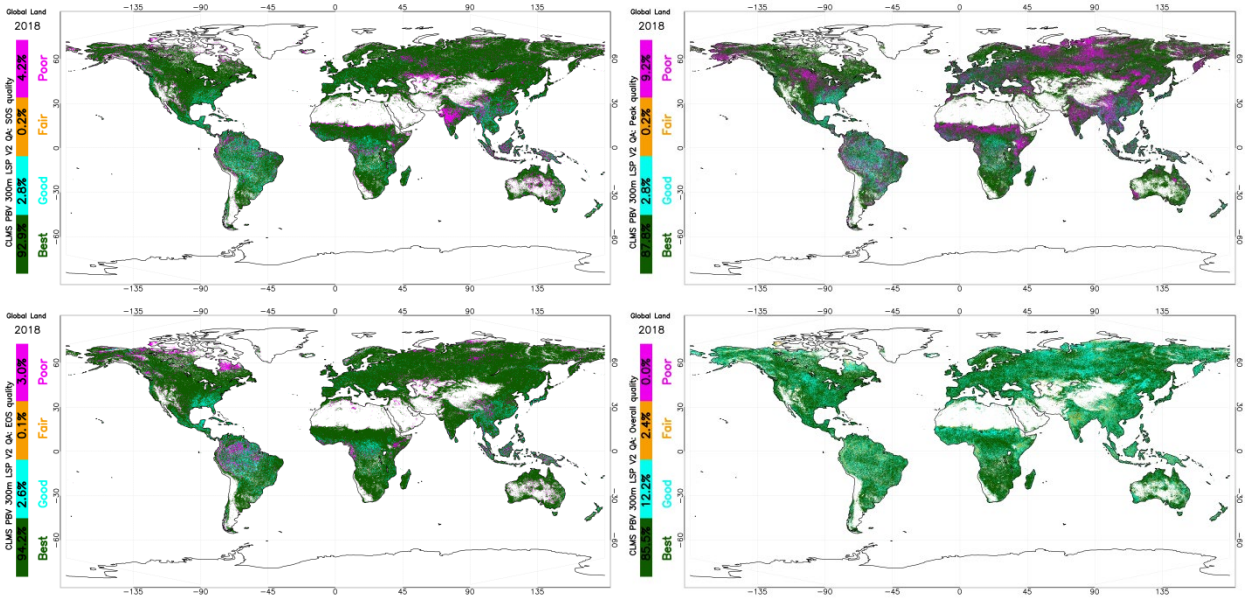


Figure 20: Global maps of CLMS PBV LSP V2 QA layers for 2018, season 1: SOS quality, Peak quality, EOS quality and Overall quality (from left to right and top to bottom).

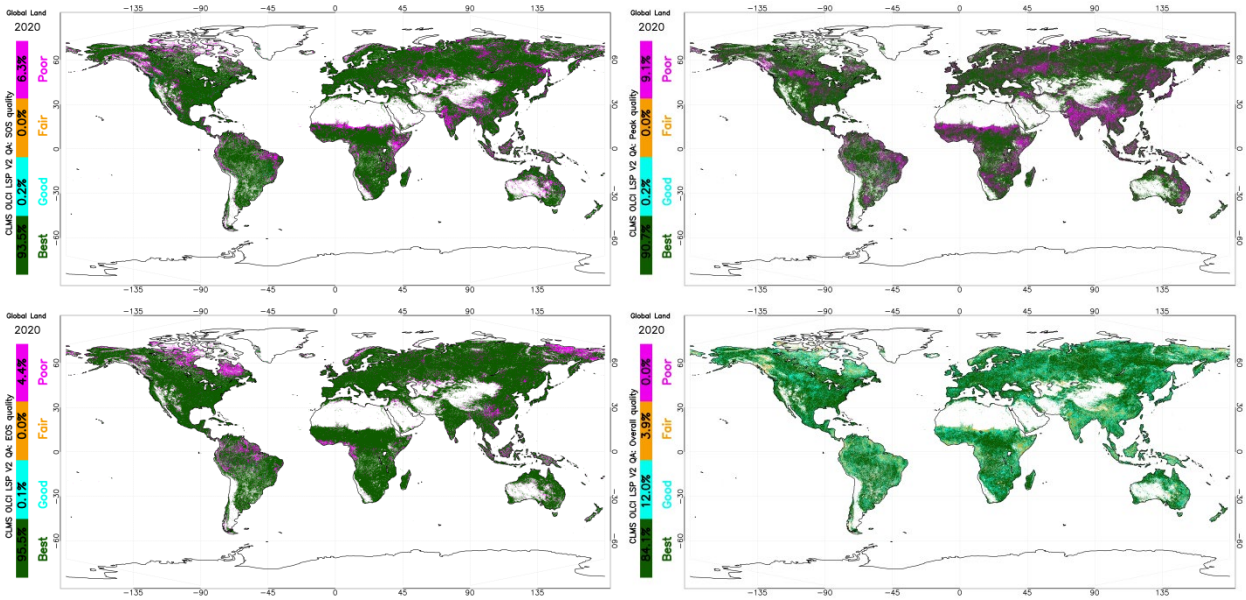


Figure 21: Global maps of CLMS OLCI LSP V2 QA layers for 2020, season 1: SOS quality, Peak quality, EOS quality and Overall quality (from left to right and top to bottom).

6.3.2 Visual inspection over subcontinental areas

In this section, a visual evaluation of the CLMS LSP V2 maps over six subcontinental areas (see Figure 3) at the full resolution is performed. Below, SOSD (Figure 22), EOSD (Figure 23), TPROD (Figure 24) and SPROD (Figure 25) maps are presented for the years 2018 (PROBA-V) and 2019 (OLCI).

Main remarks of the inspection of subcontinental CLMS LSP V2 maps at full resolution are:

- A consistent spatial distribution of phenology and productivity parameters is observed for both PROBA-V and OLCI products, reproducing the vegetation spatial patterns in a similar way.
- Exceptions are marginally found in desertic areas (e.g., AFRI), where the detected seasons may not be realistic.
- No spatial artifacts are observed.

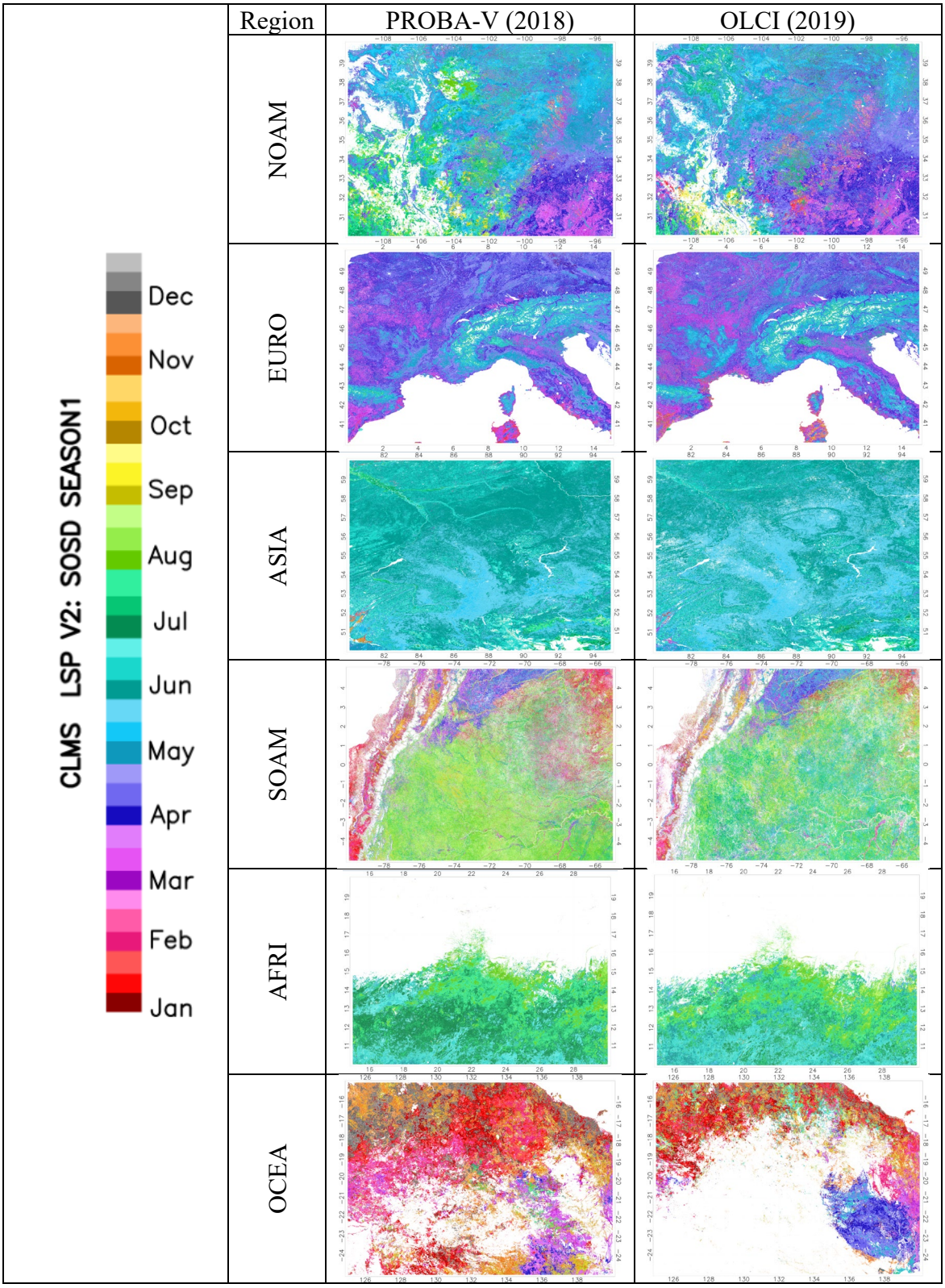


Figure 22: CLMS LSP V2 SOSD maps for season 1 over six sub-continental areas (10º latitude × 15º longitude) for 2018 (PROBA-V) and 2019 (OLCI) periods.

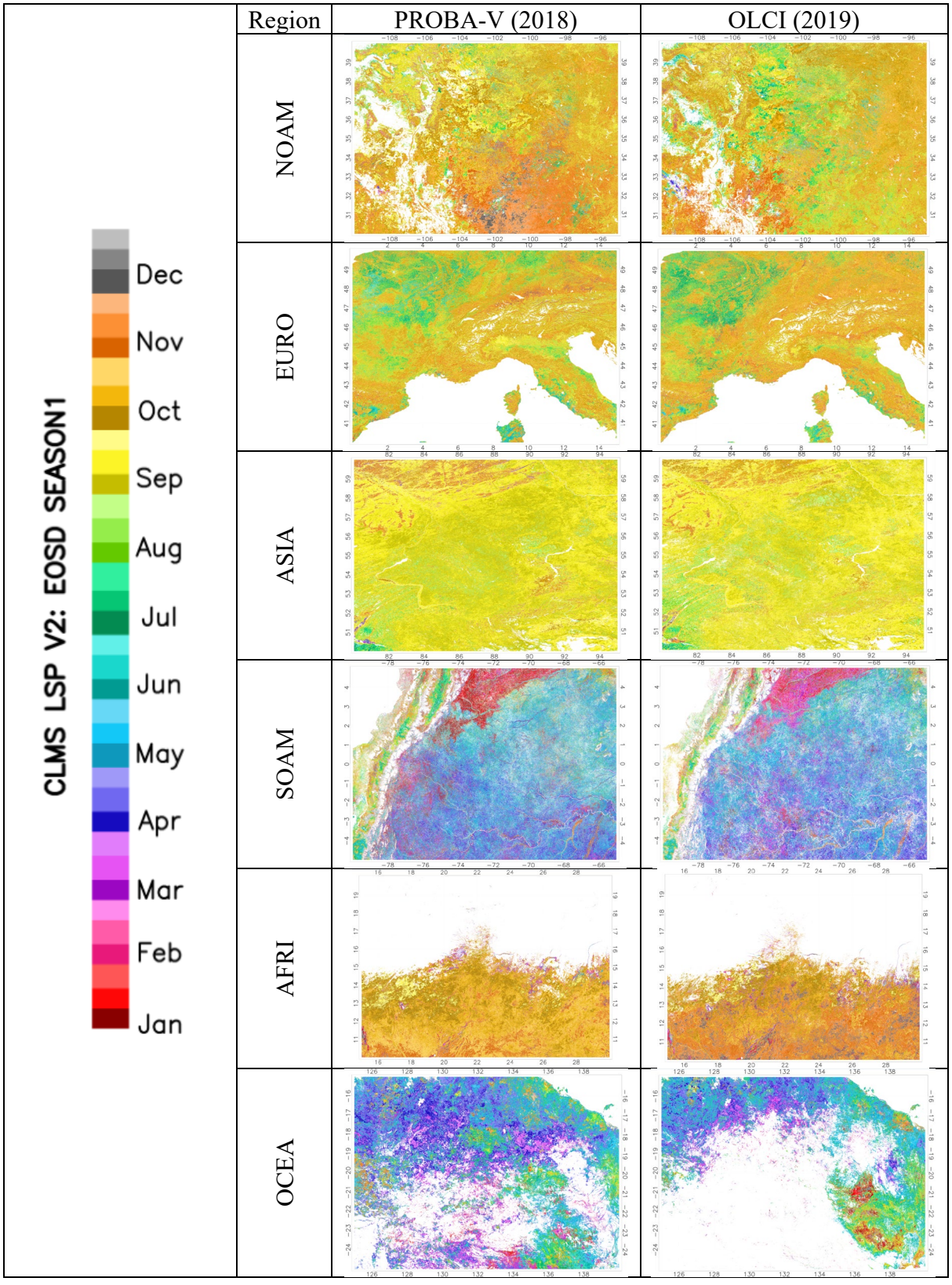


Figure 23: Same as in Figure 22 but for EOSD.

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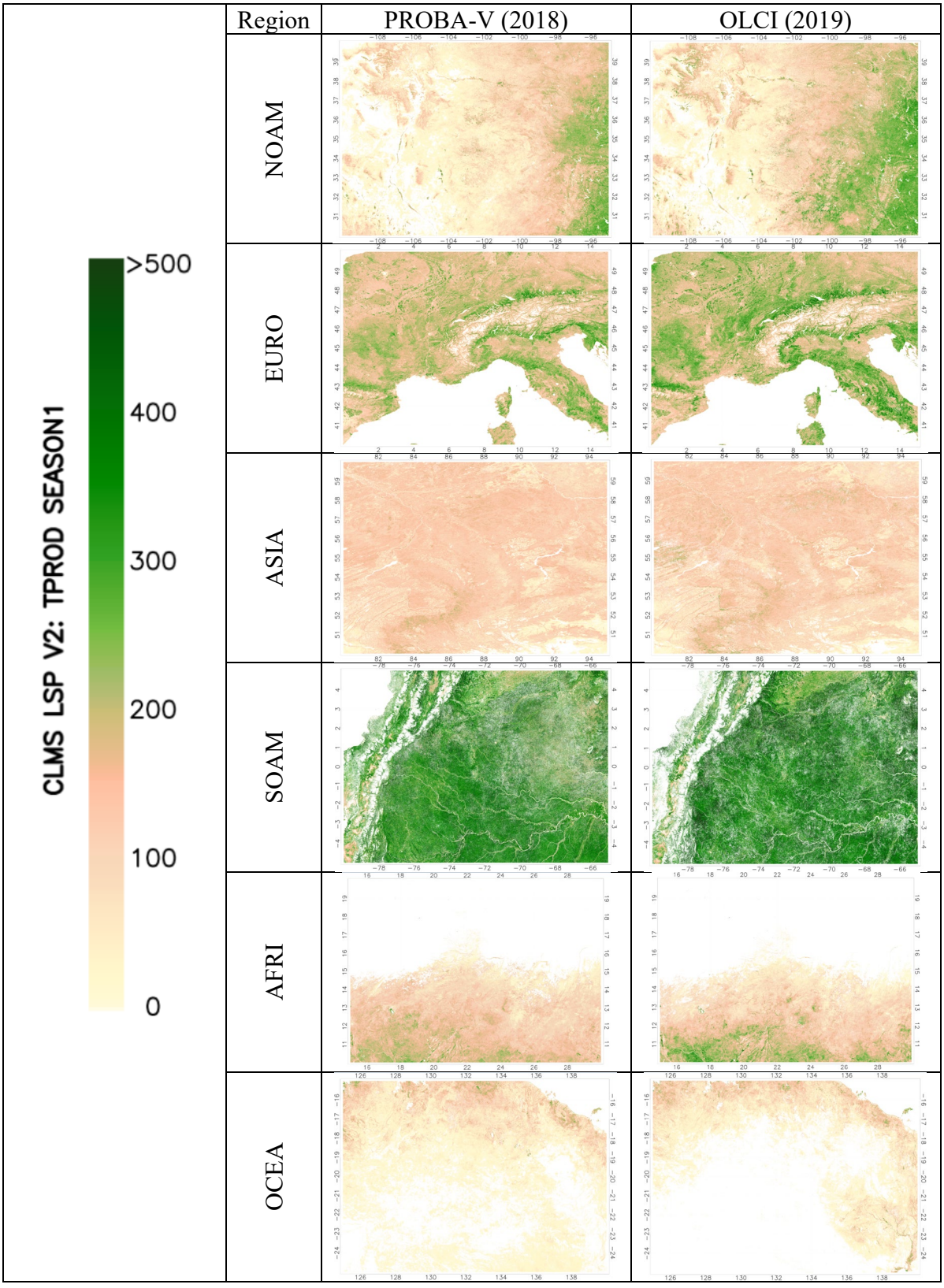

Figure 24: Same as in Figure 22 but for TPROD. Unit: $m^2 \cdot m^{-2} \cdot day$.

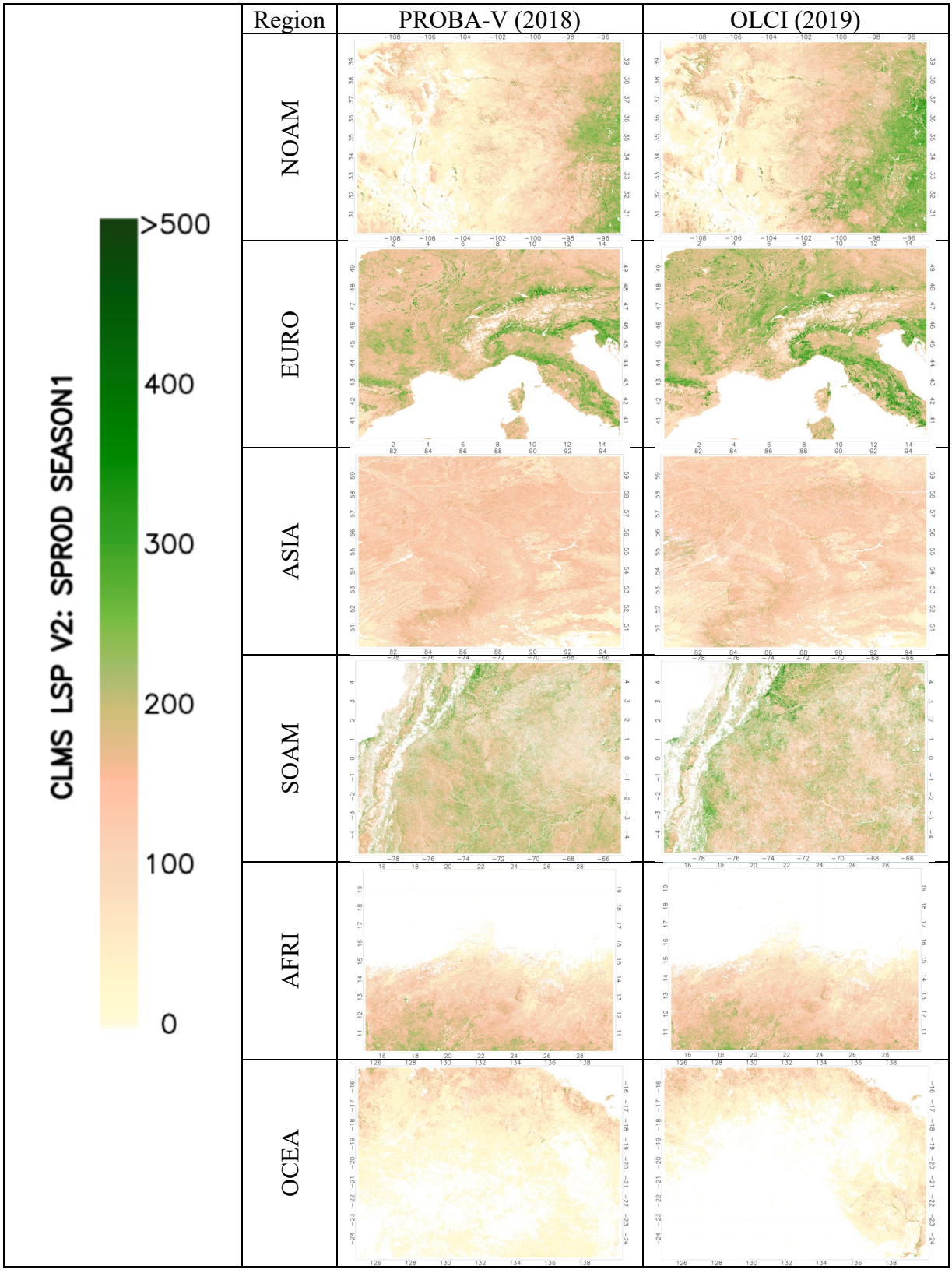

Figure 25: Same as in Figure 22 but for SPROD. Unit: $m^2 \cdot m^{-2} \cdot day$.

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6.3.3 Consistency with HR-VPP over AOIs

In this section, the spatial consistency between CLMS LSP V2 phenological (SOSD and EOSD) and productivity (SPROD and TPROD) layers and their HR-VPP equivalent is evaluated. The analysis is conducted over six AOIs (see their locations over the global map in Figure 3), centred on selected EVAL sites (see Table 13), each covering an area of 20 km x 20 km.

Figure 26 and Figure 27 illustrate, as examples, the spatial consistency results between CLMS LSP V2 and HR-VPP SOSD and EOSD for the year 2023 over the *Barrax* cultivated area and the *Białowieża National Park* DBF site. Similarly, the comparisons for SPROD and TPROD are presented in Figure 28 and Figure 29. The corresponding results for the remaining areas are provided in Annex I.

Main findings of the spatial consistency analysis between CLMS LSP V2 and HR-VPP are:

- The spatial distributions of CLMS LSP V2 SOSD, EOSD, SPROD and TPROD are consistent with those of HR-VPP, showing similar spatial patterns in most cases.
- CLMS LSP V2 generally provides similar or slightly later SOSD values compared to HR-VPP. For EOSD, CLMS LSP V2 tends to yield higher (i.e., later) values over cultivated areas, while nearly identical distributions are observed for other biomes (forests and shrublands).
- For SPROD and TPROD, CLMS LSP V2 shows systematically higher values than HR-VPP across forest cases and similar values for non-forests. Differences in the maximum PPI values established in each product (up to 5 $m^2 \cdot m^{-2}$ for CLMS LSP V2 (Table 2) and up to 3 $m^2 \cdot m^{-2}$ for HR-VPP [HR-VPP_ST&VPP_UM]) can explain the higher values of CLMS LSP V2.

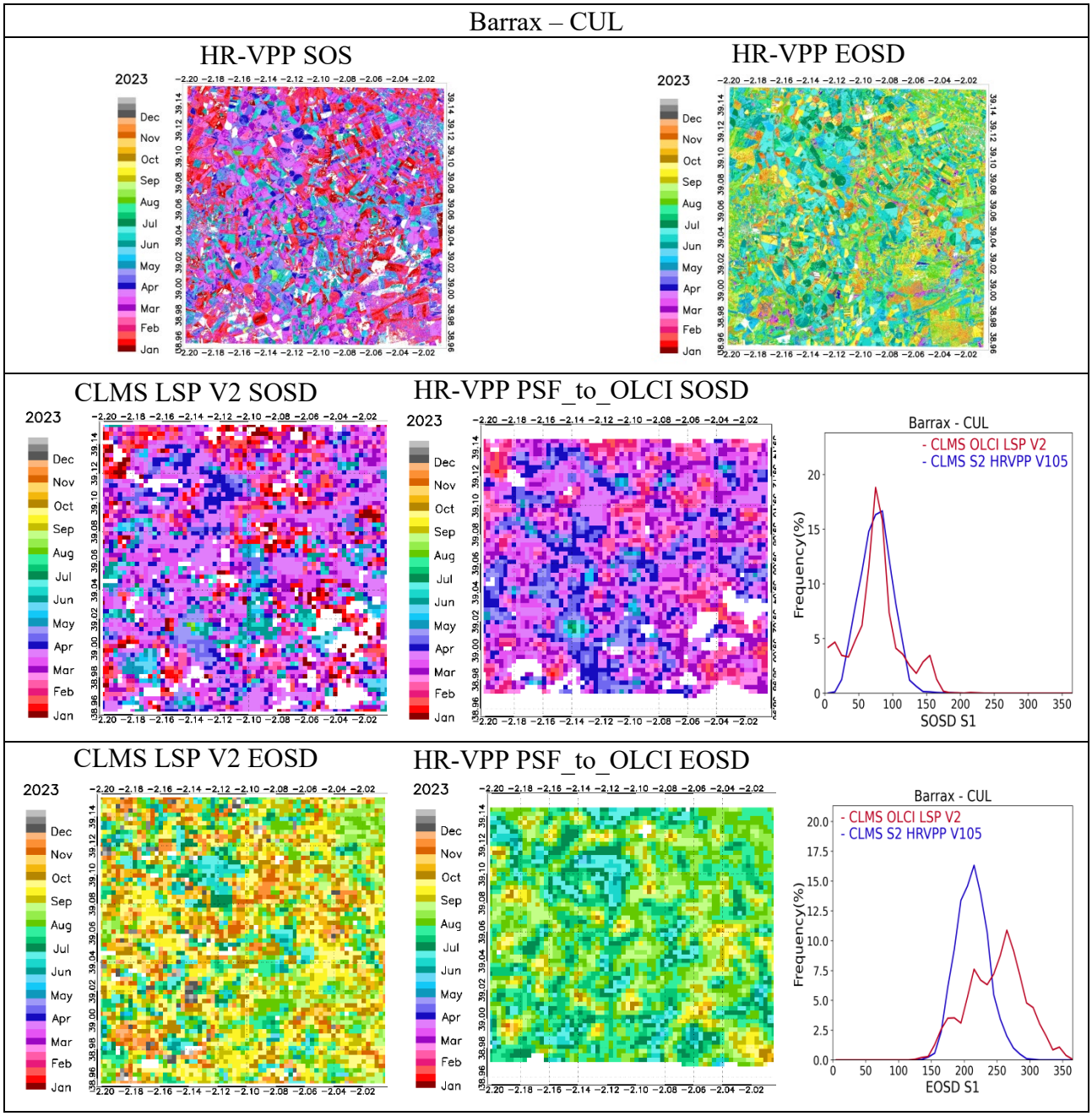

Figure 26: Spatial consistency over the *Barrax* cultivated site (20 km × 20 km) between CLMS LSP V2 and HR-VPP for season 1, 2023. Line 1: HR-VPP SOSD (left) and EOSD (right) at 10 m resolution. Line 2: CLMS LSP V2 SOSD (left), HR-VPP aggregated to 300 m using PSF (center), and their associated histograms (right). Line 3: same as in Line 2, but for EOSD.

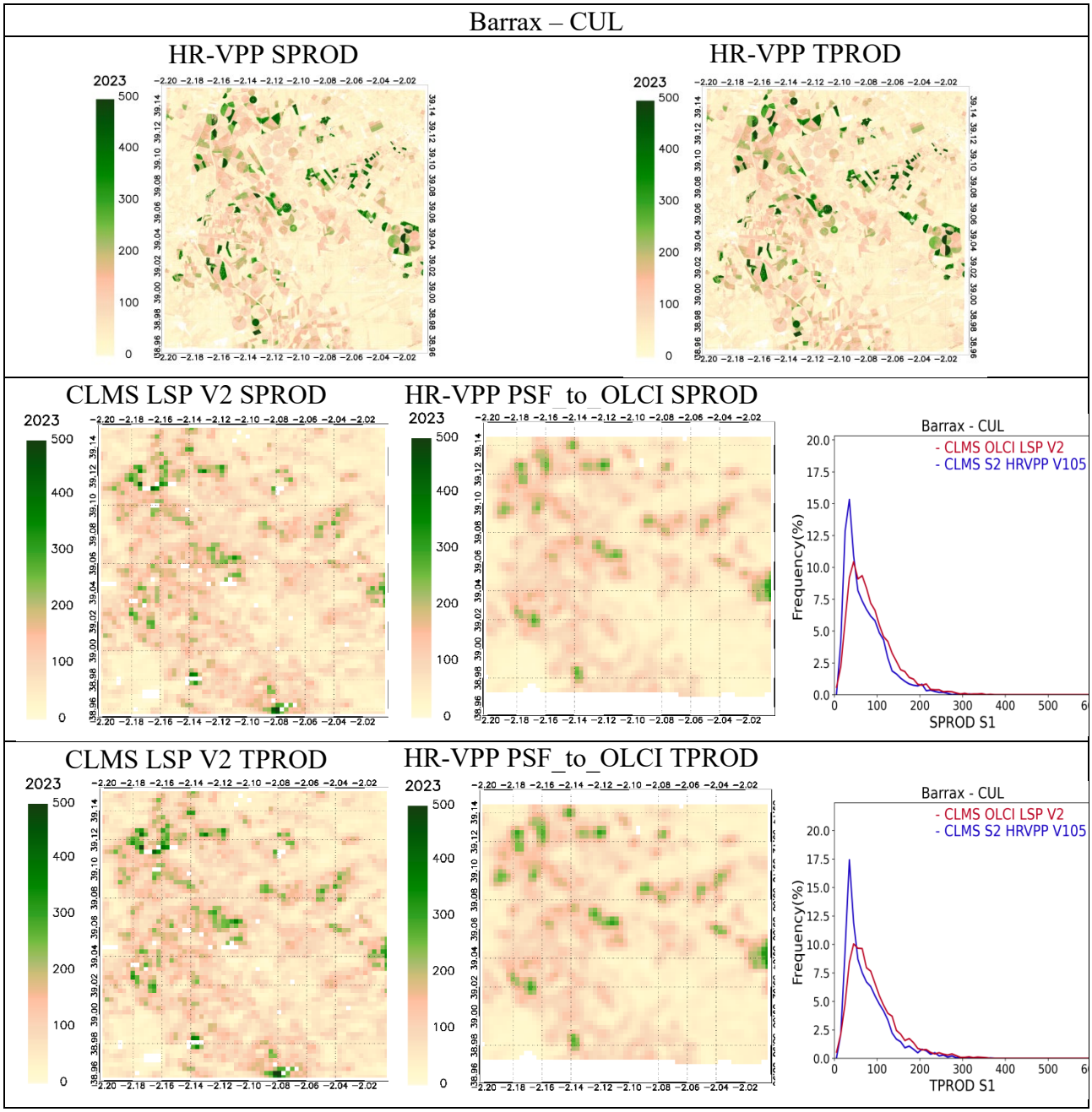

Figure 27: Spatial consistency over the *Barrax* cultivated site (20 km × 20 km) between CLMS LSP V2 and HR-VPP for season 1, 2023. Line 1: HR-VPP SPROD (left) and TPROD (right) at 10 m resolution. Line 2: CLMS LSP V2 SPROD (left), HR-VPP aggregated to 300 m using PSF (center), and their associated histograms (right). Line 3: same as in Line 2, but for TPROD.

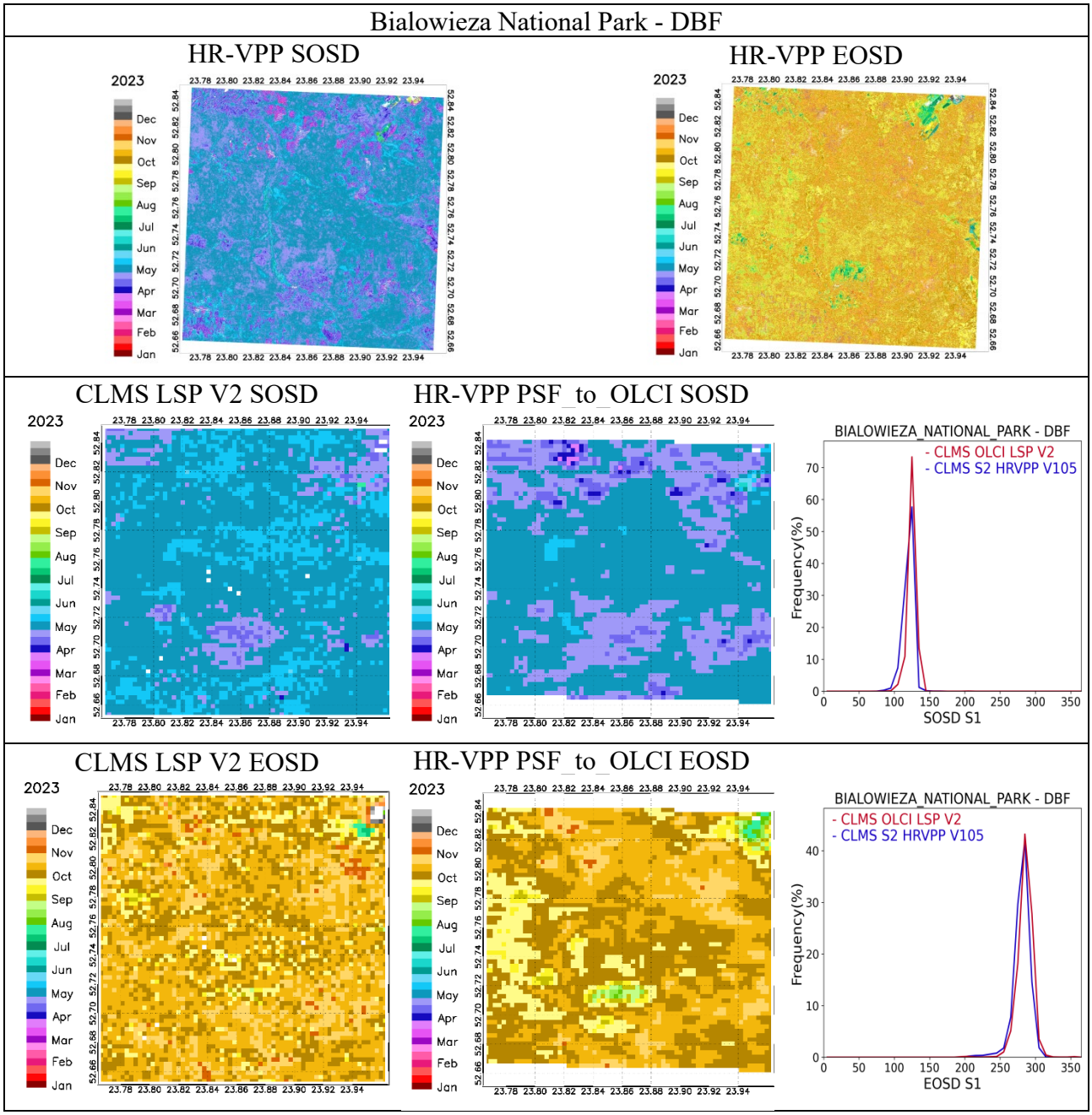

Figure 28: Same as in Figure 26 but for *Bialowieza National Park* DBF site.

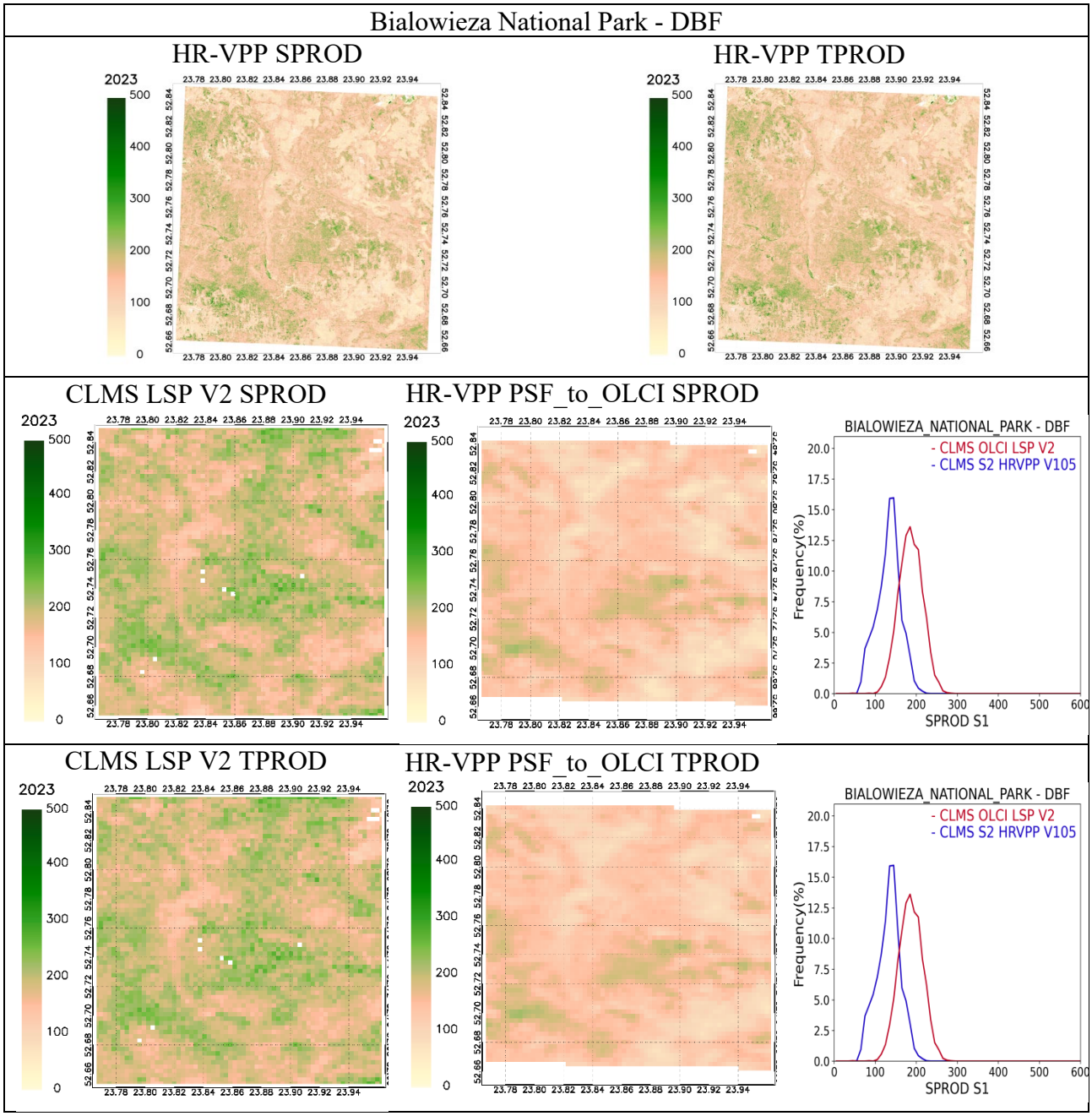

Figure 29: Same as in Figure 27 but for *Bialowieza National Park* DBF site.

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6.3.4 Consistency with global LSP products

This section assesses the spatial consistency of CLMS LSP V2 SOSD and EOSD with both NASA MODIS V6.1 and VIIRS V2 LCD products. Global maps and histograms of the differences are analysed for season 1 of year 2020, as well as the variation of SOSD and EOSD as a function of latitude. Note that only year 2020 is shown for brevity, as similar results are observed throughout the entire 2014–2024 period, regardless of the input dataset used for CLMS LSP V2 (PBV or OLCI).

6.3.4.1 Differences with MODIS LCD V6.1

The results of the spatial consistency analysis (i.e., difference maps and histograms of differences) between CLMS LSP V2 (SOSD, EOSD) and MODIS LCD V6.1 (*MidGreenup*, *MidGreendown*) are presented in Figure 30 and Figure 31. *MidGreenup* and *MidGreendown* layers are used from MODIS LCD V6.1, as they show the best agreement with CLMS LSP V2 SOSD and EOSD, respectively (see section 6.5.2.1).

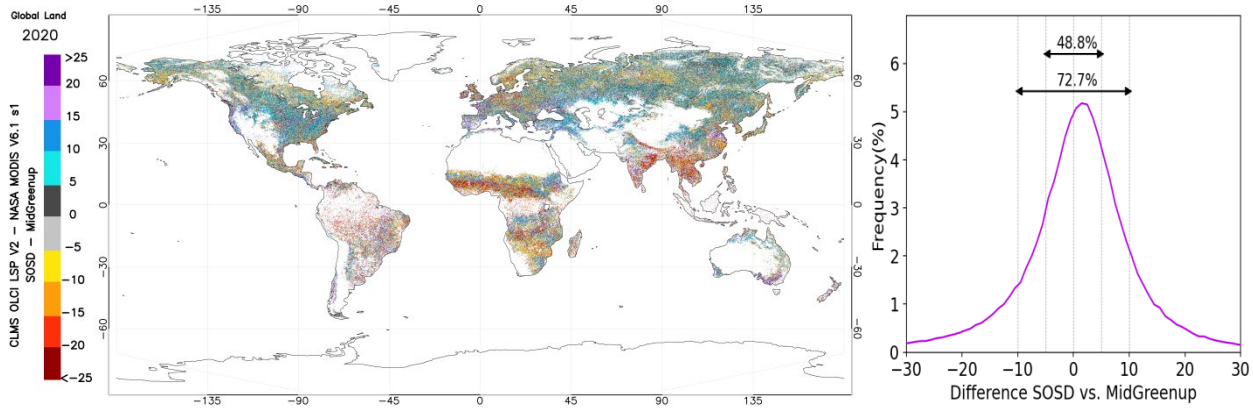

Figure 30: Global map (left) and histogram (right) of differences between CLMS LSP V2 SOSD and MODIS V6.1 *MidGreenup* for season 1 of 2020.

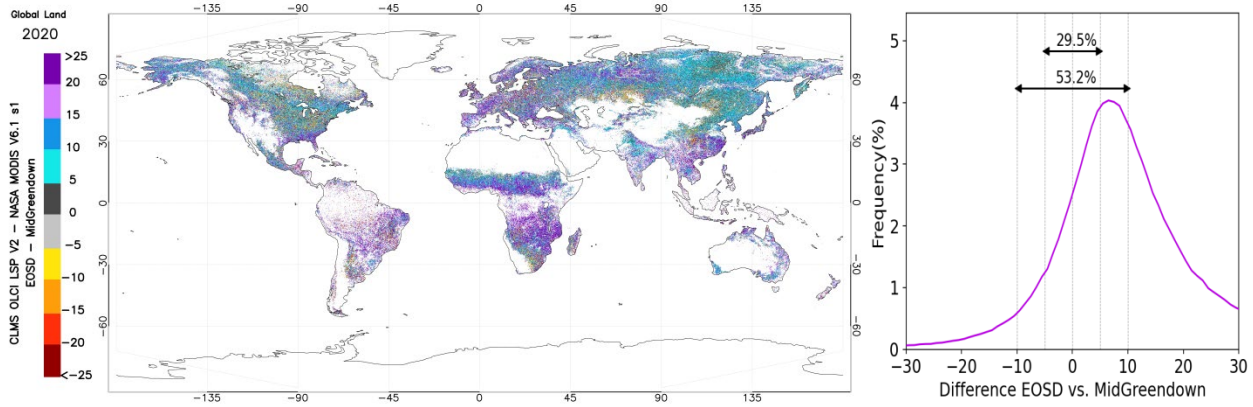

Figure 31: Global map (left) and histogram (right) of differences between CLMS LSP V2 EOSD and MODIS V6.1 *MidGreendown* for season 1 of 2020.

Main results of the spatial consistency between CLMS LSP V2 and MODIS LCD V6.1 are:

- CLMS LSP V2 SOSD and MODIS V6.1 *MidGreenup* show differences typically within ±15 days across large regions, particularly in the Northern Hemisphere. Larger discrepancies are observed over tropical areas, where MODIS *MidGreenup* generally shows later dates (>20 days). The opposite sign of differences reveals some spatial inconsistencies across some regions. The histogram of differences is centred around +2 days, indicating slightly higher values for CLMS SOSD, despite its definition at a lower threshold level than MODIS *MidGreenup* (25% for CLMS, 50% for MODIS)
- The global map of differences between CLMS LSP V2 EOSD and MODIS V6.1 *MidGreendown* shows systematically positive differences (CLMS > MODIS), as expected because of its definitions (15% for CLMS, 50% for MODIS), indicating overall good spatial consistency across most regions of the globe. The histogram of differences is centred around +4 days, with slightly higher values for CLMS EOSD, consistent with the differences in their phenological definitions.

6.3.4.2 Differences with VIIRS LCD V2

The results of the spatial consistency between CLMS LSP V2 (SOSD, EOSD) and VIIRS LCD V2 are presented in Figure 32 and Figure 33. As for MODIS, *Date_Mid_Greenup_Phase*, *Date_Mid_Senescence_Phase* VIIRS phenometrics are used, as they show the best overall agreement with CLMS SOSD and EOSD, respectively (see section 6.5.2.1).

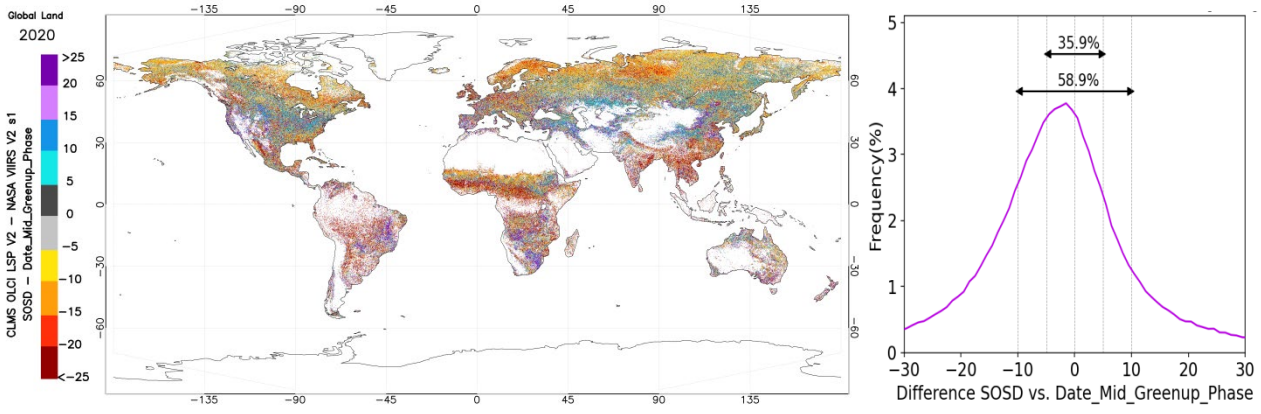

Figure 32: Global map (left) and histogram (right) of differences between CLMS LSP V2 SOSD and VIIRS V2 *Date_Mid_Greenup_Phase* for season 1 of 2020.

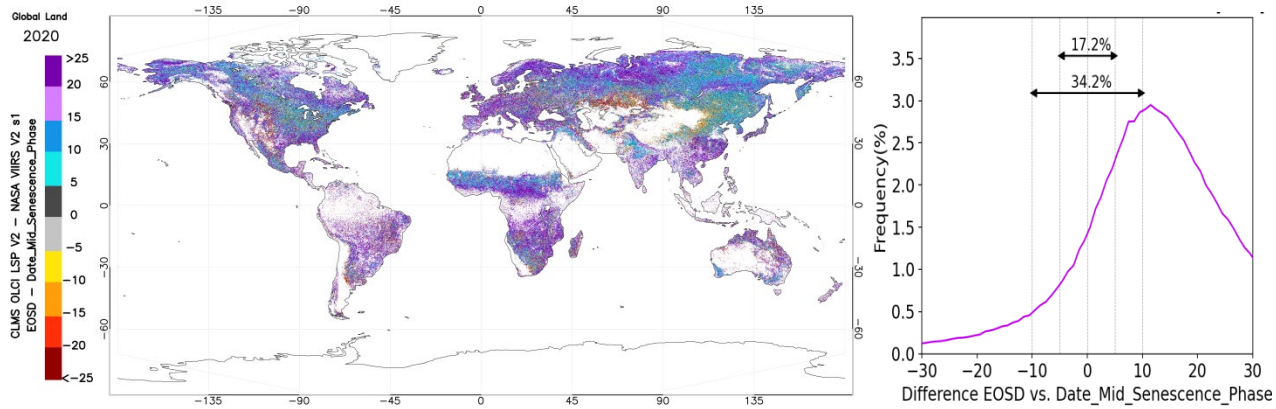

Figure 33: Global map (left) and histogram (right) of differences between CLMS LSP V2 EOSD and VIIRS V2 *Date_Mid_Senescence_Phase* for season 1 of 2020.

Main results of the spatial consistency between CLMS LSP V2 and VIIRS LCD V2 are:

- For SOSD, CLMS LSP V2 generally provides lower values (i.e., an earlier start of the season) than VIIRS V2 *Date_Mid_Greenup_Phase*, as expected from their definition differences, showing overall good spatial consistency around most parts of the globe. However, the opposite pattern is observed over temperate latitudes and certain southern regions.
- For EOSD, CLMS LSP V2 shows, as expected, higher values (i.e., later end of season) compared to the VIIRS V2 *Date_Mid_Senescence_Phase* product over most regions worldwide, showing overall good spatial consistency.
- Histograms of differences are centred around -2 and +12 days for SOSD and EOSD, respectively.

6.3.4.3 Latitudinal transect

Figure 34 shows the variation of CLMS LSP V2 SOSD and EOSD as a function of latitude along a geodesic transect from Norway (70.315ºN, 20.119ºE) to South Africa (34.872ºS, 20.119ºE), with a spatial step of 2 km. For benchmarking purposes, the latitudinal trends of MODIS LCD V6.1 and VIIRS LCD V2 *mid-greenup*, and *mid-senescence* (or *mid-greendown*) phenometrics are also included. Same as in the previous section, only results for year 2020 are presented for brevity, since similar patterns are observed throughout the entire 2014–2024 period, regardless of the input dataset used for CLMS LSP V2 (PROBA-V or OLCI).

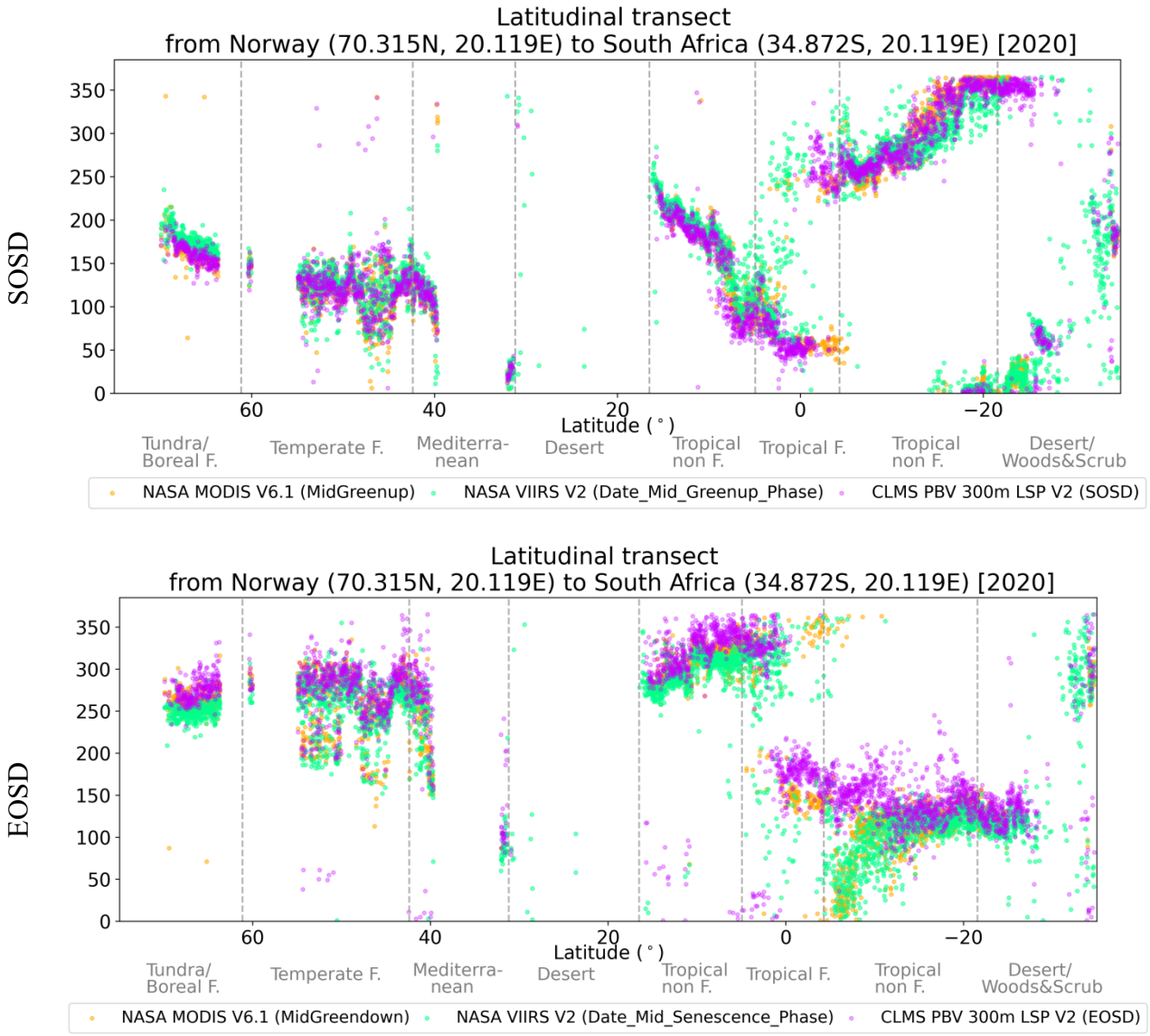

Figure 34: Latitudinal transect of season 1 SOSD (top) and EOSD (bottom) for 2020. CLMS LSP V2 values are shown as magenta points, MODIS LCD V6.1 as orange points, and VIIRS LCD V2 as green points, along the geodesic from Norway (70.315ºN, 20.119ºE) to South Africa (34.872ºS, 20.119ºE).

Main results are:

- Over Europe, all three products display similar trends for both SOSD and EOSD. As expected, SOSD decreases from north to south. The VIIRS product (green dots) shows slightly higher SOSD and lower EOSD values, indicating a shorter growing season than CLMS (magenta) and MODIS (orange).

- Over the Sahara Desert, none of the three products provide estimates, as expected. Only VIIRS shows some retrievals due to its quality flag not being applied.
- Over tropical non-forest regions of the Northern Hemisphere, all products show consistent trends: a decrease of SOSD (from July to February) and a slight increase of EOSD (from September to December) from north to south, corresponding to a growing season that extends from about 2 months near the desert border to nearly 9 months in equatorial regions.
- Larger discrepancies are observed over tropical forest around the equator for the SOSD in the VIIRS products. CLMS displays distinct two contrasting values near 0°, which is explained by the detection of two growing seasons just below the equator (see global maps in Figure 12and Figure 13).
- In the southern part of the equatorial zone, CLMS SOSD increases from September to December/January, while EOSD remains relatively stable around April/May. VIIRS and MODIS show greater variability for end-of-season dates around 12–15°S, whereas CLMS remains more stable.

6.3.5 Spatial precision

The spatial precision of CLMS LSP V2 SOSD and EOSD phenological metrics, as well as SPROD and TPROD productivity parameters, was assessed over six Regions of Interest (ROIs) covering homogeneous areas of 5 km × 5 km at full resolution (see Table 14). Two representative years from the PBV period (2017 and 2018) and two from the OLCI period (2019 and 2020) were analyzed. The standard deviation (STD) and median absolute deviation (MAD) were used as indicators of precision based on spatial analysis. Table 18 and Table 19 summarize the precision indicators (STD and MAD) for SOSD and EOSD across the six ROIs, while Table 20 and Table 21 present the same indicators for SPROD and TPROD, including relative values (in %) with respect to the mean over each area. The corresponding maps of SOSD, EOSD, SPROD, and TPROD for the 5 km × 5 km ROIs are provided in Annex II.

Table 18: Summary of the spatial precision indicators (STD and MAD) of CLMS LSP V2 SOSD (season1) evaluated over 5 km x 5 km ROIs for the 2017–2020 period.

		CLMS LSP V2 SOSD (days)							
		2017		2018		2019		2020	
ROI	Biome	STD	MAD	STD	MAD	STD	MAD	STD	MAD
Evare	EBF	53	17	59	22	34	18	34	12
Mountain Lake	DBF	4	2	2	1	3	2	3	2
Oskoba	NLF	30	12	12	2	8	2	11	4
Albufera	CUL	2	1	3	1	3	2	4	3
Cimarron N.P.	HER	14	5	12	6	4	2	18	8
Jornada	SHR	15	5	5	3	3	2	3	2

Table 19: Summary of the spatial precision indicators (STD and MAD) of CLMS LSP V2 EOSD (season1) evaluated over 5 km x 5 km ROIs for the 2017–2020 period.

		CLMS LSP V2 EOSD (days)							
		2017		2018		2019		2020	
ROI	Biome	STD	MAD	STD	MAD	STD	MAD	STD	MAD
Evare	EBF	61	34	29	12	64	18	91	36
Mountain Lake	DBF	7	4	8	5	9	6	8	6
Oskoba	NLF	51	14	42	30	12	7	8	5
Albufera	CUL	5	3	5	4	5	3	5	3
Cimarron N.P.	HER	8	5	3	2	15	11	7	6
Jornada	SHR	5	2	17	15	15	11	6	4

Table 20: Summary of the spatial precision indicators (STD and MAD) of CLMS LSP V2 SPROD (season1) evaluated over 5 km x 5 km ROIs for the 2017–2020 period.

		CLMS LSP V2 SPROD ($m^2.m^{-2}.day$)							
		2017		2018		2019		2020	
ROI	Biome	STD	MAD	STD	MAD	STD	MAD	STD	MAD
Evare	EBF	45 (29%)	24 (16%)	46 (24%)	22 (11%)	33 (22%)	20 (13%)	40 (21%)	25 (13%)
Mountain Lake	DBF	36 (13%)	24 (9%)	48 (25%)	38 (20%)	66 (30%)	56 (25%)	43 (20%)	32 (15%)
Oskoba	NLF	34 (63%)	30 (55%)	24 (46%)	18 (35%)	20 (33%)	14 (24%)	24 (30%)	17 (22%)
Albufera	CUL	22 (14%)	14 (9%)	23 (14%)	13 (8%)	32 (21%)	22 (15%)	31 (17%)	22 (12%)
Cimarron N.P.	HER	21 (22%)	16 (17%)	20 (26%)	13 (17%)	25 (38%)	18 (28%)	12 (21%)	9 (16%)
Jornada	SHR	6 (28%)	4 (18%)	3 (34%)	2 (23%)	5 (32%)	3 (23%)	8 (19%)	5 (12%)

Table 21: Summary of the spatial precision indicators (STD and MAD) of CLMS LSP V2 TPROD (season1) evaluated over 5 km x 5 km ROIs for the 2017–2020 period.

		CLMS LSP V2 TPROD ($m^2.m^{-2}.day$)							
		2017		2018		2019		2020	
ROI	Biome	STD	MAD	STD	MAD	STD	MAD	STD	MAD
Evare	EBF	83 (23%)	45 (13%)	63 (17%)	27 (7%)	62 (16%)	31 (8%)	75 (20%)	39 (10%)
Mountain Lake	DBF	36 (13%)	25 (9%)	48 (26%)	39 (21%)	69 (31%)	58 (26%)	45 (21%)	33 (15%)
Oskoba	NLF	34 (62%)	30 (55%)	24 (46%)	18 (34.7%)	20 (33%)	15 (25%)	24 (30%)	17 (22%)
Albufera	CUL	22 (14%)	14 (9%)	23 (14%)	13 (8%)	32 (21%)	22 (15%)	31 (17%)	22 (12%)
Cimarron N.P.	HER	21 (22%)	16 (17%)	20 (26%)	13 (17%)	25 (38%)	18 (28%)	12 (21%)	9 (16%)
Jornada	SHR	6 (28%)	4 (18%)	3 (34%)	2 (23%)	5 (32%)	3 (23%)	8 (19%)	5 (12%)

Main conclusions from the spatial precision analysis of CLMS LSP V2 are:

- For SOSD, the spatial precision (STD, MAD) depends on the biome type:
  - Lowest variability (i.e., better spatial precision) is observed over deciduous broadleaf forest (DBF, *Mountain Lake*), croplands (CUL, *Albufera*), and shrublands (SHR, *Jornada*), with STD below 5–15 days and MAD values around 1–5 days.
  - Intermediate variability is found for needle-leaf forest (NLF, *Oskoba*) and herbaceous (HER, *Cimarron*) areas, with STD typically between 8–30 days.
  - Highest variability (i.e., lowest spatial precision) is observed for evergreen broadleaf forest (EBF, *Evare*), with STD up to 59 days, that can be partly explained due to the natural variability of this ecosystem with hundreds of different vegetation species per hectare, along with the likely higher noise in the data due to persistent clouds in tropical forests.
  - Overall, the spatial precision (MAD) remains within 2–8 days for most biomes, except EBF where it increases up to 22 days. OLCI typically shows slightly better spatial precision than PBV.
- For EOSD, the spatial precision (STD, MAD) shows similar dependence on biome type than SOSD.
  - Deciduous broadleaf forest (DBF, *Mountain Lake*), croplands (CUL, *Albufera*), herbaceous (HER, *Cimarron*), and shrublands (SHR, *Jornada*) exhibit low to moderate variability, with STD typically below 15 days and MAD between 2–6 days. For those cases, PBV exhibits slightly better spatial precision than OLCI.
  - Moderate variability is found in needle-leaf forest (NLF, *Oskoba*), where STD decreases from 51 days (2017) to 8 days (2020), showing more spatial precision for OLCI.
  - The highest variability is observed for evergreen broadleaf forest (EBF, *Evare*), with STD up to 91 days and MAD up to 36 days, reflecting the complex and less distinct seasonality of tropical evergreen canopies.
  - Overall, spatial precision (MAD) remains within 2–6 days for most biomes, except in EBF.
- The spatial variability of SPROD and TPROD is generally moderate across biomes, with relative variability (STD%, MAD%) typically between 10–30%, except in needle-leaf forest or sparse vegetation types such as shrublands, mainly attributed to the low productivity values over this area.

6.4 ERROR EVALUATION WITH GROUND-BASED REFERENCES

This section presents the results of the error evaluation of CLMS LSP V2 SOSD and EOSD versus PEP725, TEMPO and NEON phenological observations and versus near-ground-based PhenoCam V3 datasets for the 2014–2024 period. The summary of mean Bias and RMSD per year is presented in Annex III.

6.4.1 CLMS LSP V2 vs. phenological observations

6.4.1.1 PEP725

The results of the comparison between CLMS LSP V2 SOSD and EOSD and the PEP725 dataset for the 2014–2024 period are presented in Figure 35 and Figure 36. This analysis is carried out using different phenophases from PEP725 (see Table 7). For CLMS SOSD (Figure 35), the comparison is performed against PEP725 phases 10, 11, and 223, which correspond to *first leaves separated*, *first leaves unfolded* and *leaf unfolding ≥ 50%*, respectively. CLMS EOSD (Figure 36) is compared with PEP725 phases 95 (*50% of leaves fallen*) and 205 (*autumnal colouration ≥ 50%*). The main validation metrics are summarized in Table 22.

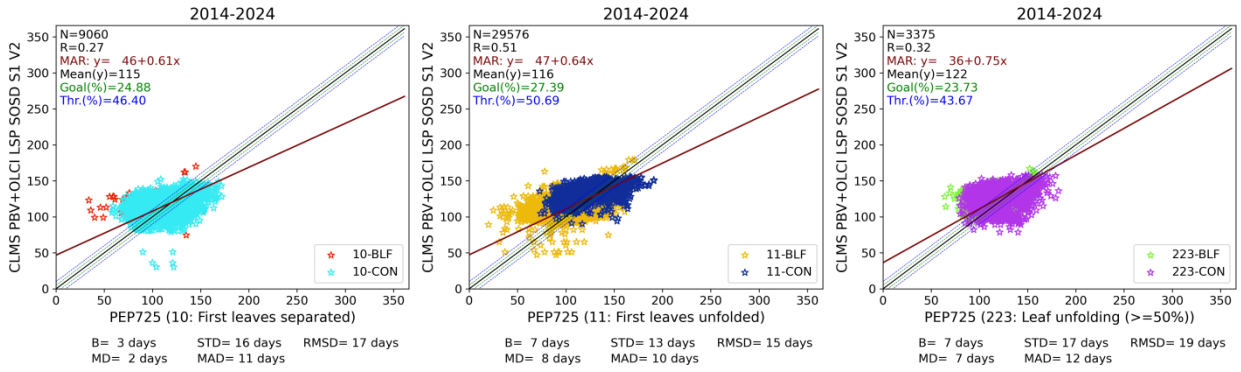

Figure 35: Scatter-plots of CLMS LSP V2 SOSD versus PEP725 phases 10 (*first leaves separated*, left), 11 (*first leaves unfolded*, middle), and 223 (*leaf unfolding ≥ 50%*, right) for season 1 over the 2014–2024 period. Green and blue lines around the 1:1 line indicate the goal and threshold requirements, respectively. BLF and CON stands for broadleaved forests and coniferous.

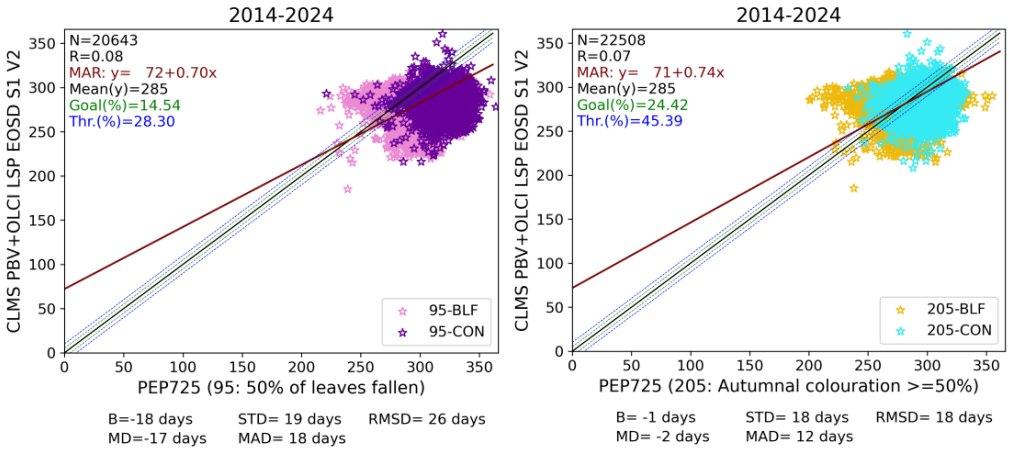

Figure 36: Scatter-plots of CLMS LSP V2 EOSD versus PEP725 phases 95 (*50% of leaves fallen*, left) and 205 (*autumnal colouration ≥ 50%*, right) for season 1 over the 2014–2024 period. Green and blue lines around the 1:1 line indicate the goal and threshold requirements, respectively. BLF and CON stands for broadleaved forests and coniferous.

Table 22: Performance statistics between CLMS LSP V2 versus PEP725 database: SOSD versus PEP725 Phases 10, 11 and 223; and EOSD versus PEP725 Phases 95 and 205. Analysis for season 1 over the 2014–2024 period.

	CLMS LSP V2 SOSD vs. PEP725			CLMS LSP V2 EOSD vs. PEP725	
	Phase 10 first leaves separated	**Phase 11** first leaves unfolded	**Phase 223** leaf unfolding ≥ 50%	**Phase 95** 50% of leaves fallen	**Phase 205** autumnal colouration ≥ 50%
N	9060	29576	3375	20643	22508
R	0.27	0.51	0.32	0.08	0.07
Bias (days)	3	7	7	-18	-1
MD (days)	2	8	7	-17	-2
STD (days)	16	13	17	19	18
MAD (days)	11	10	12	18	12
RMSD (days)	17	15	19	26	18
MAR	y=46-0.61x	y=47+0.64x	y=36+0.75x	y=72+0.70x	y=71+0.74x
% goal	24.9	27.4	23.7	14.5	24.4
% threshold	46.4	50.7	43.7	28.3	45.4

Main results of the comparison between CLMS LSP V2 versus PEP725 are:

- CLMS LSP SOSD V2 shows:
  - Positive bias with respect to PEP725 phases 10 (*first leaves separated*), 11 (*first leaves unfolded*), and 223 (*leaf unfolding ≥ 50%*) of 3, 7, and 7 days, respectively. These

small positive biases indicate CLMS SOSD (defined at 25% of seasonal amplitude) generally occurs slightly later than the PEP725 in-situ observations. PEP725 phases capture specific points in the early leaf development. Consequently, a delay of a few days can be expected, as the satellite signal integrates the reflectance of a larger area and responds only once a sufficient fraction of vegetation has developed.

  - The comparison in terms of uncertainty shows RMSD values between 15 and 19 days, with positive correlations of 0.27, 0.51 and 0.32 for phases 10, 11 and 223, respectively, being phase 11 the more populated in the database (N>29.500 observations).

- CLMS LSP EOSD V2 shows:
  - Negative bias of -18 days with PEP725 phase 95 (*50% of leaves fallen*) and -1 day with phase 205 (*autumnal coloration ≥ 50%*). These results seem to be consistent with the expected behaviour, considering the definition of the satellite EOSD, which corresponds to the moment when vegetation greenness decreases to 15% of its seasonal amplitude. As this threshold represents an earlier stage of senescence than leaf fall, the larger negative bias with phase 95 is reasonable. The near-zero bias with phase 205 suggests that the decrease to 15% greenness detected by satellite occurs nearly simultaneously with the onset of visible leaf coloration, reflecting a good temporal correspondence between satellite and in-situ observations at this stage.
  - In terms of uncertainty, a better agreement is found with PEP725 phase 205 (RMSD = 18 days) than with phase 95 (RMSD = 26 days), as expected.
- The compliance of CLMS LSP V2 SOSD and EOSD layers with user requirements is typically around 25% of cases for the goal and 45–50% for the threshold levels. The exception is the comparison of EOSD with PEP725 phase 95 (*50% of leaves fallen*), which shows lower compliance percentages.
- The analysis of B and RMSD across different (Annex III) years shows consistency throughout the entire period, with no significant differences between the PBV and OLCI years.

6.4.1.2 TEMPO

The results of the comparison between CLMS LSP V2 SOSD and EOSD and the TEMPO dataset for the 2014–2024 period are shown in Figure 37 and Figure 38. CLMS SOSD (Figure 37) is compared with TEMPO phases 0 and 1, corresponding to *germination/sprouting/bud development* and *leaf development/main shoot* (see Table 9). For CLMS EOSD (Figure 38), the comparison is performed with TEMPO phase 9 *beginning of dormancy/die off of leaves*. The main validation metrics are summarized in Table 23.

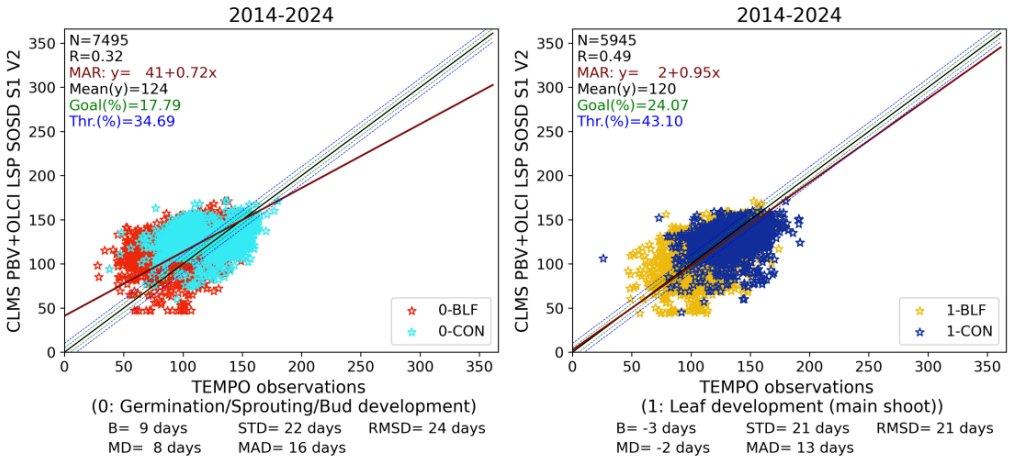

Figure 37: Scatter-plots of CLMS LSP V2 SOSD versus TEMPO phases 0 (*germination/sprouting/bud development*, left) and 1 (*leaf development/main shoo*t, right) for season 1 over the 2014–2024 period. Green and blue lines around the 1:1 line indicate the goal and threshold requirements, respectively. BLF and CON stands for broadleaved forests and coniferous.

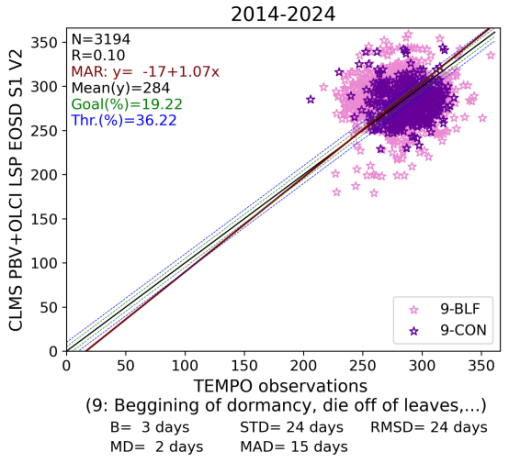

Figure 38: Scatter-plot of CLMS LSP V2 EOSD versus TEMPO phase 9 (*beginning of dormancy/die off of leaves*) for season 1 over the 2014–2024 period. Green and blue lines around the 1:1 line indicate the goal and threshold requirements, respectively. BLF and CON stands for broadleaved forests and coniferous.

Table 23: Performance statistics between CLMS LSP V2 versus TEMPO database for season 1 over the 2014–2024 period: SOSD versus TEMPO Phases 0 and 1; and EOSD versus TEMPO Phase 9.

	CLMS LSP V2 SOSD vs. TEMPO		CLMS LSP V2 EOSD vs. TEMPO
	Phase 0 germination/sprouting/ bud development	**Phase 1** leaf development/ main shoot	**Phase 9** beginning of dormancy/ die off of leaves
N	7495	5945	3194
R	0.32	0.49	0.10
Bias (days)	9	-3	3
MD (days)	8	-2	2
STD (days)	22	21	24
MAD (days)	16	13	15
RMSD (days)	24	21	24
MAR	y=41+0.72x	y=2+0.95x	y=-17-1.07x
% goal	17.8	24.1	19.2
% threshold	34.7	43.1	36.2

Main results of the comparison between CLMS LSP V2 versus TEMPO are:

- CLMS LSP V2 SOSD shows:
  - Positive bias of 9 days with TEMPO phase 0 and negative bias of -3 days with phase 1. TEMPO phase 0 typically occurs slightly earlier (germination or initial sprouting), while phase 1 represents active leaf development. Therefore, positive bias with phase 0 (germination) and small negative bias with phase 1 (leaf development) are expected for SOSD (25% threshold). These results are consistent with those obtained with PEP725, where SOSD was found to be slightly later than the earliest in-situ spring phenophases (PEP725 phenophases 10–11).
  - RMSD of 24 and 21 days are found with TEMPO phases 0 and 1, with positive correlations of 0.32 and 0.49, respectively.
- CLMS LSP V2 EOSD shows:
  - Positive low bias of +3 days relative to TEMPO phase 9 (*beginning of dormancy/die off of leaves*), which seems consistent with the CLMS definition, which sets EOSD at 15% of the seasonal amplitude. The result also agrees with the comparison against PEP725, where similar small but slightly negative bias was found compared with autumnal colouration phase.
  - An overall uncertainty (RMSD) of 24 days compared with TEMPO phase 9 and a positive correlation of 0.10.
- The compliance of CLMS LSP V2 SOSD and EOSD layers with user requirements ranges between 18–24% of cases for the goal level and 15–43% for the threshold level.

- The analysis of B and RMSD across different (Annex III) years shows similar values than for the entire 2014–2024 period, with no significant differences between the PBV and OLCI years.

6.4.1.3 NEON

The results of the comparison on CLMS LSP V2 SOSD and EOSD versus NEON dataset for the 2014–2024 period are presented in Figure 39 and Figure 40. CLMS SOSD is compared with both NEON phases *breaking* and *emerging* leaves/buds. In case of CLMS EOSD, the comparison is performed with NEON phases *colored* and *falling* leaves/buds. The main validation metrics are summarized in Table 24.

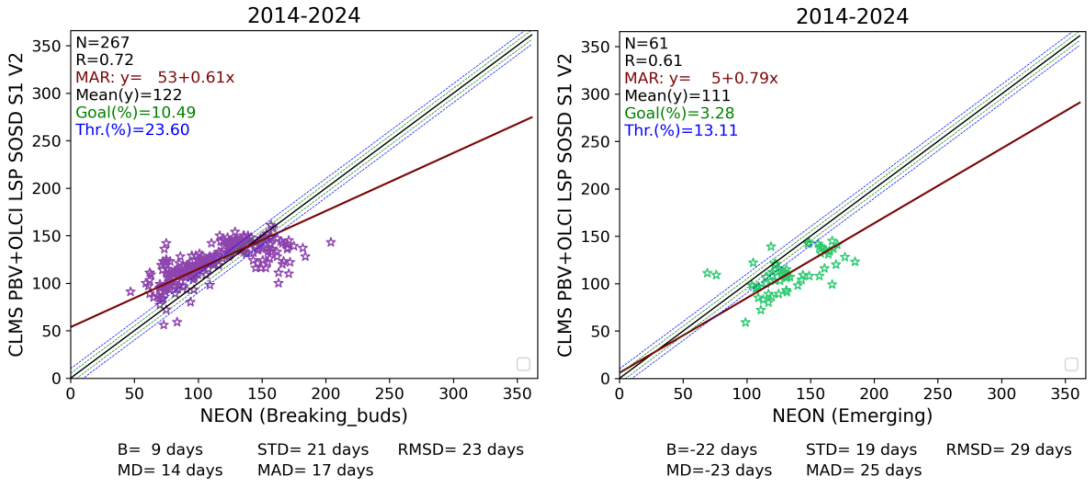

Figure 39: Scatter-plots of CLMS LSP V2 SOSD versus NEON phases *breaking* (left) and *emerging* (right) leaves/buds for season 1 over the 2014–2024 period. Green and blue lines around the 1:1 line indicate the goal and threshold requirements, respectively.

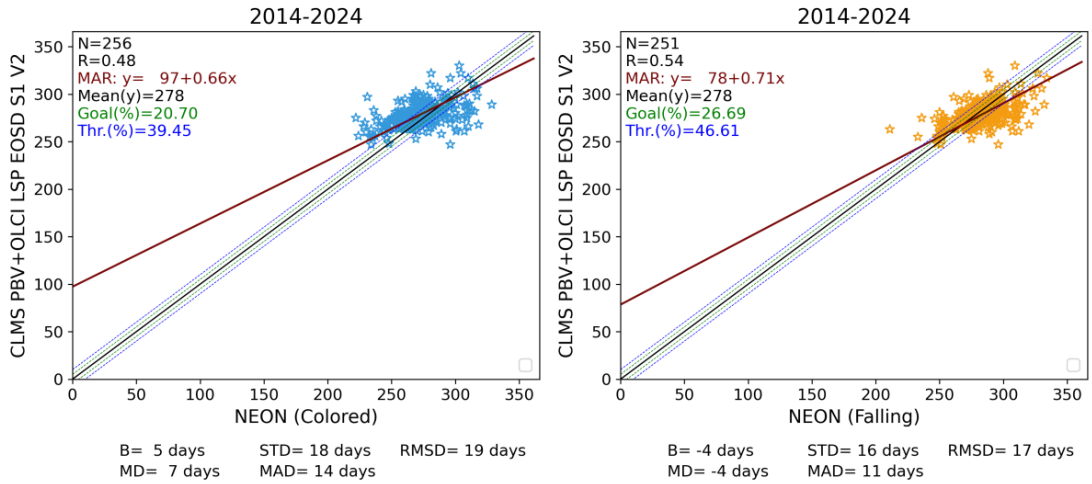

Figure 40: Scatter-plots of CLMS LSP V2 EOSD versus NEON phases *colored* (left) and *falling* (right) leaves/buds for season 1 over the 2014–2024 period. Green and blue lines around the 1:1 line indicate the goal and threshold requirements, respectively.

Table 24: Performance statistics between CLMS LSP V2 SOSD and EOSD with NEON *breaking* and *emerging*, and *colored* and *falling* leaves/buds. Comparison for season 1 over the 2014–2024 period.

	CLMS LSP V2 SOSD vs. NEON		CLMS LSP V2 EOSD vs. NEON	
	breaking leaves/buds	*emerging* leaves/buds	*colored* leaves/buds	*falling* leaves/buds
N	267	61	256	251
R	0.72	0.61	0.48	0.54
Bias (days)	9	-22	5	-4
MD (days)	14	-23	7	-4
STD (days)	21	19	18	16
MAD (days)	17	25	14	11
RMSD (days)	23	29	19	17
MAR	y=53+0.61x	y=5+0.79x	y=97+0.66x	y=78+0.71x
% goal	10.5	3.3	20.7	26.7
% threshold	23.6	13.1	39.5	46.6

Main results of the comparison between CLMS LSP V2 and NEON are:

- CLM LSP V2 SOSD shows:
  - Positive bias of +9 days with NEON *breaking* buds, indicating that the satellite SOSD (defined at 25% of maximum vegetation development) occurs slightly later than the initial bud break in-situ observations, as expected since greening is only detected once

canopy development becomes visible. This result is consistent with those obtained from PEP and TEMPO for early stages of development.
    - Negative bias of –22 days with NEON *emerging* phenophase, which in this case indicates that SOSD is anticipated to the phenological observations. It should be noted that 'emerging' observations correspond only to pine forests [NEON.DOC.014040].
    - RMSD values of 23 and 29 days are found, with remarkably positive high correlations of 0.72 and 0.61 with NEON *breaking* and *emerging* phases.
- CLMS LSP V2 EOSD shows:
    - A positive bias of +5 days with *colored* and a negative bias of -4 days with *falling*. These differences are consistent with the definitions of both the satellite and in-situ phases. Therefore, EOSD represents a date between these two NEON phenophases.
    - The slightly negative bias observed with *falling* can be explained, since the falling of leaves represents a very late phenological stage. Conversely, the positive bias with *colored* can be explained since visible leaf coloration may precede measurable greenness decline in satellite data.
    - Overall, these results are consistent with those found for PEP725 and TEMPO datasets, which also showed small biases ranging from −18 to +3 days depending on the specific phenophase definition.
    - Overall uncertainties (RMSD) of 19 and 17 days are found, with also remarkably positive correlations of around 0.5.
- The compliance of CLMS LSP V2 SOSD layers with user requirements ranges between 3–11% of cases for the goal and up to 24% for the threshold. Improved performance is observed for EOSD, with compliance values between 21–27% for the goal and 40–47% for the threshold.
- The year-to-year evolution of B and RMSD (Annex III) shows some significant differences in certain cases (e.g., RMSD versus NEON *emerging*, with RMSD values ranging between 19 and 43 days). This is mainly due to differences in sampling related to NEON data availability, with more observations acquired in recent years compared to the earliest years.

6.4.2 CLMS LSP V2 vs. near-ground-based PhenoCam V3

The results of the comparison on CLMS LSP V2 SOSD and EOSD versus PhenoCam V3 near-ground-based dataset for the 2014–2024 period are presented in Figure 41 and Figure 42. This analysis is performed with PhenoCam V3 start and end of seasons defined at 10%, 25% and 50%. The main validation metrics are summarized in Table 25.

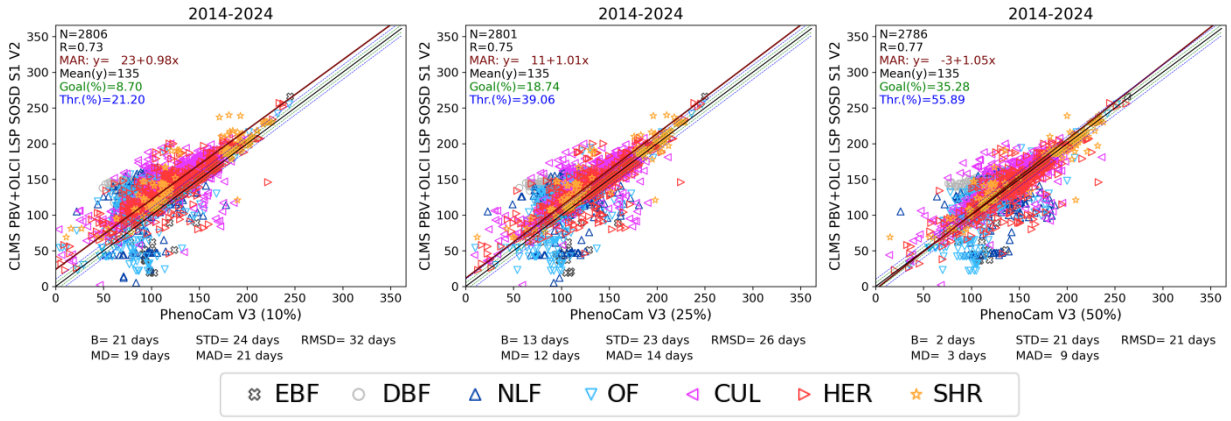

Figure 41: Scatter-plots of CLMS LSP V2 SOSD versus PhenoCam V3 start of season defined at 10% (left), 25% (middle), and 50% (right) for season 1 over the 2014–2024 period.

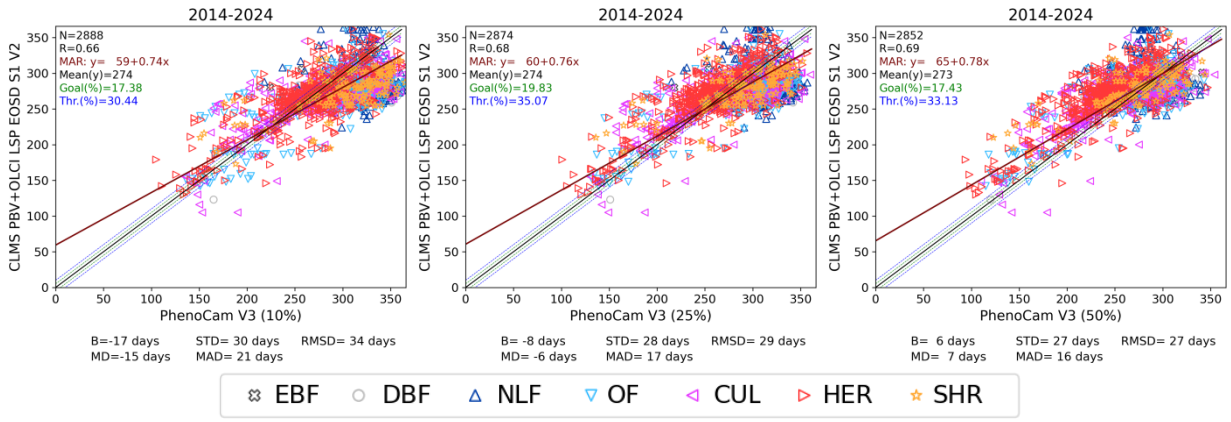

Figure 42: Scatter-plots of CLMS LSP V2 EOSD versus PhenoCam V3 end of season defined at 10% (left), 25% (middle), and 50% (right) for season 1 over the 2014–2024 period.

Main results of the comparison between CLMS LSP V2 and PhenoCam V3 are:

- CLMS LSP V2 SOSD shows the best agreement in terms of accuracy (B, MD) and uncertainty (RMSD) with PhenoCam V3 start of season defined at 50% threshold (Bias = 2 days), compared to larger delays with 25% (Bias = 14 days) and 10% (Bias = 21 days) threshold levels. Although CLMS SOSD is defined at a 25% threshold, it appears delayed with respect to PhenoCam V3 near-ground observations, which may be explained by the different definitions and spectral domains of GCC and PPI indices.
- CLMS LSP V2 EOSD (defined at 15%) falls between PhenoCam phases defined at 25% (B = -8 days) and 50% (B = 6 days) threshold levels, while larger differences are found with the theoretically closest 10% phase (B = −17 days). This indicates that CLMS EOSD is

anticipated compared with PhenoCam V3 GCC estimates. The lowest overall uncertainties are found with the 25% and 50% PhenoCam phases (RMSD = 29 and 27 days, respectively).

- The combination of positive SOSD bias and negative EOSD bias results in a shorter season in CLMS LSP V2 compared to PhenoCam V3 near-ground observations.

Table 25: Performance statistics between CLMS LSP V2 SOSD and EOSD with PhenoCam V3 equivalent phases defined at 10%, 25% and 50% for season 1 over the 2014–2024 period.

	CLMS LSP V2 SOSD vs. PhenoCamV3			CLMS LSP V2 EOSD vs. PhenoCamV3		
	10%	25%	50%	10%	25%	50%
N	2806	2801	2786	2888	2874	2852
R	0.73	0.75	0.77	0.66	0.68	0.69
Bias (days)	21	13	2	-17	-8	6
MD (days)	19	12	3	-15	-6	7
STD (days)	24	23	21	30	28	27
MAD (days)	21	14	9	21	17	16
RMSD (days)	32	26	21	34	29	27
MAR	y=23+0.98x	y=11+1.01x	y=-3+1.05x	y=59+0.74x	y=60+0.76x	y=65+0.78x
% goal	8.7	18.7	35.3	17.4	19.8	17.4
% threshold	21.2	39.1	55.9	30.4	35.1	33.1

6.5 ERROR EVALUATION WITH SATELLITE-BASED REFERENCES

6.5.1 CLMS LSP V2 vs. HR-VPP over Europe

The overall comparison between CLMS LSP V2 SOSD and EOSD versus HR-VPP equivalent layers over the European sampling EVAL is presented in this section (Figure 43, Table 26). Main results are:

- Both products show a strong correspondence, with high correlations (R ≥ 0.85) and RMSD values of 19 and 24 days for SOSD and EOSD, respectively.
- CLMS LSP V2 shows a positive bias (i.e., a slight delay) for both SOSD (B = 12 days, MD = 11 days) and EOSD (B = 9 days, MD = 5 days) compared with HR-VPP. The somewhat larger delay in SOSD results in a slightly shorter season in CLMS LSP V2, estimated at around 3 days in mean or 6 days in median.
- Conformity testing shows that the threshold level is met in 41% of cases for SOSD and 45% for EOSD (19% and 25% considering goal level).

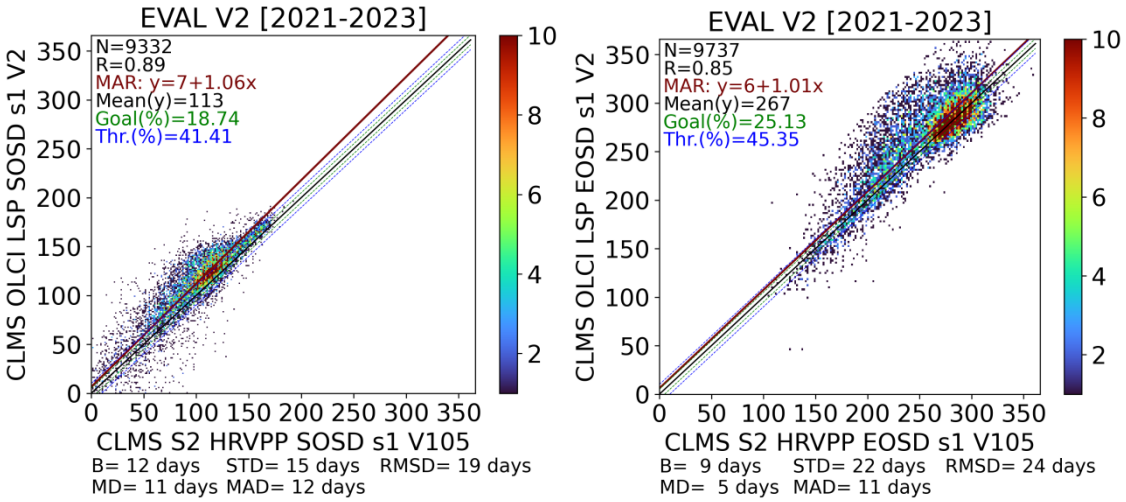

Figure 43: Scatter-plots of CLMS LSP V2 SOSD and EOSD versus HR-VPP equivalent layers over EVAL sites for season 1 during the 2021–2023 period.

Table 26: Performance statistics between CLMS LSP V2 SOSD and EOSD versus HR-VPP equivalent layers. Computation over EVAL sites for season 1 during the 2021–2023 period.

	CLMS LSP V2 vs. HR-VPP	
	SOSD (days)	EOSD (days)
N	9332	9737
R	0.89	0.85
Bias (days)	12	9
MD (days)	11	5
STD (days)	15	22
MAD (days)	12	11
RMSD (days)	19	24
MAR	y=7+1.06x	y=6+1.01x
% goal	18.7	25.1
% threshold	41.4	45.4

The overall comparison between CLMS LSP V2 TPROD and SPROD and HR-VPP equivalent layers over the European sampling EVAL is presented in Figure 44, with main statistics summarized in Table 27. Main results are:

- Both products show strong agreement, with high correlations (R > 0.90).

- CLMS LSP V2 TPROD and SPROD are systematically higher than HR-VPP by around 23% in mean or 21% in median, despite CLMS LSP V2 having a slightly shorter growing season compared with HR-VPP.

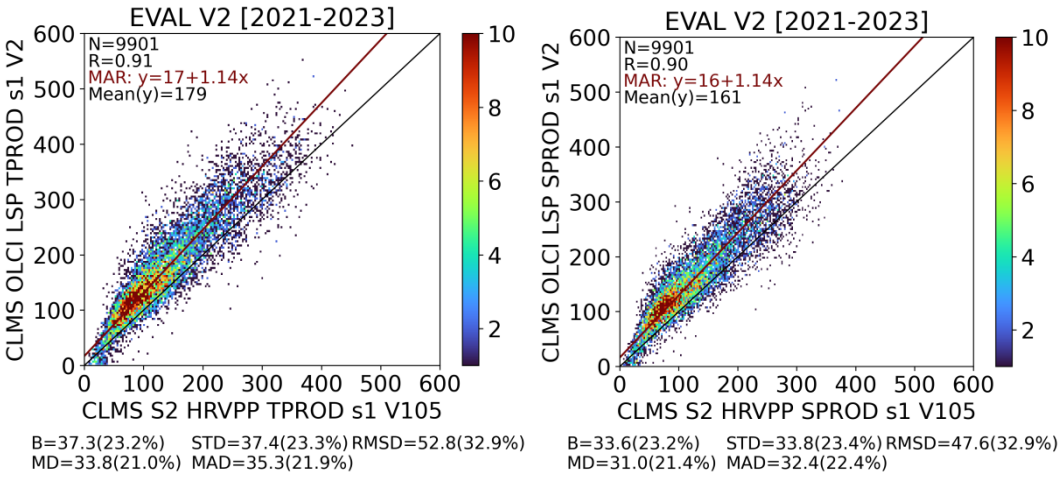

Figure 44: Scatter-plots of CLMS LSP V2 TPROD and SPROD versus HR-VPP equivalent layers over EVAL sites for season 1 during the 2021–2023 period.

Table 27: Performance statistics between CLMS LSP V2 TPROD and SPROD versus HR-VPP equivalent layers. Computation over EVAL sites for season 1 during the 2021–2023 period.

	CLMS LSP V2 vs. HR-VPP	
	TPROD ($m^2 \cdot m^{-2} \cdot day$)	SPROD ($m^2 \cdot m^{-2} \cdot day$)
N	9901	9901
R	0.91	0.90
Bias (m^2m^{-2})	37.3 (23.2%)	33.6 (23.2%)
MD (m^2m^{-2})	33.8 (21.0%)	31.0 (21.9%)
STD (m^2m^{-2})	37.4 (23.3%)	33.8 (23.4%)
MAD (m^2m^{-2})	35.3 (21.9%)	32.4 (22.4%)
RMSD (m^2m^{-2})	52.8 (32.9%)	47.6 (32.9%)
MAR	y=17+1.14x	y=16+1.14x

6.5.2 CLMS LSP V2 vs. global products

6.5.2.1 SOSD and EOSD

This section presents the analysis of CLMS LSP V2 SOSD and EOSD phenometrics with the MODIS V6.1 and VIIRS V2 LCD products. The analysis focuses on season 1, as only a limited number of high-quality pixels were available for season 2 in both reference datasets. Scatter-plots and validation metrics were computed for best quality pixels over LANDVAL V2 sites at 1 km^2 resolution, considering both the overall case and the aggregation by major biome types. It should be noted that the SBA class is not included, since fewer than ten concomitant pixels were available between product pairs, as LSP products do not provide estimates over desert regions.

- **CLMS LSP V2 vs. MODIS LCD V6.1**

The scatter-plots between CLMS OLCI LSP and MODIS V6.1 for SOSD (Figure 45) and EOSD (Figure 46) are presented for the 2014-2024 period in this section. The corresponding performance statistics are summarized in Table 28. The three MODIS phenometrics *Greenup*, *MidGreenup* and *Maturity* are intercompared with SOSD, while *Dormancy*, *MidGreendown* and *Senescence* are compared with EOSD. MODIS phenophases are defined at the corresponding 15%, 50% and 90% of the EVI2 amplitude, respectively. To assess the consistency of CLMS LSP V2 with MODIS V6.1 when using different CLMS input datasets, Annex IV presents results for two separate periods: 2014–2018 (PROBA-V) and 2019–2024 (Sentinel-3/OLCI).

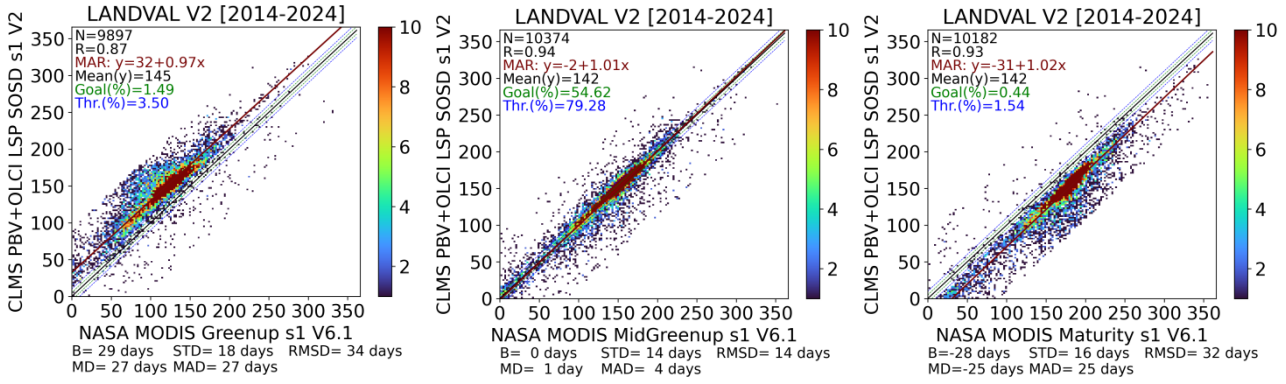

Figure 45: Scatter-plots of CLMS LSP V2 SOSD versus MODIS V6.1 *Greenup* (left), *MidGreenup* (middle), and *Maturity* (right) over LANDVAL V2 sites for the 2014–2024 period (season 1).

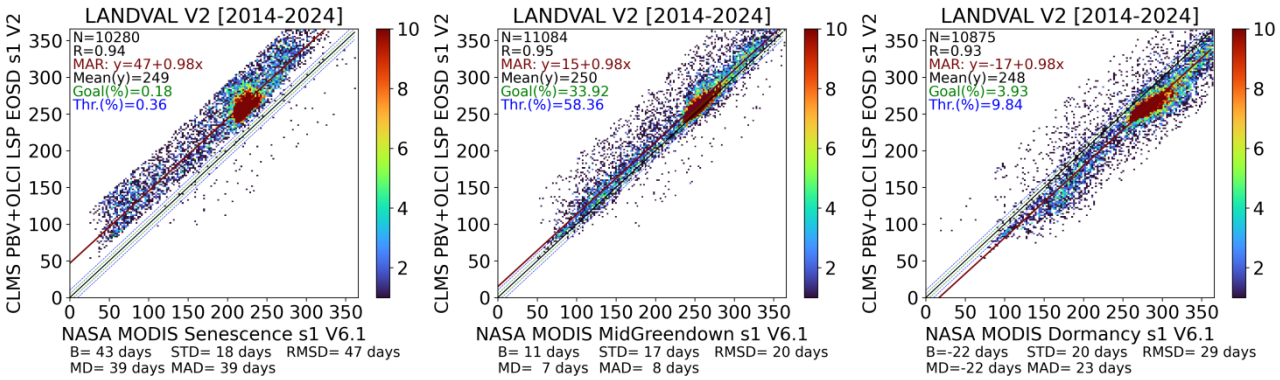

Figure 46: Scatter-plots of CLMS LSP V2 EOSD versus MODIS V6.1 *Senescence* (left), *MidGreendown* (middle), and *Dormancy* (right) over LANDVAL V2 sites for the 2014–2024 period (season 1).

Table 28: Performance statistics between CLMS LSP V2 SOSD and EOSD with MODIS V6.1 *Greenup*/*MidGreenup*/*Maturity* and *Dormancy*/*MidGreendown*/*Senescence* phenometrics. Computation over LANDVAL V2 sites for the 2014–2024 period (season 1).

	CLMS OLCI LSP SOSD (25%) vs. MODIS V6.1			CLMS OLCI LSP EOSD (15%) vs. MODIS V6.1		
	Greenup (15%)	MidGreenup (50%)	Maturity (90%)	Senescence (90%)	MidGreendown (50%)	Dormancy (15%)
N	9897	10374	10182	10280	11084	10875
R	0.87	0.94	0.93	0.94	0.95	0.93
Bias	29	0	-28	43	11	-22
MD	27	1	-25	39	7	-22
STD	18	14	16	18	17	20
MAD	27	4	25	39	8	23
RMSD	34	14	32	47	20	29
MAR	y=32-0.87x	y=-2+1.01x	y=-31+1.02x	y=47+0.98x	y=15+0.98x	y=-17+0.98x
% goal	1.5	54.6	0.4	0.2	33.9	3.9
% thr.	3.5	79.3	1.6	0.4	58.4	9.9

Main results of the comparison of CLMS LSP V2 versus MODIS LCD V6.1 (2014-2024) are:

- Start of season (SOSD, defined for 25% of the PPI seasonal amplitude):
  - The best correspondence is obtained with the MODIS V6.1 *MidGreenup* phenometric (RMSD = 14 days, R=0.94, 79.3% of cases within the threshold requirement), even

though a stronger agreement might be expected with *Greenup* (RMSD = 34 days, R=87, 3.5% of cases within threshold) based on the definition of their threshold levels.
    - CLMS SOSD shows no bias with respect to MODIS *MidGreenup* (defined at 50%), while positive and negative biases of +29 days and -28 days are observed when compared with *Greenup* (15%) and *Maturity* (90%), respectively.
    - Almost identical results are obtained in the comparison with MODIS when using different CLMS input datasets (PBV or OLCI, see Annex IV).
- End of season (EOSD, defined at 15% of the PPI seasonal amplitude):
    - The best overall agreement is found with MODIS *MidGreendown* (RMSD = 20 days, R=0.95, 58.4% of cases within threshold requirement).
    - CLMS EOSD shows a slight delay compared to MODIS *MidGreendown* (B = +11 days) and an earlier timing compared to MODIS *Dormancy* (B = -22 days), meaning CLMS EOSD typically falls between both MODIS phenophase dates.
    - As for SOSD, similar results are obtained when using different CLMS input datasets (PBV or OLCI, see Annex IV).

The analysis of CLMS LSP V2 SOSD versus MODIS V6.1 by biome type is presented in Figure 47, using MODIS *MidGreenup* phase, which showed the best overall agreement with CLMS SOSD. Table 29 reports the corresponding performance statistics for both MODIS V6.1 *MidGreenup* and *Greenup,* representing respectively the phenophase with the highest statistical agreement and the one most consistent with the CLMS product definition.

Main results of the comparison between CLMS LSP V2 SOSD and MODIS V6.1 are:

- The comparison of CLMS SOSD with MODIS *MidGreenup* shows very good agreement (R > 0.90 for most cases), with B and MD within 0–3 days for all biomes except for EBF (B = -15 days, MD = -10 days). The percentage of cases within the goal requirement is typically higher than 50%, except for EBF (21.9%) and CUL (40%).
- A systematic positive bias of CLMS SOSD relative to MODIS *Greenup* (Table 29, left side) is observed for most biomes, with B and MD ranging between 14 and 32 days. The proportion of cases within the goal uncertainty requirement remains below 2% for all biomes, except for EBF (10.7%).

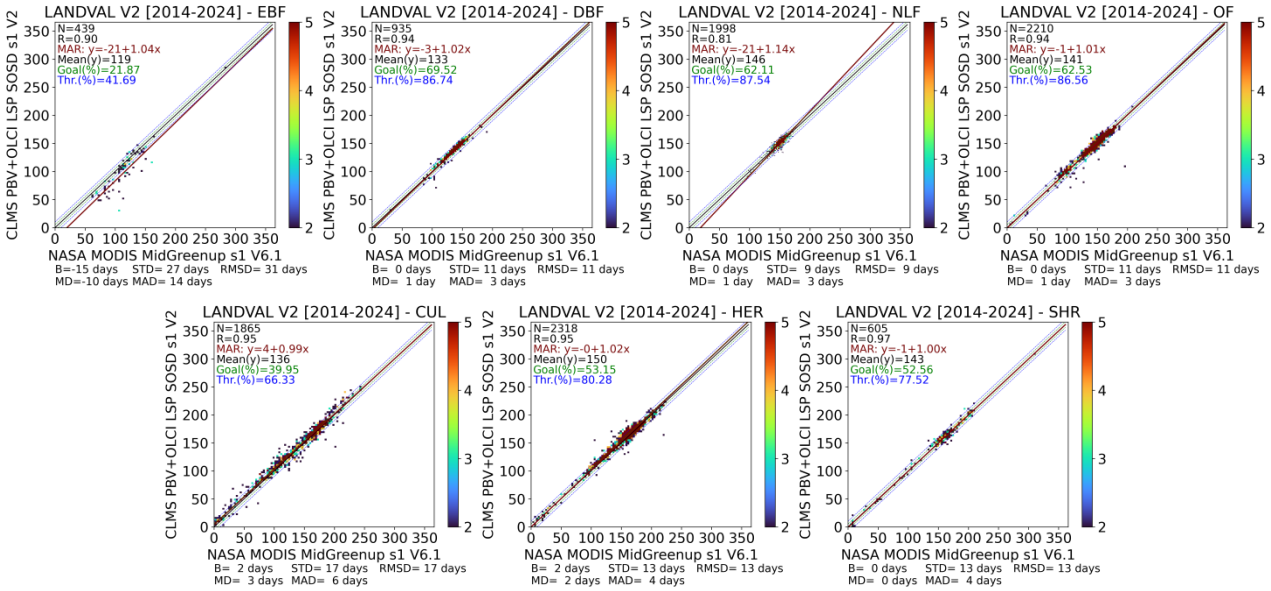

Figure 47: Scatter-plots of CLMS LSP V2 SOSD versus MODIS V6.1 *MidGreenup* per aggregated biome type. Computation over LANDVAL V2 sites for the 2014–2024 period (season 1).

Table 29: Performance statistics between CLMS LSP V2 SOSD with MODIS V6.1 *Greenup* and *MidGreenup* phenometrics. Computation over LANDVAL V2 sites for the 2014–2024 period (season1).

	CLMS LSP V2 SOSD (25%) vs.													
	MODIS V6.1 *Greenup* (15%)							MODIS V6.1 *MidGreenup* (50%)						
	EBF	DBF	NLF	OF	CUL	HER	SHR	EBF	DBF	NLF	OF	CUL	HER	SHR
N	441	923	1981	2137	1711	2163	537	439	935	1998	2210	1865	2318	605
R	0.89	0.95	0.66	0.85	0.92	0.84	0.90	0.90	0.94	0.81	0.94	0.95	0.95	0.97
Bias	14	25	29	29	30	32	31	-15	0	0	0	2	2	0
MD	19	25	27	27	29	28	28	-10	1	1	1	3	2	0
STD	29	9	15	15	19	20	21	27	11	9	11	17	13	13
MAD	23	25	27	27	30	29	28	14	3	3	3	6	4	4
RMSD	32	27	33	33	36	38	38	31	11	9	11	17	13	13
MAR	y=17 +0.98x	y=30 +0.96x	y=53 +0.80x	y=39 +0.91x	y=33 +0.98x	y=38 +0.96x	y=31 +1.01x	y=-21 +1.04x	y=-3 +1.02x	y=-21 +1.14x	y=-1 +1.01x	y=4 +0.99x	y=0 +1.02x	y=-1 +1.00x
% goal	10.7	0.5	0.8	0.8	1.8	1.1	1.5	21.9	69.5	62.1	62.5	40.0	53.2	52.6
%threshold	18.1	2.7	2.8	2.0	3.4	3.1	3.4	41.7	86.7	87.5	86.6	66.3	80.3	77.5

The analysis of CLMS LSP V2 EOSD versus MODIS V6.1 per biome type is presented in Figure 48 using MODIS *MidGreendown* phenophase, which provides the best agreement with CLMS EOSD.

Table 30 summaryzes the performance statistics considering both MODIS V6.1 *MidGreendown* and *Dormancy* phenophases, the latter being more consistent with the CLMS definition.

Main results of the comparison between CLMS LSP V2 EOSD and MODIS V6.1 are:

- As observed for the overall comparison, CLMS EOSD shows better agreement with MODIS *MidGreendown* than with MODIS *Dormancy*, regardless of the biome type (except for EBF).
- CLMS EOSD typically exhibits slight positive differences with MODIS *MidGreendown* (B < 14 days), except for EBF, where a larger positive bias of 22 days is found. The percentage of cases within the goal requirement generally exceeds 30%, except for EBF (11.1%) and CUL (26.4%).
- Compared to MODIS *Dormancy*, CLMS EOSD shows a systematic negative bias across all biome types, with B and MD values typically ranging between 15 and 25 days earlier. The percentage of cases within goal requirements remains below 5% for all biomes, except EBF (8.4%).

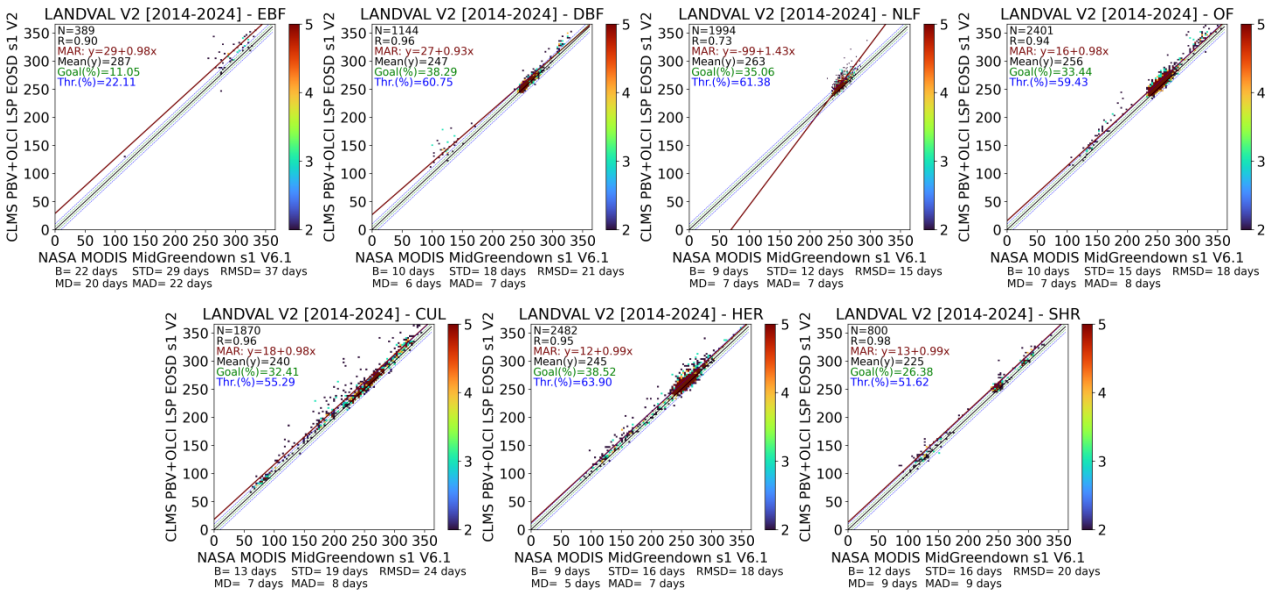

Figure 48: Scatter-plots of CLMS LSP V2 EOSD versus MODIS V6.1 *MidGreendown* per aggregated biome type. Computation over LANDVAL V2 sites for the 2014–2024 period (season 1).

Table 30: Performance statistics between CLMS LSP V2 EOSD with MODIS V6.1 *Dormancy* and *MidGreendown* phenometrics. Computation over LANDVAL V2 sites for the 2014–2024 period (season 1).

	CLMS LSP V2 EOSD (15%) vs.													
	MODIS V6.1 *Dormancy* (15%)							MODIS V6.1 *MidGreendown* (50%)						
	EBF	DBF	NLF	OF	CUL	HER	SHR	EBF	DBF	NLF	OF	CUL	HER	SHR
N	357	1116	1980	2327	1825	2479	787	389	1144	1994	2401	1870	2482	800
R	0.90	0.96	0.73	0.93	0.93	0.91	0.96	0.90	0.96	0.73	0.94	0.96	0.95	0.98
Bias	-12	-25	-21	-22	-17	-23	-25	22	10	9	10	13	9	12
MD	-17	-27	-20	-21	-20	-23	-26	20	6	7	7	7	5	9
STD	32	16	13	15	25	21	21	29	18	12	15	19	16	16
MAD	26	27	20	22	23	24	27	22	7	7	8	8	7	9
RMSD	34	30	25	27	30	31	33	37	21	15	18	24	18	20
MAR	y=11 +0.92x	y=-28 +1.01x	y=7 +0.90x	y=-18 +0.98x	y=2 +0.92x	y=-23 +1.00x	y=-42 +1.06x	y=29 +0.98x	y=27 +0.93x	y=-99 +0.1.43x	y=16 +0.98x	y=18 +0.98x	y=12 +0.99x	y=13 +0.99x
% goal	8.4	3.6	4.3	3.0	4.1	4.4	2.4	11.1	38.3	35.1	33.4	32.4	38.5	26.4
% threshold	17.4	6.8	10.4	8.4	10.2	11.5	7.6	22.1	60.8	61.4	59.4	55.3	63.9	51.6

- **CLMS LSP V2 vs. VIIRS LCD V2**

The scatter-plots between CLMS LSP V2 SOSD (Figure 49) and EOSD (Figure 50) and the VIIRS LCD V2 phenophases are presented in this section. The main performance statistical metrics are summarized in Table 31. The three VIIRS V2 *Onset_Greenness_Increase*, *Date_Mid_Greenup_Phase* and *Onset_Greenness_Maximum* phenometrics are intercompared with SOSD, while *Onset_Greeness_Decrease*, *Date_Mid_Senescence_Phase* and *Onset_Greeness_Minimum* are compared with EOSD. To assess the consistency of CLMS LSP V2 with VIIRS V2 when using different CLMS input datasets, Annex V presents results for two distinct periods: 2014–2018 (PROBA-V) and 2019–2023 (Sentinel-3/OLCI).

The comparison of CLMS LSP V2 and VIIRS V2 during 2014–2023 shows:

- Start of season (SOSD, defined for 25% of the PPI seasonal amplitude):
  - The best correspondence was found with VIIRS *Mid_Greenup_Phase* (RMSD = 18 days, 69.7% of cases within threshold requirement) despite of the fact that the best agreement could be expected with *Onset_Greeness_Increase* (RMSD = 30 days, 23.9% of cases within threshold) according to its definition.
  - CLMS SOSD shows low negative bias (B = -4 days, MD = -2 days) with VIIRS *Mid_Greenup_Phase*, whereas positive bias (B = 15 days, MD = 16 days) is found

with *Onset_Greeness_Increase*. Large negative bias (B = -24 days) is found with *Onset_Greenness_Maximum,* as expected.

- End of season (EOSD, defined at 15% of the PPI seasonal amplitude):
  - CLMS falls between VIIRS *Date_Mid_Senescence_Phase* (B = 15 days, MD = 11 days) and *Onset_Greeness_Minimum* (B = -11 days, MD = -13 days).
  - Slightly better overall agreement is found in with the *Date_Mid_Senescence_Phase* (R = 0.83, RMSD = 31 days) compared to *Onset_Greeness_Minimum* (R = 0.77, RMSD = 35 days). Worse uncertainty is found with VIIRS *Onset_Greeness_Decrease* (RMSD = 49 days), as expected.

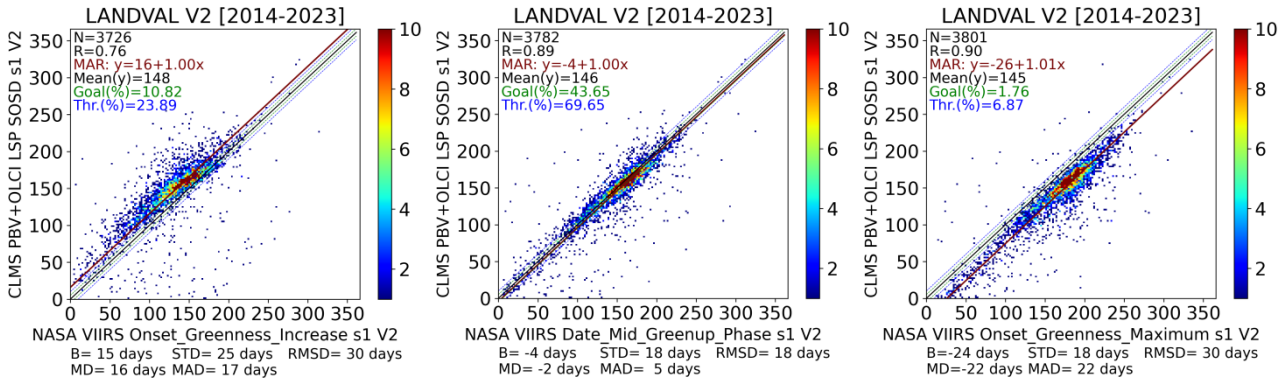

Figure 49: Scatter-plots of CLMS LSP V2 SOSD versus VIIRS V2 *Onset_Greenness_Increase* (left), *Date_Mid_Greenup_Phase* (middle), and *Onset_Greenness_Maximum* (right) per aggregated biome type. Computation over LANDVAL V2 sites during 2014–2023 (season 1).

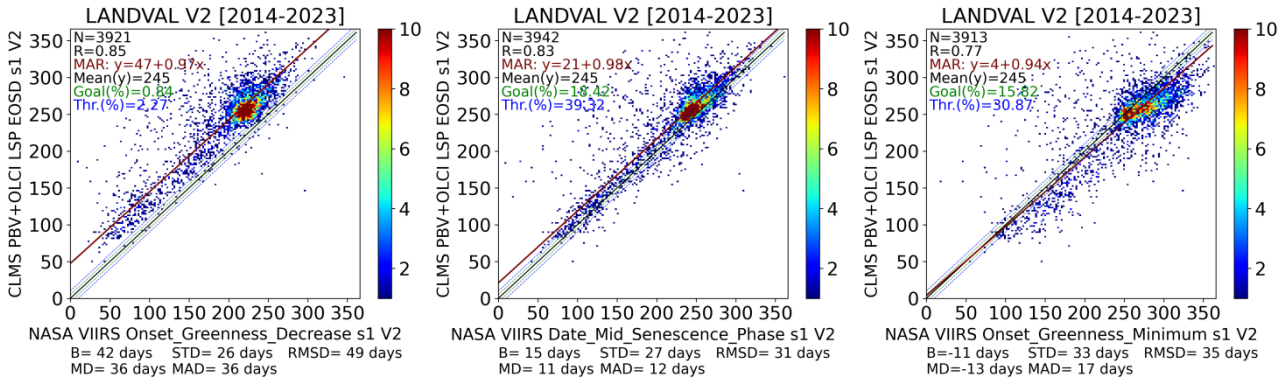

Figure 50: Scatter-plots of CLMS LSP V2 EOSD versus VIIRS V2 *Onset_Greeness_Decrease* (left), *Date_Mid_Senescence_Phase* (middle), and *Onset_Greeness_Minimum* (right) per aggregated biome type. Computation over LANDVAL V2 sites during 2014–2023 (season 1).

Table 31: Performance statistics between CLMS LSP V2 SOSD and EOSD with VIIRS V2 *Onset_Greenness_Increase*/*Date_Mid_Greenup_Phase*/*Onset_Greenness_Maximum* and *Onset_Greeness_Decrease*/*Date_Mid_Senescence_Phase*/*Onset_Greeness_Minimum* phenometrics. Computation over LANDVAL V2 sites during 2014-2023 (season 1).

	CLMS LSP V2 SOSD (25%) vs. VIIRS V2			CLMS LSP V2 EOSD (15%) vs. VIIRS V2		
	Onset_ Greenness_ Increase	*Date_Mid_ Greenup _Phase*	*Onset_ Greenness_ Maximum*	*Onset_ Greeness_ Decrease*	*Date_Mid_ Senescence_ Phase*	*Onset_ Greeness_ Minimum*
N	3726	3782	3801	3921	3942	3913
R	0.76	0.89	0.90	0.85	0.83	0.77
Bias	15	-4	-24	42	15	-11
MD	16	-2	-22	36	11	-13
STD	25	18	18	26	27	33
MAD	17	5	22	36	12	17
RMSD	30	18	30	49	31	35
MAR	y=16-1.00x	y=-4+1.00x	y=-26+1.01x	y=47+0.97x	y=21+0.98x	y=4+0.94x
% goal	10.8	43.7	1.8	0.8	18.4	15.8
% thr.	23.9	69.7	6.9	2.3	39.3	30.9

The analysis of CLMS LSP V2 SOSD versus VIIRS V2 by biome type is presented in Figure 51, using *Date_Mid_Greenup_Phase* VIIRS V2 phenometric, which provides the best overall agreement with CLMS. Table 32 summarizes the performance statistics for both *Date_Mid_Greenup_Phase* and *Onset_Greeness_Increase* VIIRS V2, the latter being the phenophase that best matches CLMS in terms of threshold definition. The SBA class is not displayed due to the limited number of concomitant samples (<10) between both products, while the EBF class is omitted because VIIRS V2 does not provide valid best-quality retrievals over these areas.

Main conclusions of the comparison of CLMS LSP V2 SOSD versus VIIRS V2 by biome type are:

- The comparison of CLMS SOSD with VIIRS *Date_Mid_Greenup_Phase* shows a slightly earlier start of season (B and MD within -9 and -1 days) except for cultivated (B and MD = 0 days). The percentage of cases within the threshold requirement exceeds 70% for cultivated and forest biomes, and is around 60% for herbaceous and shrubland areas.
- A systematic positive bias of CLMS SOSD relative to VIIRS *Onset_Greenness_Increase* (Table 32, left side) is observed across all biomes, with B and MD ranging between 9 and 21 days.

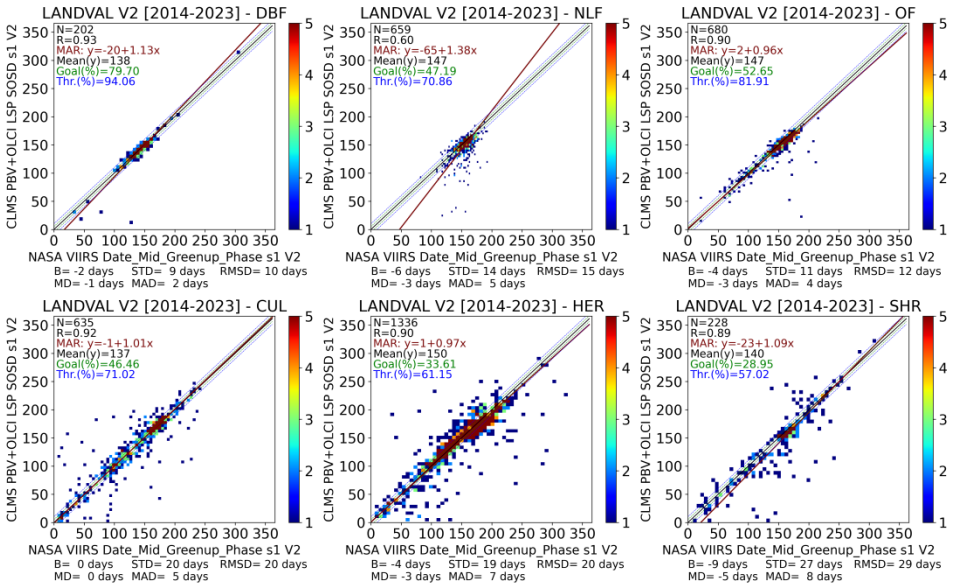

Figure 51: Scatter-plots of CLMS LSP V2 SOSD versus VIIRS V2 *Date_Mid_Greenup_Phase* per aggregated biome type. Computation over LANDVAL V2 sites during 2014–2023 (season 1).

Table 32: Performance statistics between CLMS LSP V2 SOSD with VIIRS V2 *Onset_Greenness_Increase* and *Date_Mid_Greenup_Phase*. Computation over LANDVAL V2 sites during 2014–2023 (season 1).

	CLMS LSP V2 SOSD vs. VIIRS V2											
	Onset_Greenness_Increase						*Date_Mid_Greenup_Phase*					
	DBF	NLF	OF	CUL	HER	SHR	DBF	NLF	OF	CUL	HER	SHR
N	201	659	678	608	1320	218	202	659	680	635	1336	228
R	0.82	0.64	0.82	0.73	0.78	0.73	0.93	0.60	0.90	0.92	0.90	0.89
Bias	14	15	14	19	15	9	-2	-6	-4	0	-4	-9
MD	15	16	14	21	14	14	-1	-3	-3	0	-3	-5
STD	15	14	16	32	28	39	9	14	11	20	19	27
MAD	15	16	15	23	16	16	2	5	4	5	7	8
RMSD	21	21	22	37	32	40	10	15	12	20	20	29
MAR	y=9 +01.05x	y=9 +1.05x	y=38 +0.82x	y=6 +1.11x	y=22 +0.95x	y=-8 +1.13x	y=-20 +1.13x	y=-65 +1.38x	y=2 +0.96x	y=-1 +1.01x	y=1 +0.97x	y=-23 +1.09x
% goal	4.0	8.5	9.7	2.8	17.6	8.3	79.7	47.2	52.7	46.5	33.6	29.0
% threshold	16.4	21.1	27.1	8.7	31.8	24.3	94.1	70.9	81.9	71.0	61.2	57.0

The analysis of CLMS LSP V2 EOSD versus VIIRS V2 per biome type is presented in Figure 52, using VIIRS V2 *Date_Mid_Senescence_Phase*, which provides the best overall agreement with CLMS EOSD. Table 33 presents the performance statistics considering both VIIRS V2 *Date_Mid_Senescence_Phase* and *Onset_Greeness_Minimum* phases. Note that the EBF class is excluded due to the absence of valid *best-quality* VIIRS V2 retrievals.

Main conclusions of the comparison of CLMS LSP V2 EOSD versus VIIRS V2 by biome type are:

- As observed for the overall case, CLMS EOSD generally falls between VIIRS V2 *Date_Mid_Senescence_Phase* (B and MD from 8 to 22 days) and *Onset_Greeness_Minimum* (B and MD ranging from -26 to -5 days) across all biome types.
- Similar performance statistics (e.g., R, RMSD, MAD) are observed when comparing CLMS EOSD with both VIIRS V2 phenophases, although with opposite signs in the differences. However, the comparison with *Onset_Greeness_Minimum* shows notably poorer results for the DBF biome.

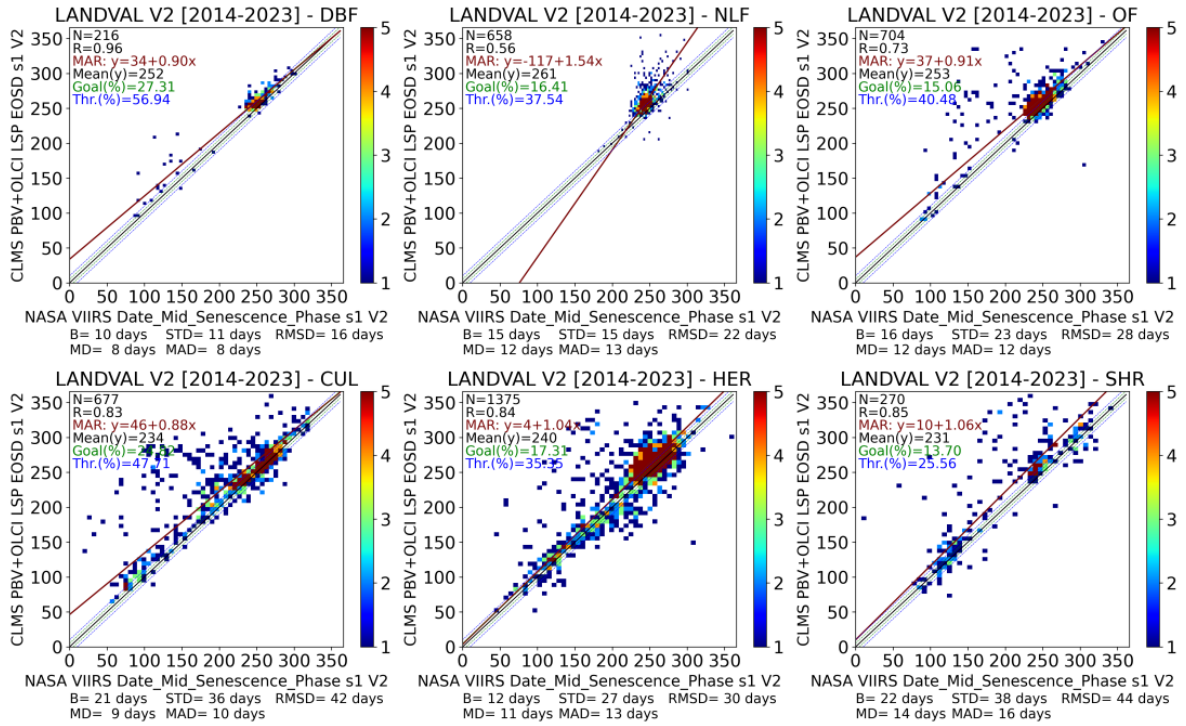

Figure 52: Scatter-plots of CLMS LSP V2 EOSD versus VIIRS V2 Date_Mid_Senescence_Phase per aggregated biome type. Computation over LANDVAL V2 sites during 2014–2023 (season 1).

Table 33: Performance statistics between CLMS LSP V2 EOSD with VIIRS V2 *Date_Mid_Senescence_Phase* (left) and *Onset_Greeness_Minimum* (right). Computation over LANDVAL V2 sites during 2014–2023 (season 1).

	CLMS LSP V2 EOSD vs. VIIRS V2											
	Date_Mid_Senescence_Phase						*Onset_Greeness_Minimum*					
	DBF	NLF	OF	CUL	HER	SHR	DBF	NLF	OF	CUL	HER	SHR
N	216	658	704	677	1375	270	216	657	703	669	1372	251
R	0.96	0.56	0.73	0.83	0.84	0.85	0.94	0.55	0.73	0.79	0.74	0.81
Bias	10	15	16	21	12	22	-26	-10	-9	-5	-14	-8
MD	8	12	12	9	11	14	-26	-11	-9	-16	-14	-12
STD	11	15	23	36	27	38	14	17	24	42	36	42
MAD	8	13	12	10	13	16	26	13	11	23	20	21
RMSD	16	22	28	42	30	44	30	20	25	43	39	43
MAR	y=34 +0.90x	y=-117 +1.54x	y=37 +0.91x	y=46 +0.88x	y=4 +1.04x	y=10 +1.06x	y=8 +0.88x	y=-30 +1.07x	y=29 +0.86x	y=38 +0.82x	y=-18 +1.01x	y=-40 +1.13x
% goal	27.3	16.4	15.1	24.8	17.3	13.7	1.9	17.8	24.6	8.1	17.0	13.6
% threshold	56.9	37.5	40.5	47.7	35.4	25.6	3.7	37.3	45.1	19.9	30.9	27.9

6.5.2.2 Productivity

- CLMS LSP V2 vs. CLMS GPP

The comparison between CLMS LSP V2 SPROD/TPROD and CLMS GPP is performed to assess the consistency of vegetation productivity estimates. Since the sum of CLMS 10-daily GPP is expressed in $kgC \cdot m^{-2}$ and CLMS SPROD/TPROD in $m^2 \cdot m^{-2} \cdot day$, only correlation and regression statistics are reported. SPROD and TPROD are computed as the sum of daily PPI values between SOSD and EOSD, and the corresponding CLMS GPP was aggregated over the same period for comparability. The analysis was conducted at 1 km resolution (aggregating 3×3 CLMS PBV/OLCI 300 m pixels) for the 2014–2024 period (first seasons).

The scatter-plots of CLMS LSP V2 SPROD and TPROD versus the aggregated values of CLMS GPP values are displayed in Figure 53, with main performance statistics summarized in Table 34. The key finding is:

- CLMS LSP V2 productivity parameters shows strong correlations with temporally aggregated CLMS GPP (R = 0.74 for SPROD and R = 0.84 for TPROD), indicating a robust relationship between PPI-based and carbon flux-based productivity estimates. CLMS GPP is more related,

as expected, with TPROD because it corresponds to the seasonal integral of the PPI signal including the base (background) level, whereas SPROD measures only the seasonal excess above that base.

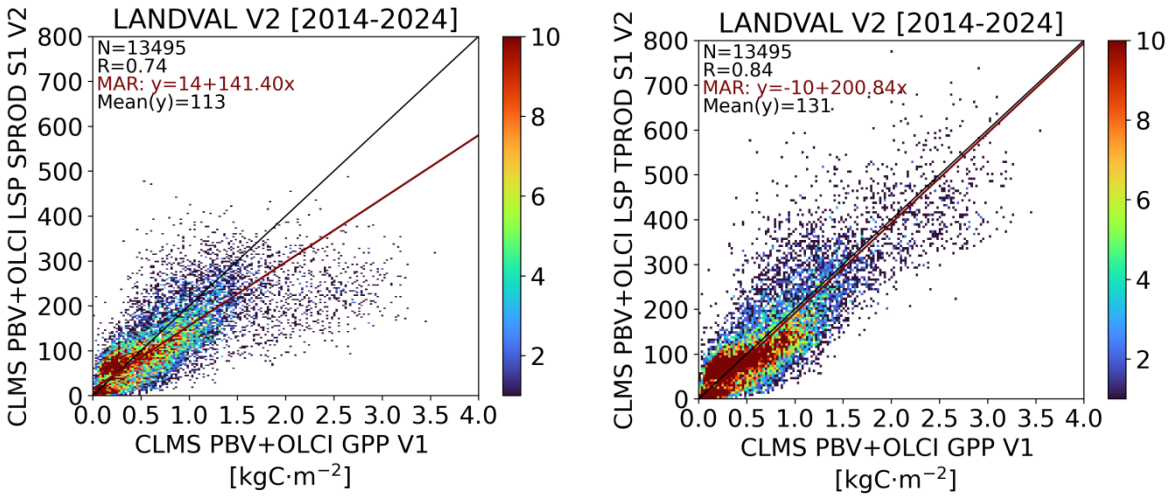

Figure 53: Scatter-plots of CLMS LSP V2 SPROD (left) and TPROD (right) ($m^2 \cdot m^{-2} \cdot day$) versus the sum of CLMS GPP values ($kgC \cdot m^{-2}$) computed over the same season length. Computation over LANDVAL V2 sites during 2014–2024 (season 1).

Table 34: Performance statistics between CLMS LSP V2 SPROD and TPROD ($m^2 \cdot m^{-2} \cdot day$) versus CLMS GPP values ($kgC \cdot m^{-2}$) for the same season length. Computation over LANDVAL V2 sites during 2014-2024 (season 1).

	CLMS LSP V2 SPROD/TPROD vs. CLMS GPP (between SOSD and EOSD)	
	SPROD vs. GPP	TPROD vs. GPP
N	13495	13495
R	0.74	0.84
MAR	y=14+141x	y=-10+201x

6.5.3 Inter-annual precision

To assess the inter-annual precision of the products, median absolute deviations (MAD) of SOSD and EOSD between consecutive years were computed over LANDVAL V2 sites, excluding croplands. Figure 54 and Figure 55 display the inter-annual precision indicators for CLMS LSP V2 SOSD and

EOSD, respectively, over the 2014–2024 period. For benchmarking, equivalent results from NASA MODIS V6.1 and VIIRS V2 phenometrics are also shown.

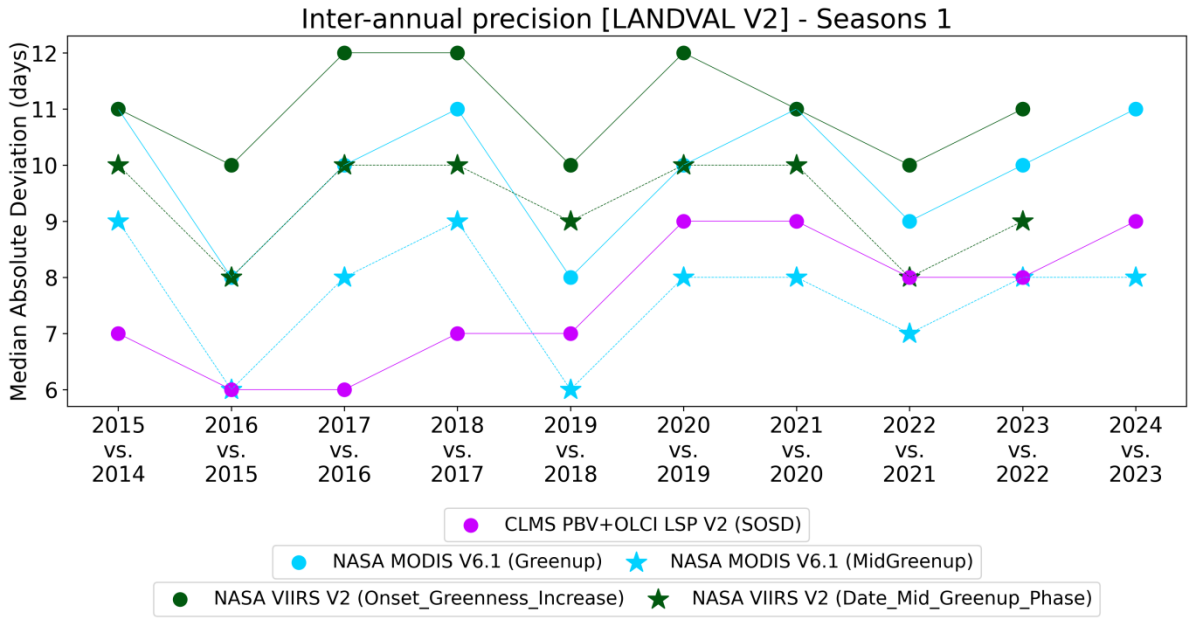

Figure 54: Median absolute deviation (MAD) of consecutive years for the 2014–2024 period over LANDVAL sites for CLMS LSP V2 (pink), MODIS V6.1 (blue), and VIIRS V2 (green) phenophases related to the start of the season (SOSD).

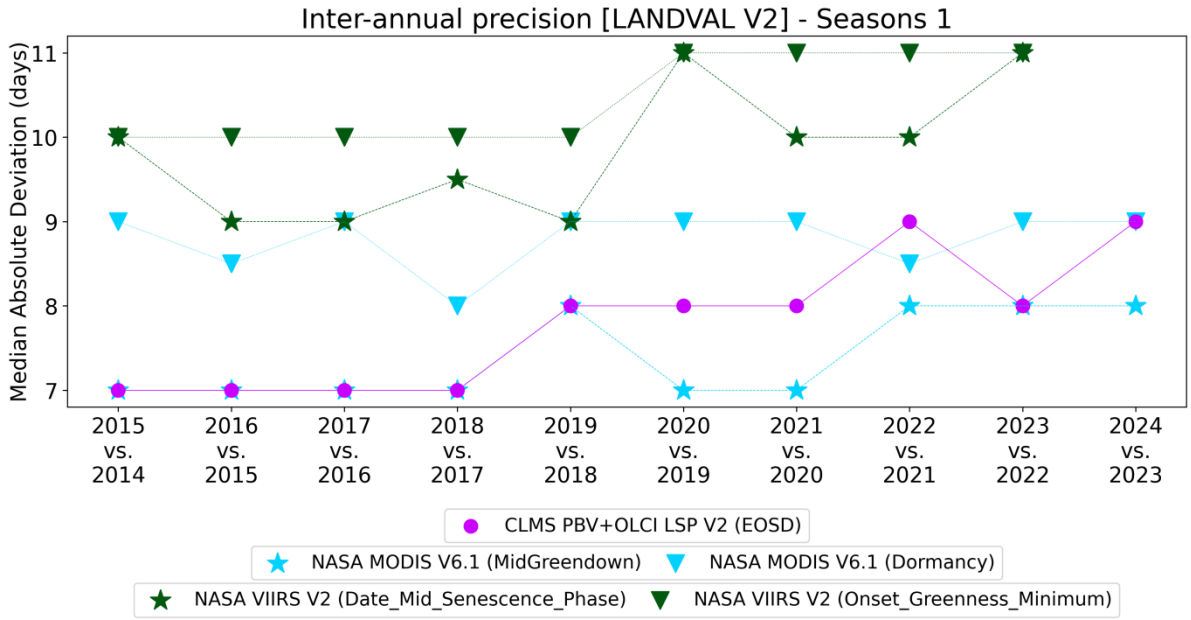

Figure 55: Same as in Figure 54, but for the end of the season (EOSD).

Main findings of the inter-annual precision analysis for SOSD (Figure 54):

- CLMS LSP V2 shows MAD of 6–7 days during the PBV period and 8–9 days during the OLCI period, with no impact from the sensor transition (2018–2019), where MAD is 7 days.
- Compared with other products:
    - Better precision than MODIS V6.1 *Greenup* (MAD 8–11 days) and similar to MODIS V6.1 *MidGreenup* (MAD 6–9 days).
    - Better than VIIRS V2 *Onset_Greenness_Increase* (MAD 8–10 days) and *Date_Mid_Greenup_Phase* (MAD 10–12 days).

Main conclusions of the inter-annual precision analysis for EOSD (Figure 55) are:

- CLMS LSP V2 shows slightly better inter-annual precision for the PBV period (MAD = 7 days) compared with the OLCI period (MAD = 8–9 days). As for SOSD, the sensor transition (2018–2019) has no noticeable impact (MAD = 8 days).
- Compared with other products:
    - Similar precision to MODIS V6.1 *MidGreendown* (MAD 7–8 days) and *Dormancy* (MAD 8–9 days).
    - Better than VIIRS V2 *Onset_Greenness_Minimum* and *Date_Mid_Senescence_Phase* (MAD 9–11 days).

7 SUMMARY AND CONCLUSIONS

The quality assessment of CLMS LSP V2 products was carried out for the 2014–2024 period following the methodology described in the CLMS Product Quality Assurance Document [CGLOPS1_PQAD]. Two main validation approaches were applied: direct validation, comparing satellite products with in situ observations, and indirect validation, involving product intercomparison. Several criteria of performance are analyzed, including the visualization of the product content, and the evaluation of the product completeness, spatial consistency and error (accuracy, precision and uncertainty).

Direct validation of SOSD and EOSD relied on PEP725, TEMPO, and NEON phenological datasets, complemented by PhenoCam V3 observations derived from digital cameras. Indirect validation involved both qualitative and quantitative comparisons of SOSD/EOSD with global satellite products (MODIS LCD V6.1 and VIIRS LCD V2) and quantitative comparisons of TPROD/SPROD with CLMS GPP. Additionally, spatial consistency and agreement with equivalent HR-VPP layers were assessed over Europe. The consistency between PBV and OLCI retrievals during the overlap year (2019) was also evaluated to quantify the impact of the sensor transition on the CLMS LSP V2 algorithm.

Main findings are:

Consistency between estimates from PBV and OLCI

- The transition from PBV to OLCI shows strong overall consistency for the overlap year (2019), with almost same spatial global distributions and high correlations for all phenological and productivity metrics.
- Minor systematic differences appear (almost no bias for SOSD and OLCI slightly later for EOSD, with marginally longer season of ~1-2 days), but these remain within requirements and lead only to small increases in SPROD and TPROD.
- Overall, no significant discontinuities are introduced by the sensor change and the temporal evolution of mean SOSD and OLCI values across the 2014–2024 period is consistent with no significant variations.

Product completeness

- CLMS LSP V2 provides a high proportion of valid season 1 data (75–80%), comparable to VIIRS V2 and higher than MODIS V6.1. CLMS improves coverage over tropical regions, whereas VIIRS exhibits better completeness in some sparsely vegetated areas.
- After applying quality flags, CLMS retains most data (~95%), while VIIRS shows a strong reduction in valid pixels (~30%), and MODIS filters out only a small fraction (<2%).

- Season 2 retrievals are sparse for all products, with CLMS maintaining similar completeness across PROBA-V and OLCI periods.

Spatial consistency

- The visual inspection of global maps of all CLMS LSP V2 layers confirms that the products exhibit globally consistent and spatially coherent patterns across the globe, seasons and phenological parameters. The QA layer additionally shows that most global pixels achieve *good* or *best* quality for all phenological phases.
- Visual inspection of CLMS LSP V2 maps at subcontinental scale shows a consistent and realistic spatial distribution of phenology and productivity parameters. Minor exceptions occur in desert or sparsely vegetated regions (more on PBV), particularly in Australia and parts of the Sahara. Overall, no spatial artifacts are observed, indicating high spatial coherence of the products.
- The spatial consistency analysis with HR-VPP shows that CLMS LSP V2 SOSD, EOSD, SPROD, and TPROD exhibit spatial patterns largely consistent with HR-VPP across all examined AOIs. CLMS shows slightly delayed SOSD and slightly later EOSD over cultivated areas, while other biomes show near-identical distributions. SPROD and TPROD values are systematically higher in CLMS LSP V2, reflecting the higher maximum PPI observed in the product.
- The spatial consistency analysis with MODIS LCD V6.1 shows that CLMS LSP V2 SOSD and EOSD are generally in good agreement with *MidGreenup* and *MidGreendown*, respectively. Differences are typically within ±15 days for SOSD, with slightly higher CLMS values (+2 days on average), and systematically positive for EOSD (+4 days on average). Larger discrepancies occur in some areas and tropical regions.
- The spatial consistency analysis with VIIRS LCD V2 shows overall good spatial consistency with CLMS LSP V2 SOSD, which is generally earlier than VIIRS *Date_Mid_Greenup_Phase* (histogram centred at -2 days), while EOSD is later than *Date_Mid_Senescence_Phase* (+12 days), reflecting the expected differences in phenological definitions.
- The latitudinal transect confirms that CLMS LSP V2 captures realistic SOSD and EOSD trends from Norway to South Africa, with SOSD decreasing north-to-south and EOSD showing the expected seasonal progression. CLMS values are largely consistent with MODIS V6.1 and VIIRS V2, though minor deviations occur in tropical forests and near the equator, where CLMS detects two distinct growing seasons.
- Spatial precision of CLMS LSP V2 depends on biome type: SOSD/EOSD show low variability in croplands, shrublands, and deciduous forests, moderate in needle-leaf forests and herbaceous areas, and highest in evergreen broadleaf forests. SPROD/TPROD show moderate variability (10–30%), higher in low-productivity regions.

Error evaluation with ground-based references

- CLMS LSP V2 SOSD and EOSD show generally good agreement with in-situ phenology datasets (PEP725, TEMPO, NEON) across 2014–2024. Summary of Bias, RMSD and conformity testing is provided in Figure 56.
- SOSD tends to occur slightly later than early leaf development phases and earlier than leaf development (>50%), while EOSD is close to the leaf coloration and slightly earlier than the leaf fall, consistent with the 25% and 15% seasonal amplitude thresholds used in the satellite product. The exception was NEON *emerging* (pine forests, few observations) where CLMS SOSD is largely anticipated to the observations. Biases typically range from -3 to +9 days for SOSD (except for NEON *emerging*) and from -4 to +5 days for EOSD (except for *50% leaves fallen* where a larges bias of -18 days is observed as expected). RMSD values of 15–29 days are found, with moderate to high correlations (R ~ 0.3–0.7).
- Compliance with user requirements shows percentages within threshold typically between 30% and 50% for both SOSD and EOSD (except for NEON emerging).
- Overall, the results indicate that CLMS LSP V2 captures the timing of phenological transitions with reasonable accuracy and is consistent across multiple ground-based datasets. The temporal evolution of the annual bias indicates that accuracy remains stable throughout the 2014–2024 time series, with some exceptions due to differences in sampling applied each year.

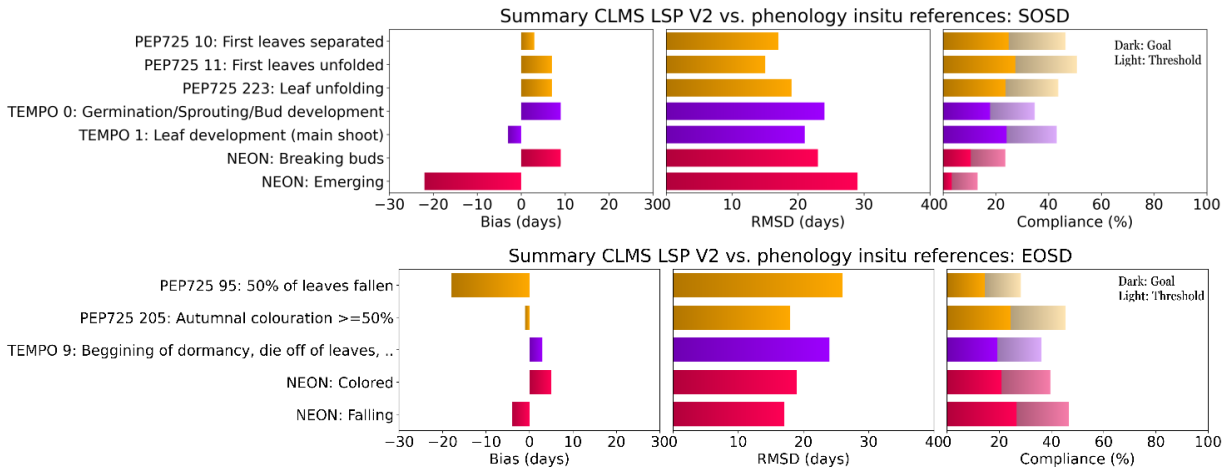

Figure 56: Summary of Bias, RMSD and requirements conformity test of CLMS LSP V2 SOSD (top) and EOSD (bottom) versus different phenophases of PEP725, TEMPO and NEON in-situ datasets. Comparison for season 1 over the 2014–2024 period.

- CLMS LSP V2 shows good agreement with PhenoCam V3 near-ground observations over 2014–2024 (see summary in Figure 57).

- SOSD aligns best with the 50% PhenoCam start-of-season threshold (Bias ~ 2 days), while EOSD falls between the 25% and 50% PhenoCam end-of-season thresholds (Bias ~ −8 to +6 days). Overall, CLMS SOSD tends to be slightly later and EOSD slightly earlier than PhenoCam V3 observations, resulting in a marginally shorter growing season, with moderate uncertainties (RMSD ≈ 21–34 days) and correlations of 0.66–0.77.

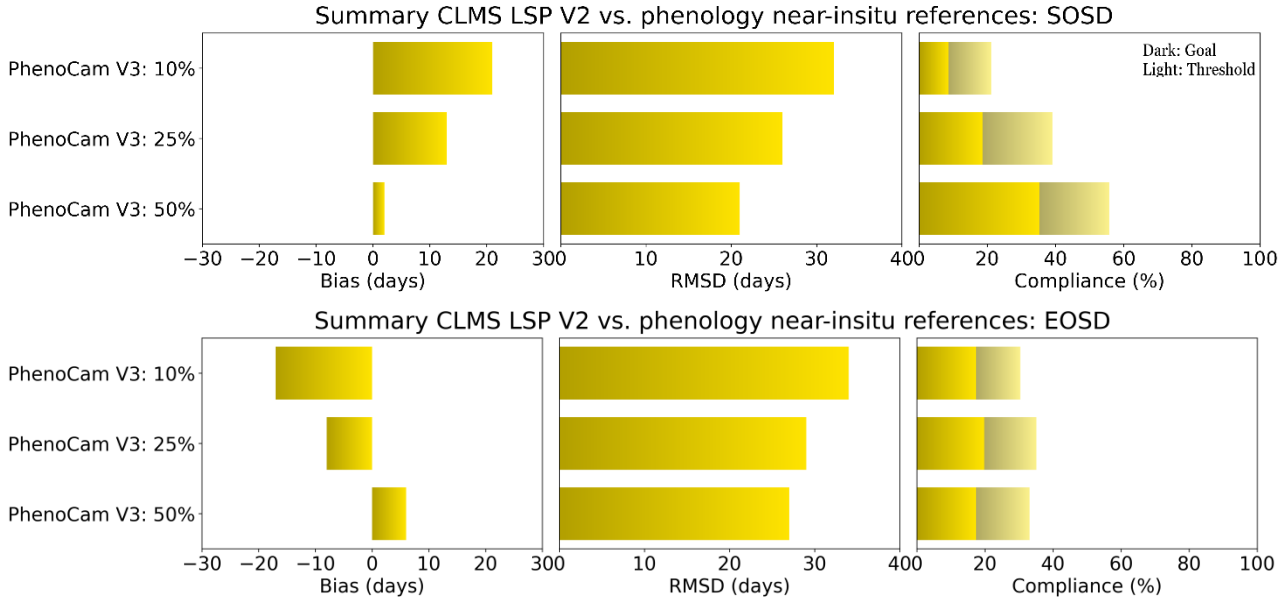

Figure 57: Summary of Bias, RMSD and user requirements conformity test of CLMS LSP V2 SOSD (top) and EOSD (bottom) versus different phenophases of Phenocam V3 near-ground dataset. Comparison for season 1 over the 2014–2024 period.

- In summary, CLMS LSP V2 shows good agreement with multiple ground-based datasets, capturing SOSD and EOSD reliably with small biases and moderate uncertainties, despite differences arising from satellite threshold definitions compared to in-situ observations of various phenological stages.

Error evaluation with satellite-based references

- CLMS LSP V2 shows strong consistency with HR-VPP over Europe (see summary in Figure 58), with high correlations ($R \geq 0.85$ for SOSD/EOSD; $R > 0.90$ for SPROD/TPROD). SOSD and EOSD exhibit slight positive biases (delays of 9–12 days), resulting in a marginally shorter growing season, while productivity metrics (SPROD and TPROD) are systematically higher than HR-VPP by ~21–23%.
- CLMS LSP V2 shows strong agreement with MODIS LCD V6.1 across the 2014–2024 period Europe (see summary in Figure 58).

- SOSD (25% threshold) aligns best with MODIS *MidGreenup* (50% threshold), exhibiting negligible bias, high correlation (R = 0.94), and RMSD of 14 days.
- EOSD (15% threshold) corresponds best with MODIS *MidGreendown*, with a slight positive bias (+11 days), R = 0.95, and RMSD = 20 days.
- Agreement varies by biome: SOSD and EOSD show minor biases with MODIS *MidGreenup* and *MidGreendown* (<15 days) for most biomes except EBF, which exhibits larger discrepancies. Compared to MODIS *Greenup* or *Dormancy* phases, CLMS SOSD is systematically later and EOSD earlier.
- Overall, CLMS LSP V2 reliably captures phenology trends, with best alignment to MODIS phases representing mid-season stages.

- CLMS LSP V2 shows strong agreement with VIIRS LCD V2 (2014–2023, see summary in Figure 58).
  - SOSD (25% threshold) aligns best with VIIRS *Date_Mid_Greenup_Phase* (RMSD = 18 days, 70% within threshold), with minor negative bias (-4 days)
  - EOSD (15% threshold) generally falls between VIIRS *Date_Mid_Senescence_Phase* (B = 15 days) and *Onset_Greeness_Minimum* (B = -11 days).
  - By biome, SOSD shows slightly earlier VIIRS *Date_Mid_Greenup_Phase* for most forests and grasslands, with cultivated areas matching closely; EOSD shows consistent positioning between *Date_Mid_Senescence_Phase* and *Onset_Greeness_Minimum*.

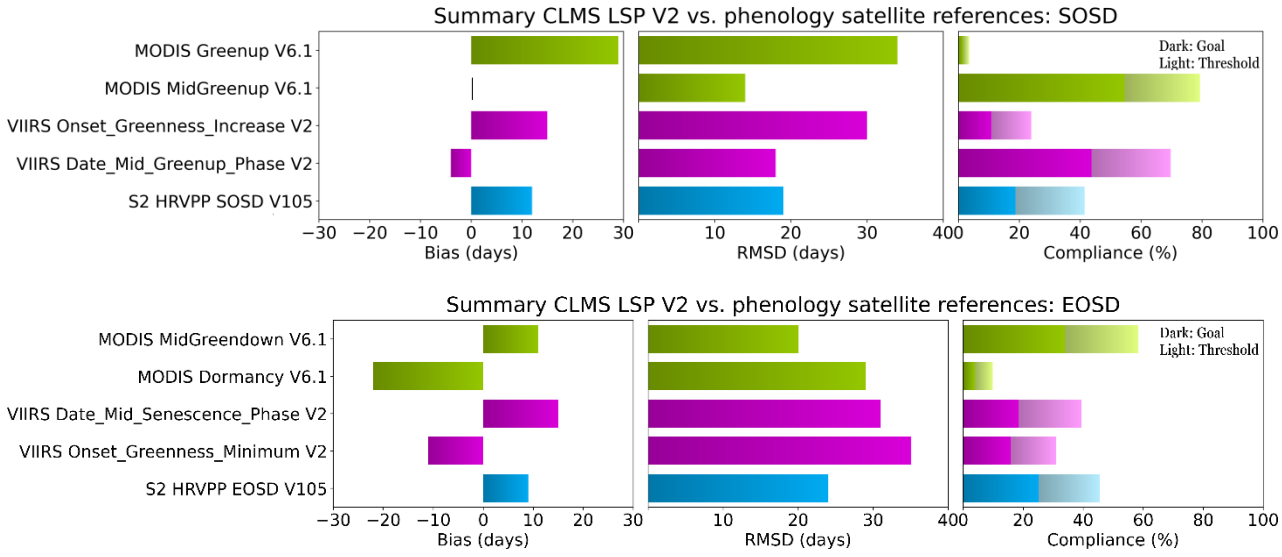

Figure 58: Summary of Bias, RMSD and user requirements conformity test of CLMS LSP V2 SOSD (top) and EOSD (bottom) versus relevant phenophases of global (MODIS LCD V6.1 and VIIRS LCD V2) and European (HR-VPP) satellite datasets. Comparison for season 1 over the 2014–2024 period.

- CLMS LSP V2 productivity metrics (SPROD and TPROD) show strong consistency with CLMS GPP over the same seasonal period (2014–2024). SPROD correlates with GPP at R = 0.74 and TPROD at R = 0.84, indicating that PPI-based seasonal productivity closely reflects carbon flux-based estimates.
- CLMS LSP V2 shows high inter-annual precision, with median absolute deviations (MAD) of 6–7 days for SOSD and 7 days for EOSD during the PBV period, and 8–9 days for both during the OLCI period, unaffected by the 2018–2019 sensor transition. Precision is generally better than VIIRS V2, or comparable to MODIS V6.1 phenometrics, indicating robust year-to-year consistency.

Concluding remarks

The quality of CLMS LSP V2 (2014–2024) was evaluated using direct validation (comparison with in-situ datasets: PEP725, TEMPO, NEON, and PhenoCam V3) and indirect validation (comparison with global satellite products MODIS LCD V6.1, VIIRS LCD V2, HR-VPP, and CLMS GPP). Performance metrics included inter-sensor consistency, product completeness, spatial consistency, and error evaluation (accuracy, precision, and uncertainty) along with the conformity with requirements.

CLMS LSP V2 estimates from PBV and OLCI show high spatial and temporal consistency, with only minor differences and no significant discontinuities. OLCI and PBV provide similar SOSD whilst OLCI shows slightly later EOSD than PBV, resulting in marginally longer seasons (~1-2 days). Global maps are spatially coherent, with a high proportion of season 1 data (75–80%) and sparse season 2 retrievals.

Direct validation with in-situ datasets (PEP725, TEMPO, NEON, PhenoCam V3) confirms that SOSD tends to be slightly later and EOSD slightly earlier than observations, resulting in marginally shorter seasons. Biases range from -3 to +9 days for SOSD (except for NEON *emerging*) and -4 to +5 days for EOSD, with moderate to high correlations (R ~ 0.3–0.77) and RMSD of 15–34 days. Conformance with requirements typically lies around 30–50% with threshold level.

Product intercomparison shows high agreement with HR-VPP and similar spatial patterns ($R \geq 0.85$ for SOSD/EOSD; $R > 0.90$ for SPROD/TPROD), with slight positive biases for SOSD/EOSD and productivity metrics 21–23% higher. Comparison with MODIS LCD V6.1 shows that SOSD best matches *MidGreenup* (Bias = 0 days, RMSD = 14 days) and EOSD aligns with *MidGreendown* (Bias = +11 days, RMSD = 20 days). Comparison with VIIRS LCD V2 shows SOSD closely matching *Date_Mid_Greenup_Phase* (Bias = -4 days, RMSD = 18 days), while EOSD falls between VIIRS Date_Mid_Senescence_Phase (Bias = +15 days, RMSD = 31 days) and Onset_Greeness_Minimum (Bias = -11 days, RMSD = 35 days). Latitudinal trends and biome-dependent variability of CLMS LSP V2 are also realistic and comparable with MODIS LCD V6.1 and VIIRS LCD V2. SPROD and

TPROD also correlate strongly with CLMS GPP (R = 0.74 and 0.84). Inter-annual precision is high, with MAD of 6–7 days (PBV) and 8–9 days (OLCI) for SOSD, and 7–9 days for EOSD. No impact of the 2018–2019 sensor transition is observed, and inter-annual precision is comparable or better than MODIS V6.1 and VIIRS V2 phenometrics.

In summary, CLMS LSP V2 derived from PBV (2014–2018) and OLCI (2019–2024) is a robust and consistent product for global monitoring of vegetation phenology and productivity, with high spatial coherence and realistic latitudinal and biome-dependent variability. The PBV–OLCI sensor transition introduces no significant discontinuities. SOSD and EOSD are reliably captured with small biases and moderate uncertainties against both satellite and in-situ reference datasets. SPROD and TPROD provide consistent seasonal productivity estimates, strongly correlated with CLMS GPP. High inter-annual precision supports multi-year monitoring of land surface phenological dynamic from space.

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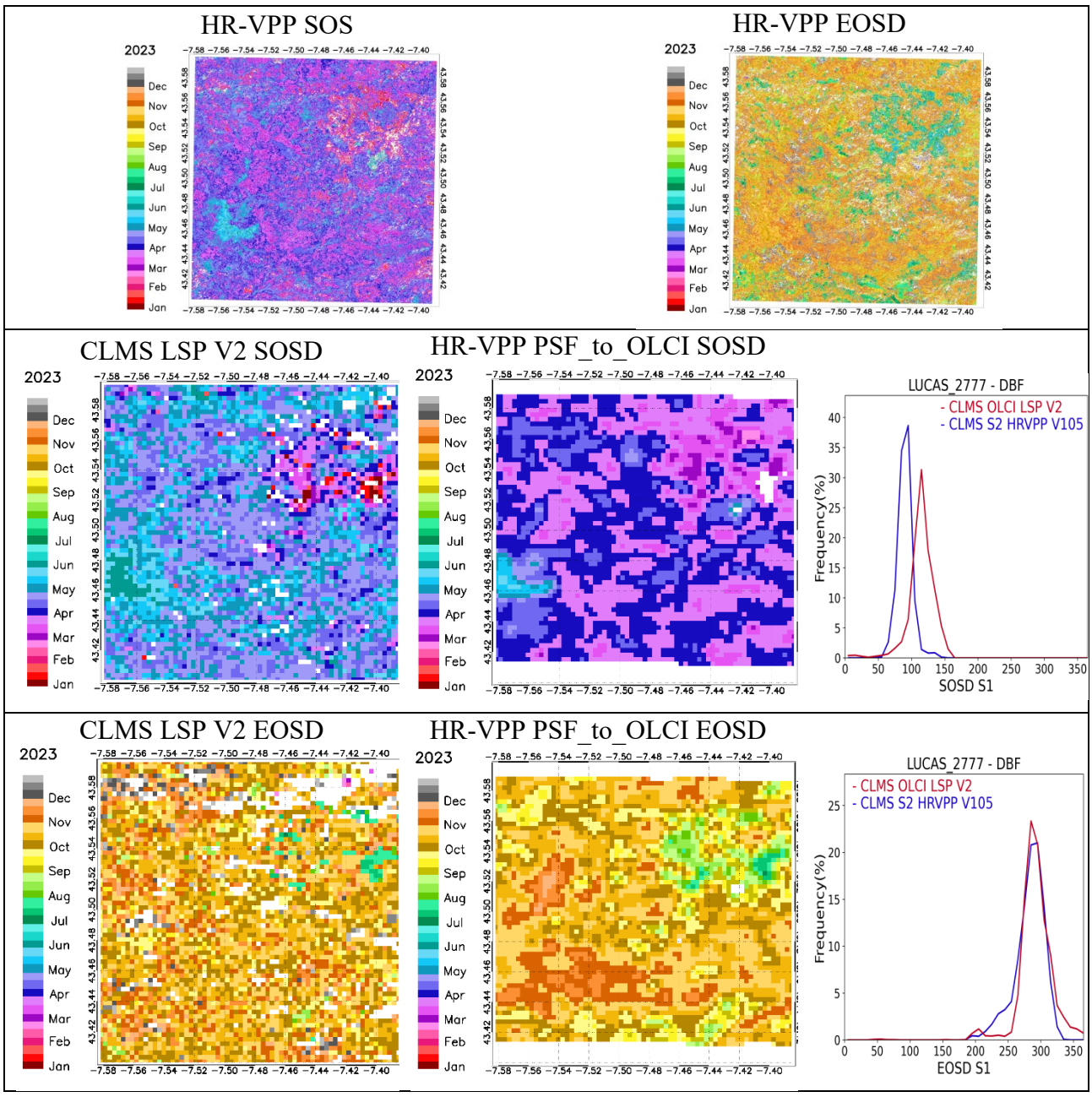
ACKNOWLEDGEMENTS

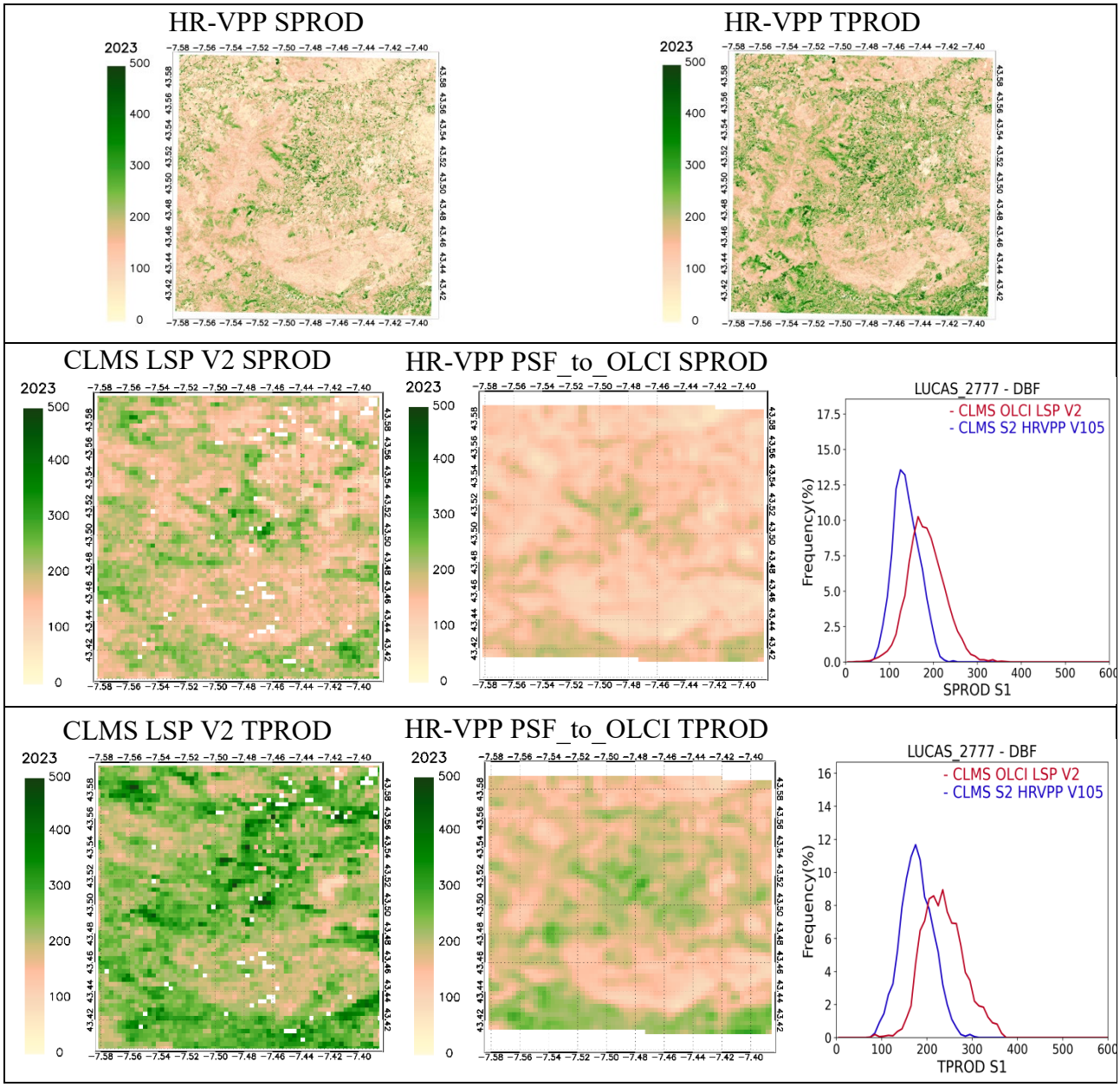
We gratefully acknowledge the in-situ datasets supplied by the PEP725, TEMPO, NEON, and PhenoCam networks.

- PEP725 data were provided by the members of the PEP725 project.
- TEMPO French public data are subject to the Open Data Act of October 8, 2016 and all this data are available under an Open License 2.0
- NEON material is based in part upon work supported by the NEON program, sponsored by the U.S. National Science Foundation and operated under cooperative agreement by Battelle.
- Phenocam data used in this research were provided by the PhenoCam Network, which has been supported by the National Science Foundation, the Long-Term Agroecosystem Research network which is supported by the United States Department of Agriculture, the U.S. Department of Energy, the U.S. Geological Survey, the Northeastern States Research Cooperative, and the USA National Phenology Network. We thank the PhenoCam Network collaborators, including site PIs and technicians, for publicly sharing the data that were used in this study.

ANNEX I: SPATIAL CONSISTENCY BETWEEN CLMS LSP V2 AND HR-VPP OVER AOIS

LUCAS 2777 - DBF





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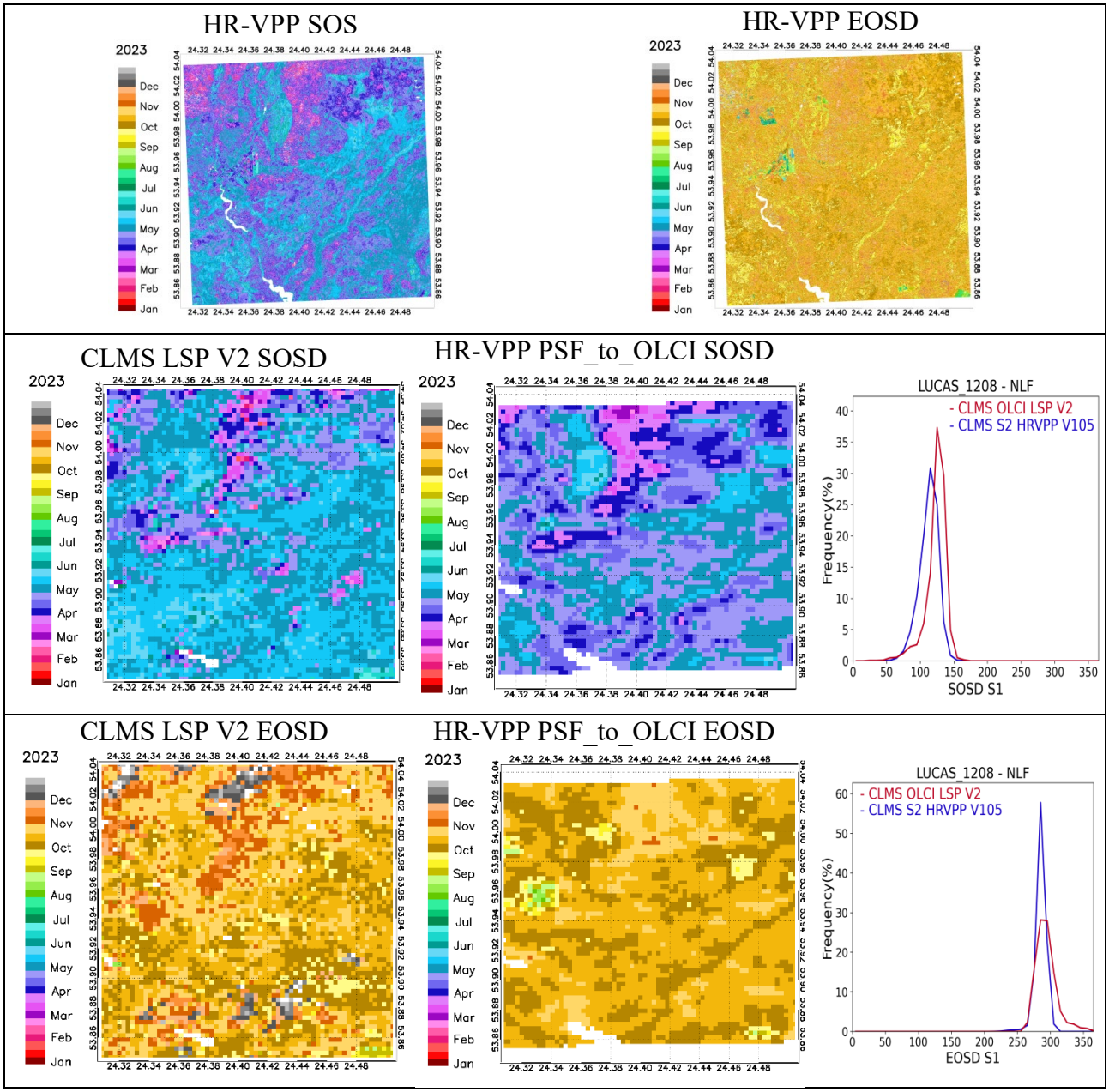
LUCAS 1208 – NLF



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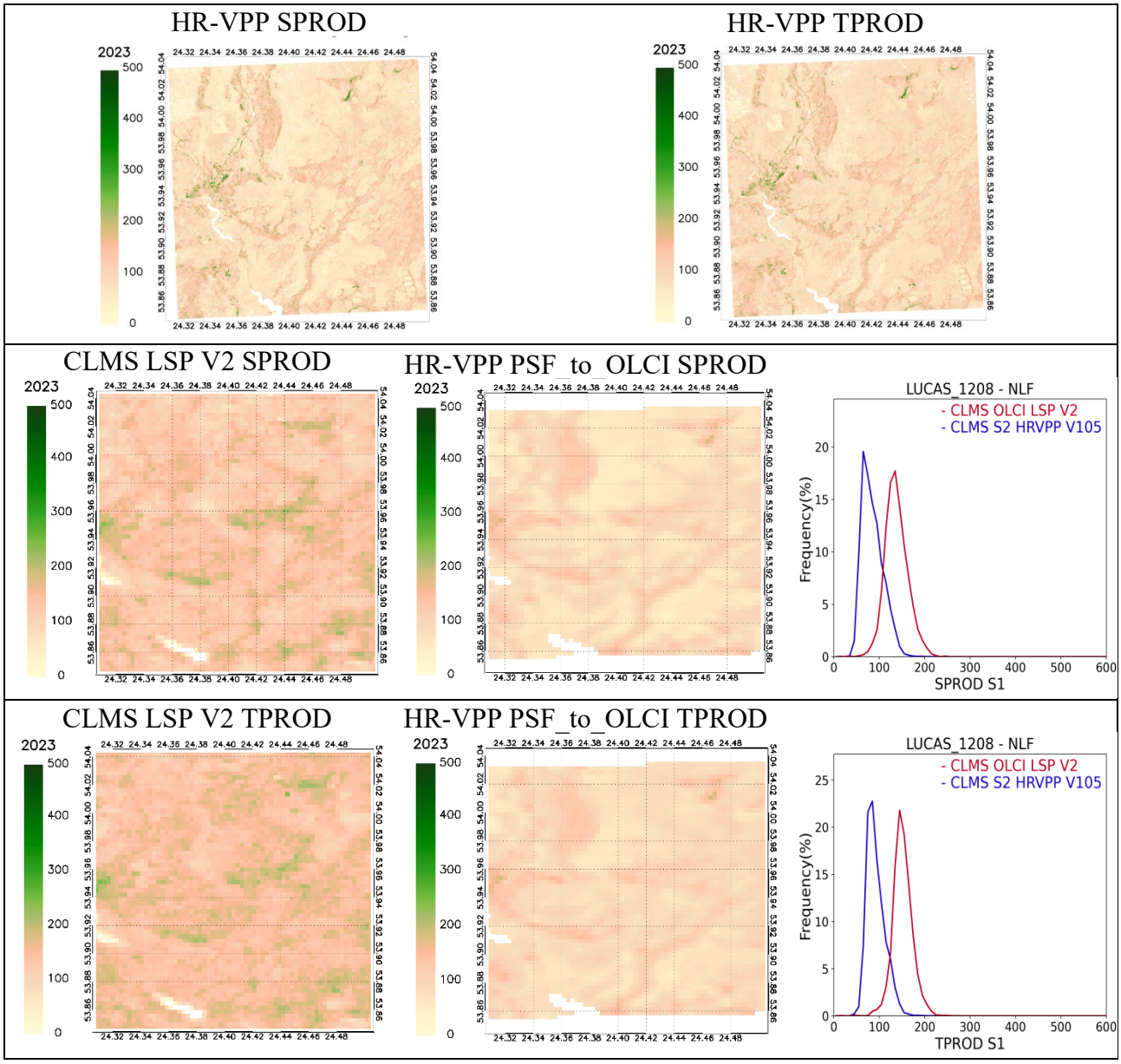
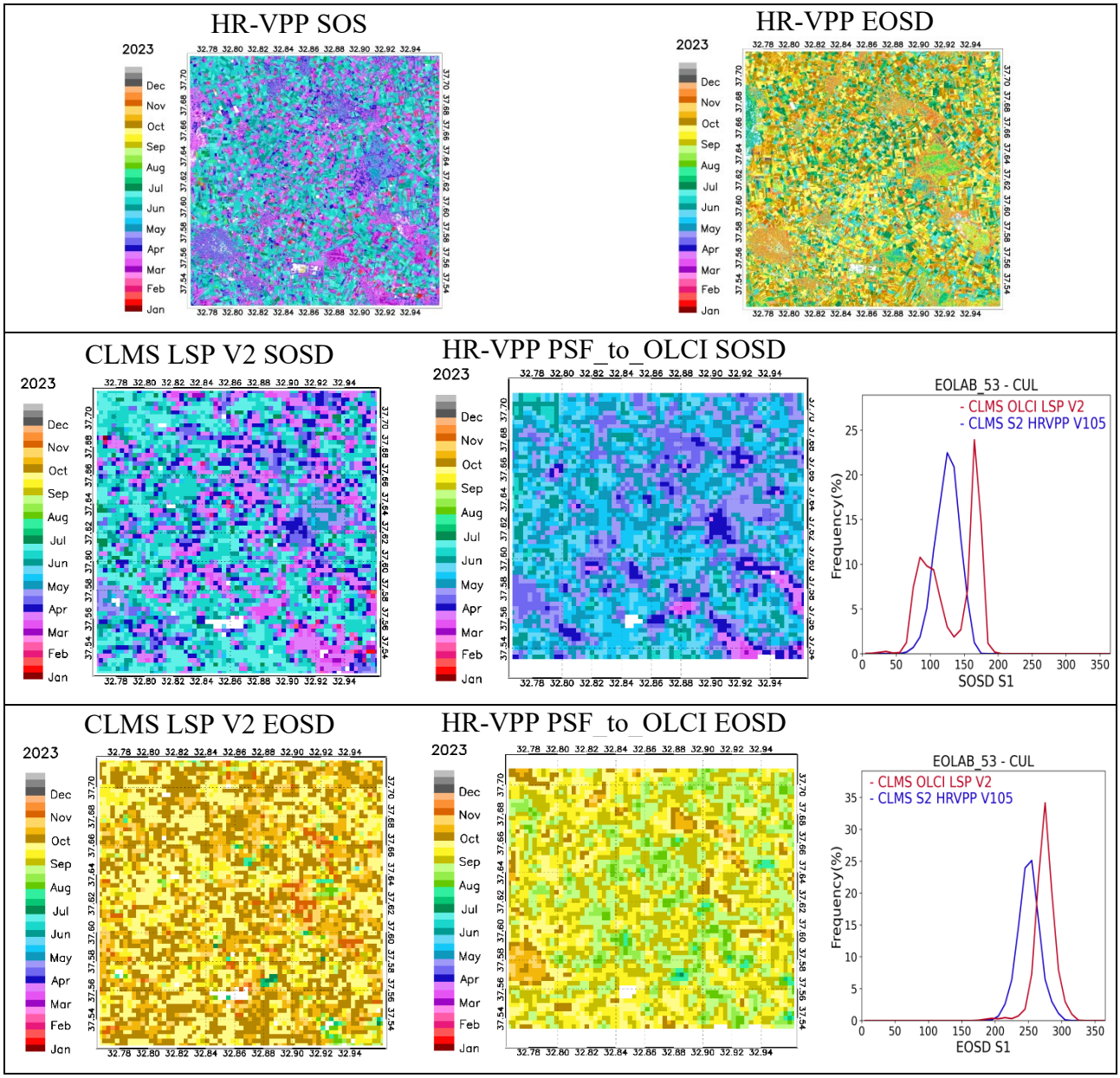

HR-VPP SPROD
HR-VPP TPROD
CLMS LSP V2 SPROD
HR-VPP PSF_to_OLCI SPROD
LUCAS_1208 - NLF
- CLMS OLCI LSP V2
- CLMS S2 HRVPP V105
Frequency(%)
SPROD S1
CLMS LSP V2 TPROD
HR-VPP PSF_to_OLCI TPROD
LUCAS_1208 - NLF
- CLMS OLCI LSP V2
- CLMS S2 HRVPP V105
Frequency(%)
TPROD S1

EOLAB 53 - CUL



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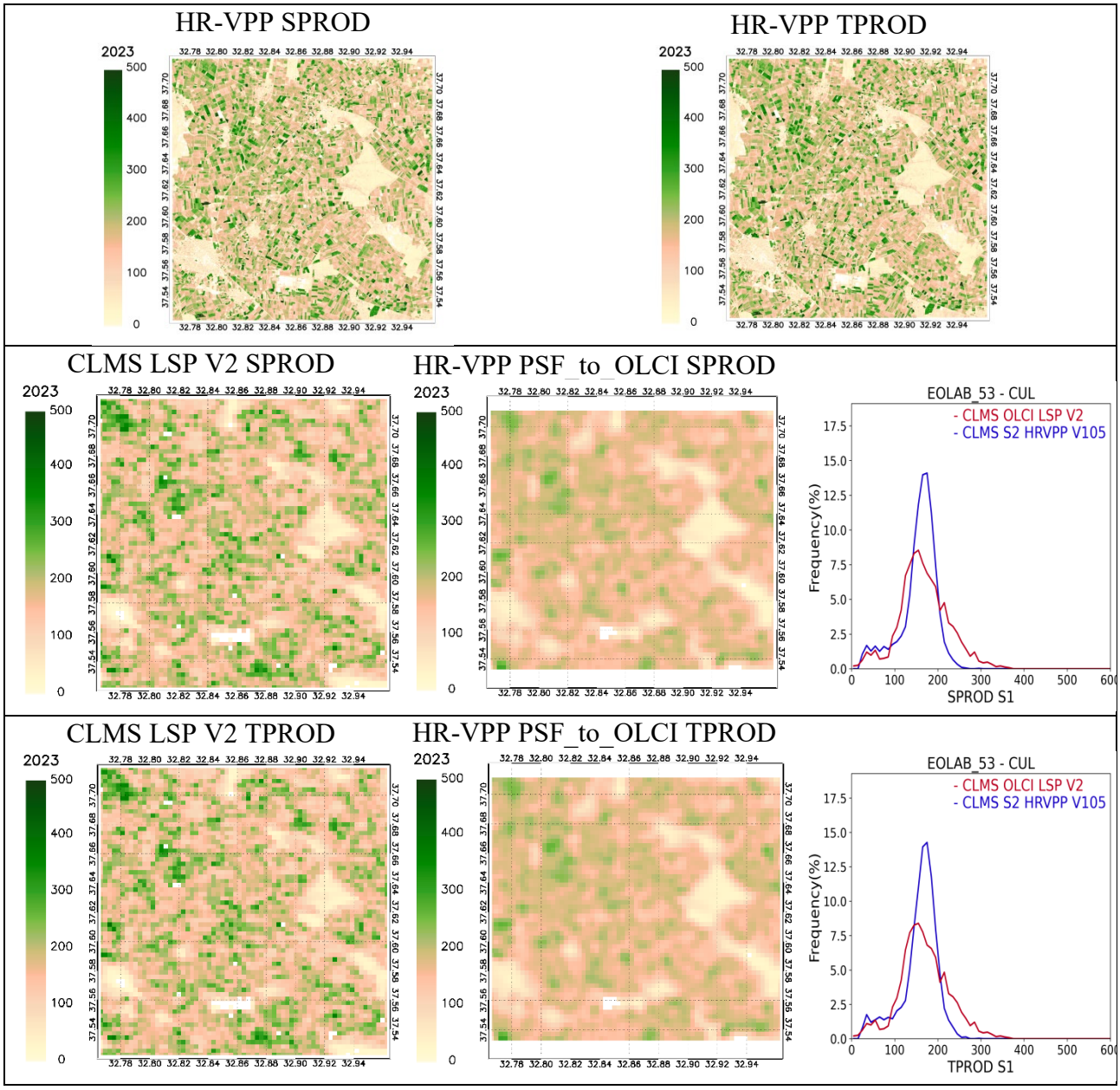

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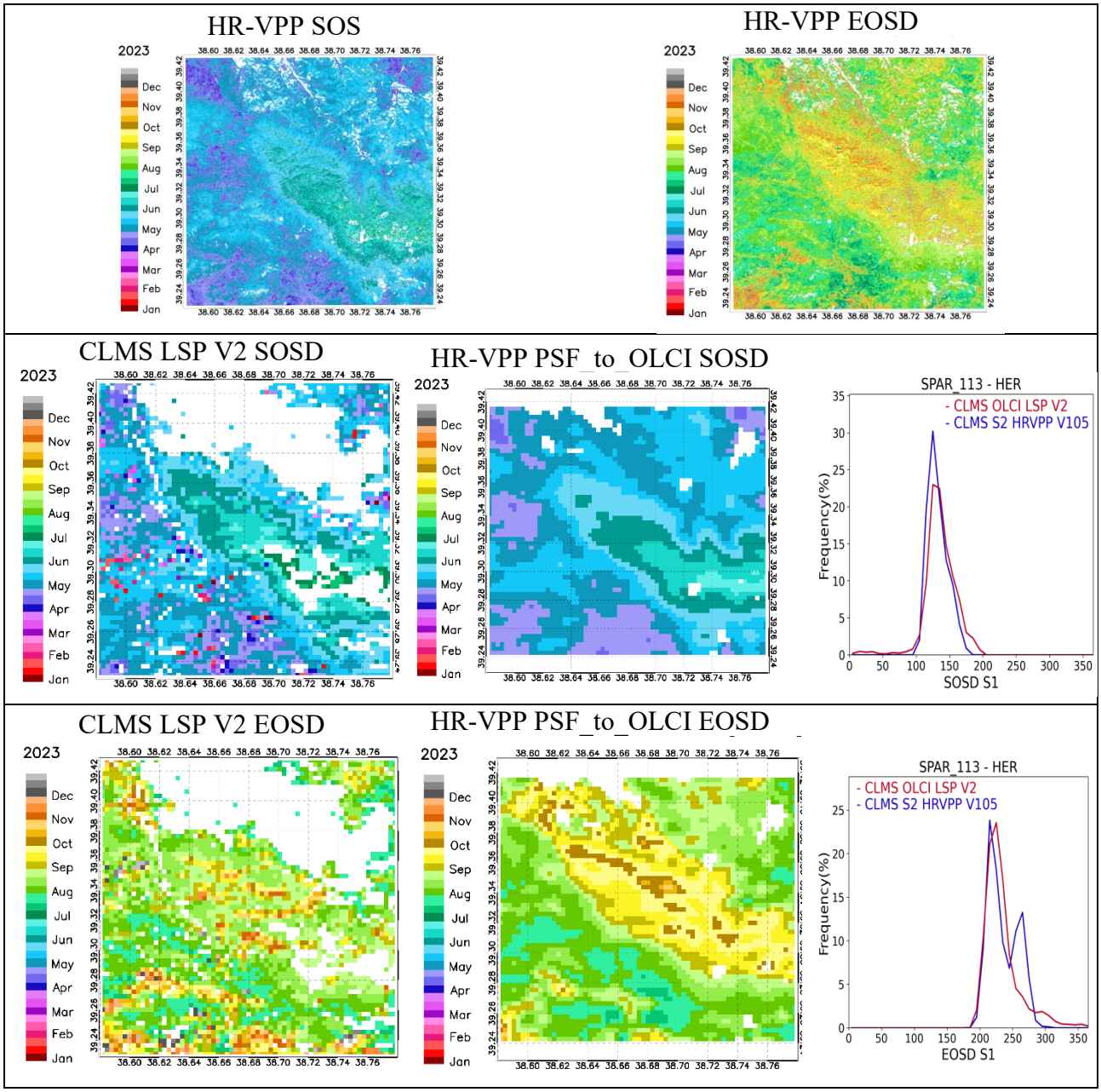
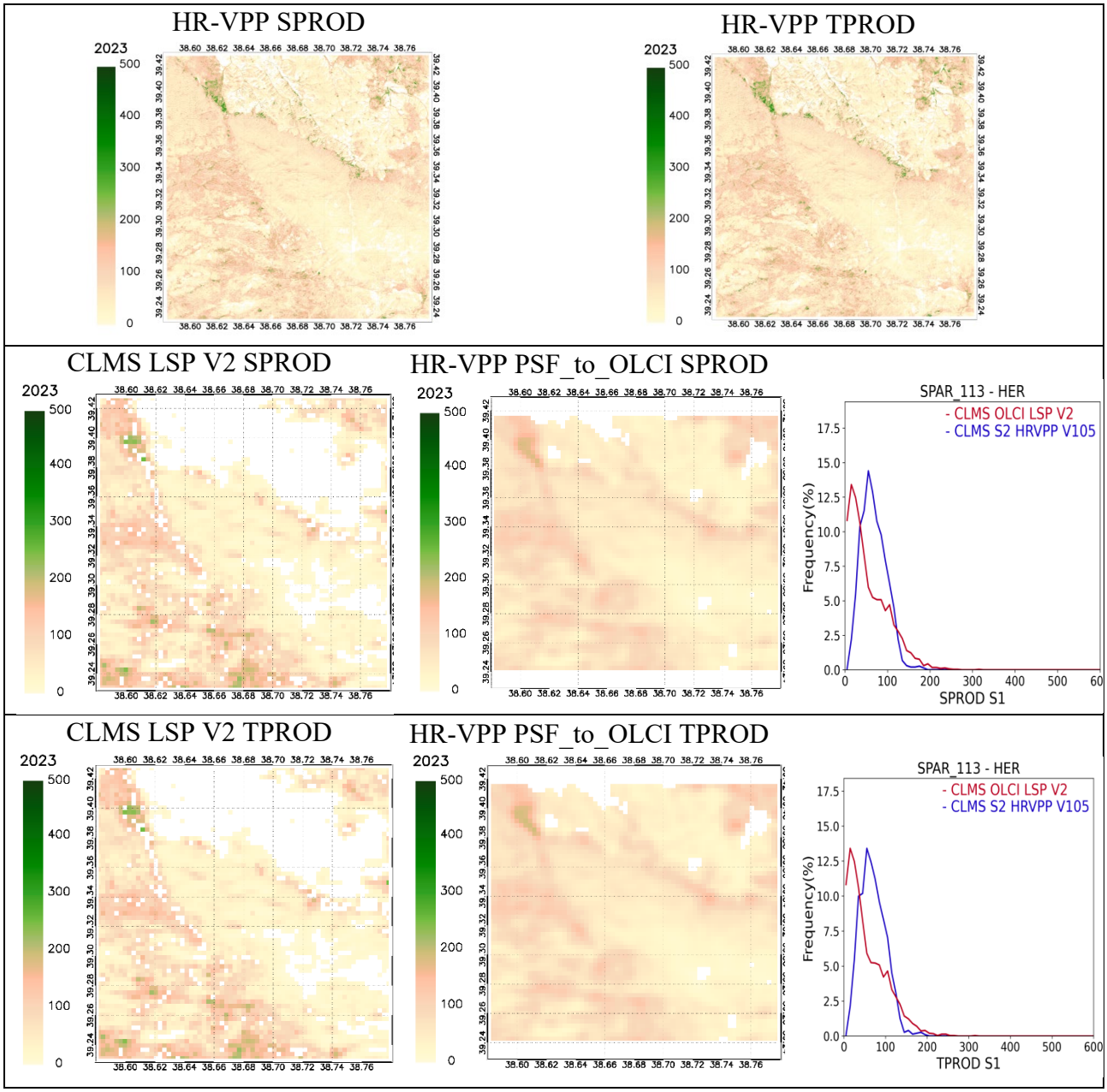
SPAR 113 - SHR

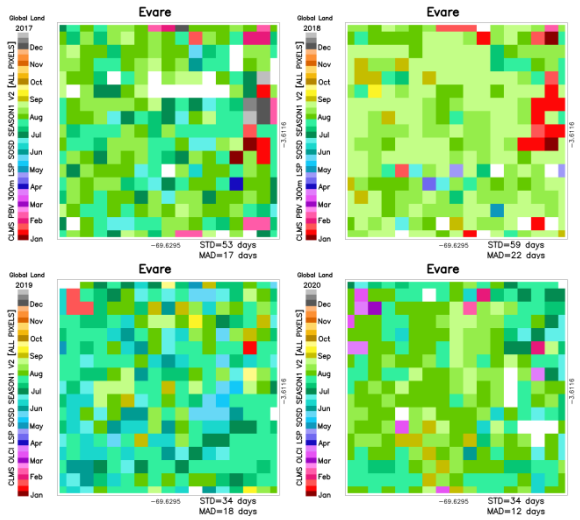

HR-VPP SOS
HR-VPP EOSD
CLMS LSP V2 SOSD
HR-VPP PSF_to_OLCI SOSD
SPAR_113 - HER
- CLMS OLCI LSP V2
- CLMS S2 HRVPP V105
Frequency(%)
SOSD S1
CLMS LSP V2 EOSD
HR-VPP PSF_to_OLCI EOSD
SPAR_113 - HER
- CLMS OLCI LSP V2
- CLMS S2 HRVPP V105
Frequency(%)
EOSD S1


HR-VPP SPROD
HR-VPP TPROD
CLMS LSP V2 SPROD
HR-VPP PSF_to_OLCI SPROD
SPAR_113 - HER
- CLMS OLCI LSP V2
- CLMS S2 HRVPP V105
Frequency(%)
SPROD S1
CLMS LSP V2 TPROD
HR-VPP PSF_to_OLCI TPROD
SPAR_113 - HER
- CLMS OLCI LSP V2
- CLMS S2 HRVPP V105
Frequency(%)
TPROD S1

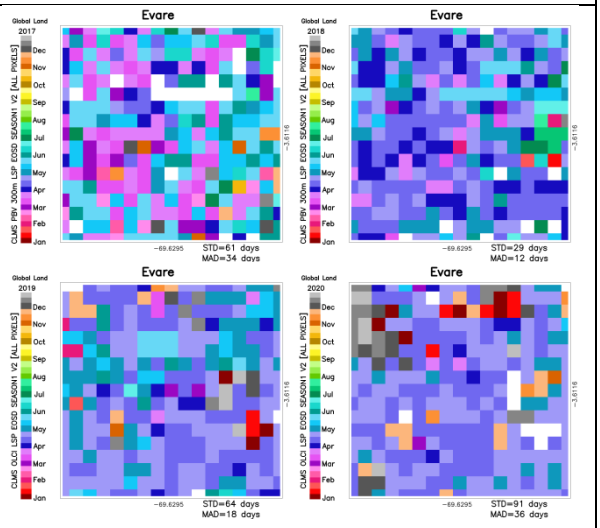
ANNEX II: MAPS AND SPATIAL PRECISION OF CLMS LSP V2 OVER SELECTED ROIS

Evare – EBF (2017-2020)

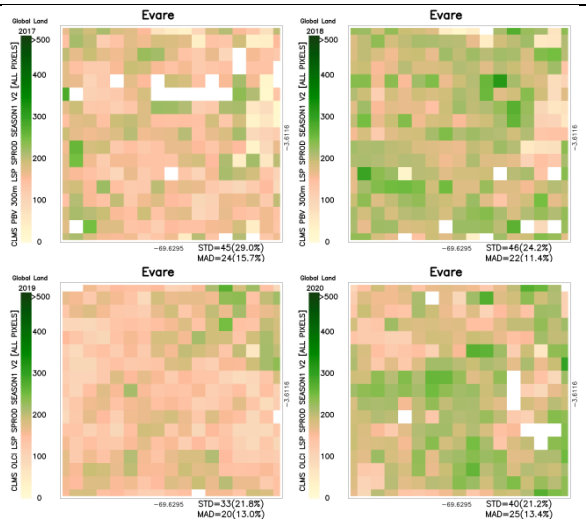
SOSD



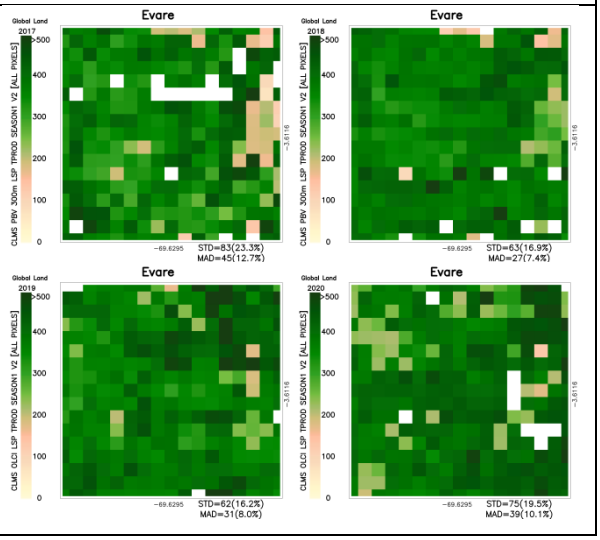
EOSD



SPROD

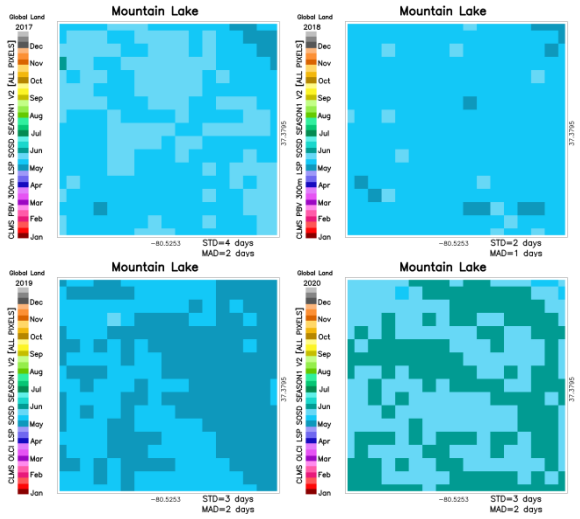


TPROD

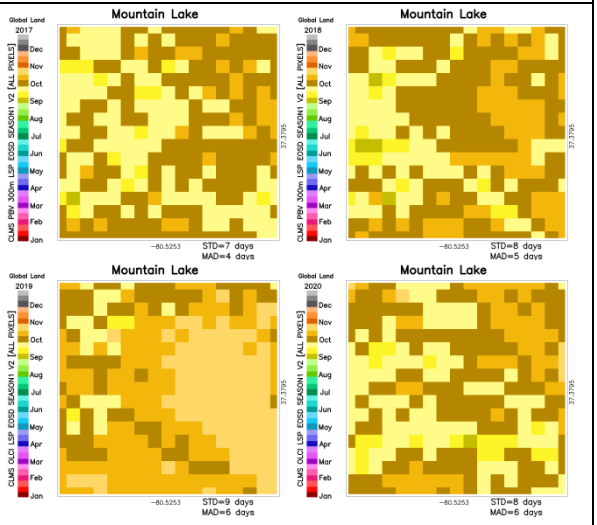


Mountain Lake - DBF (2017-2020)

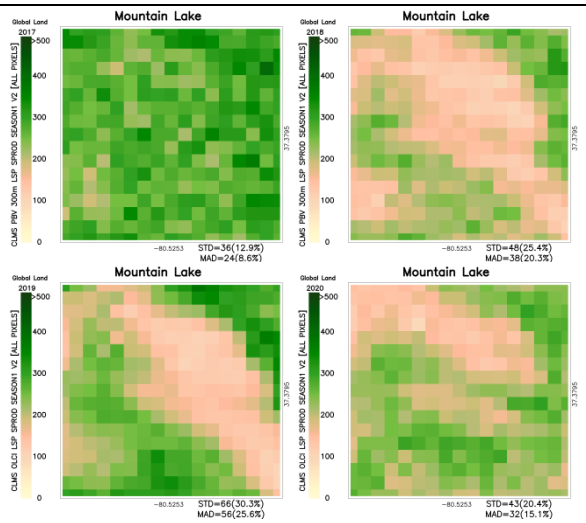
SOSD



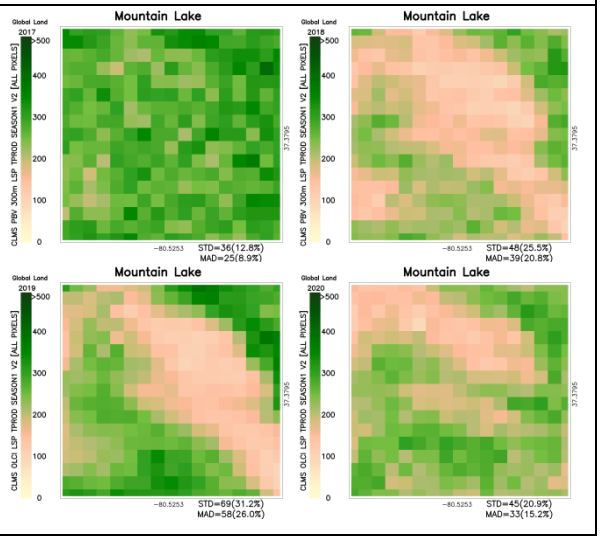
EOSD



SPROD

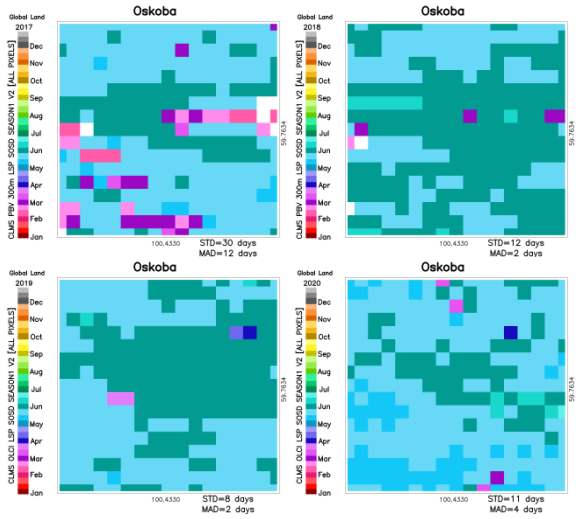


TPROD

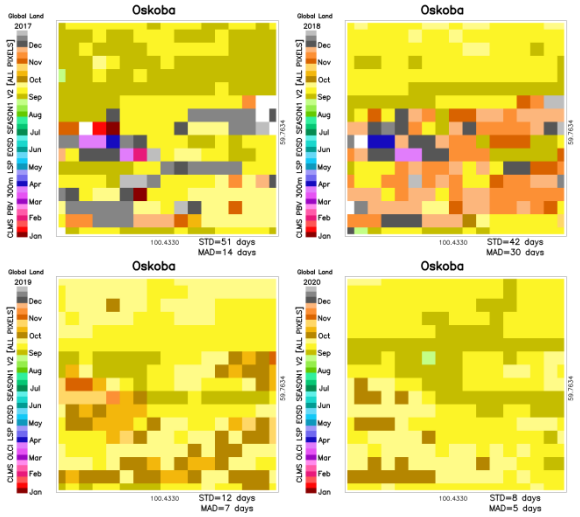


Oskoba - NLF (2017-2020)

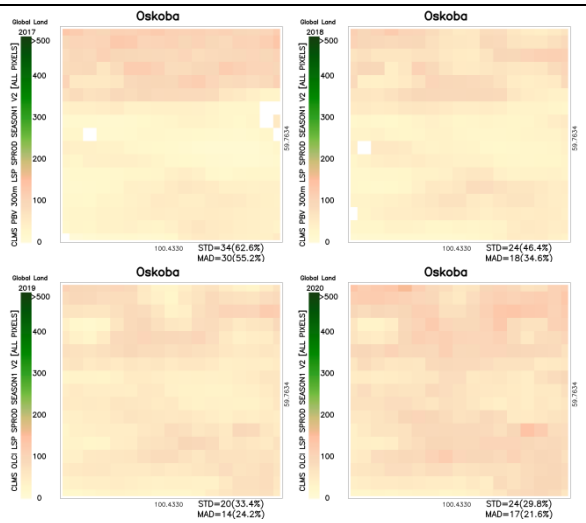
SOSD



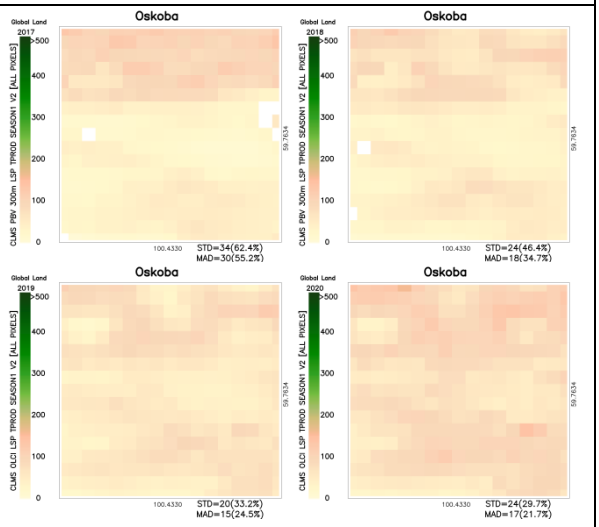
EOSD



SPROD

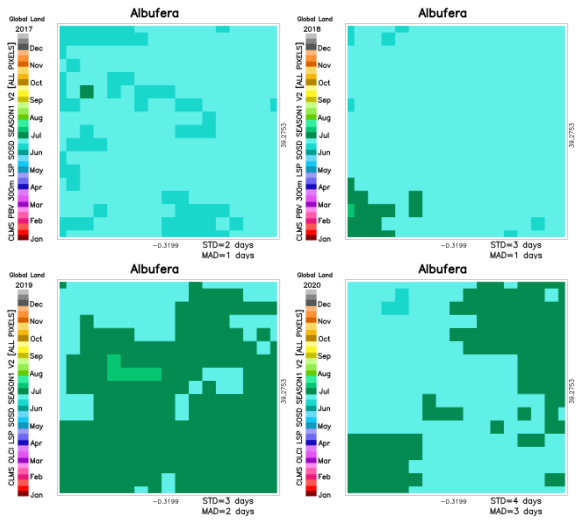


TPROD

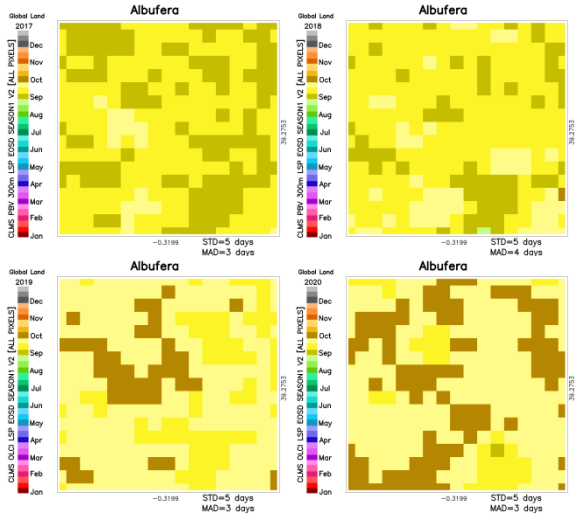


Albufera - CUL (2017-2020)

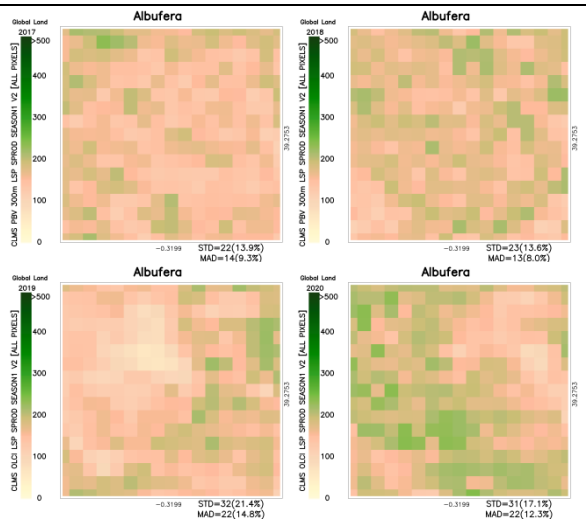
SOSD



EOSD



SPROD



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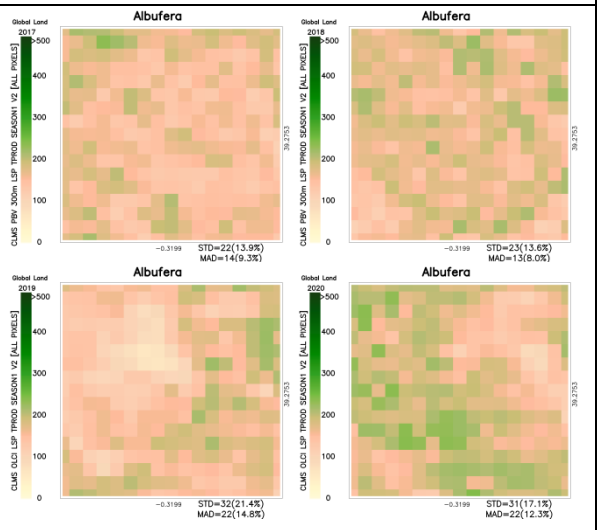
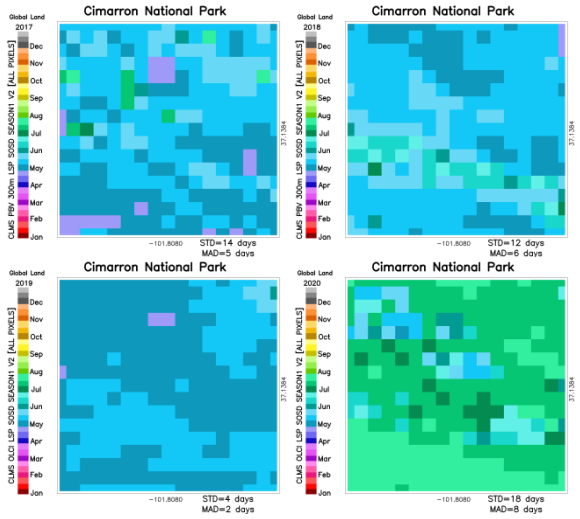
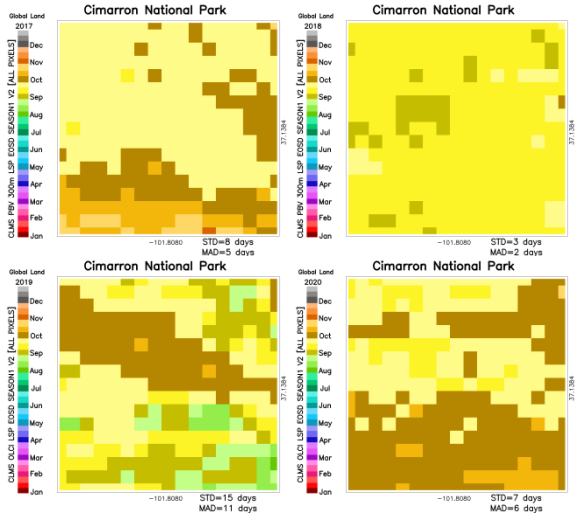


Cimarron National Grassland Park - HER (2017-2020)

SOSD



EOSD



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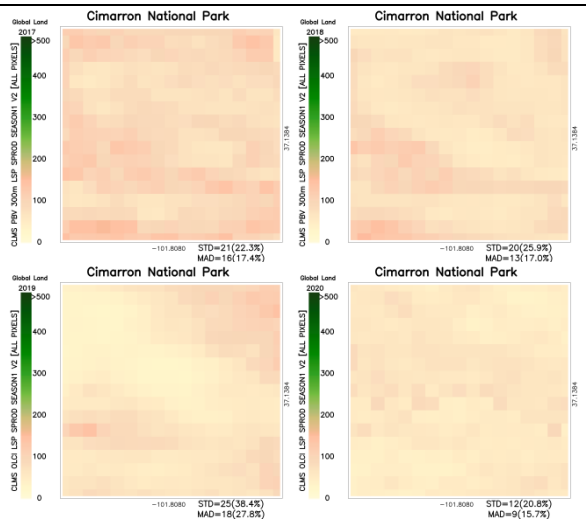

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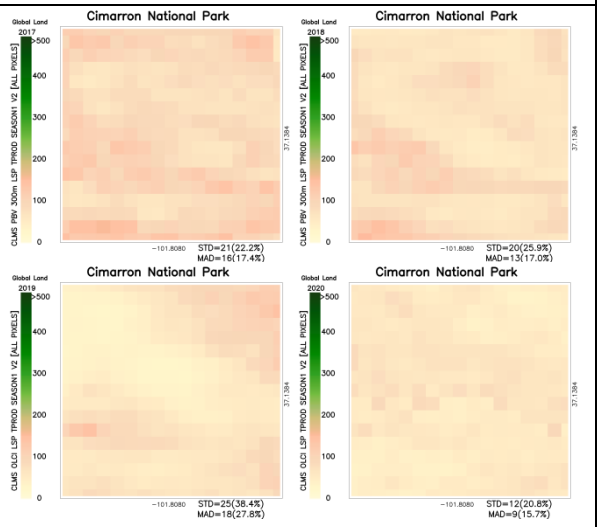
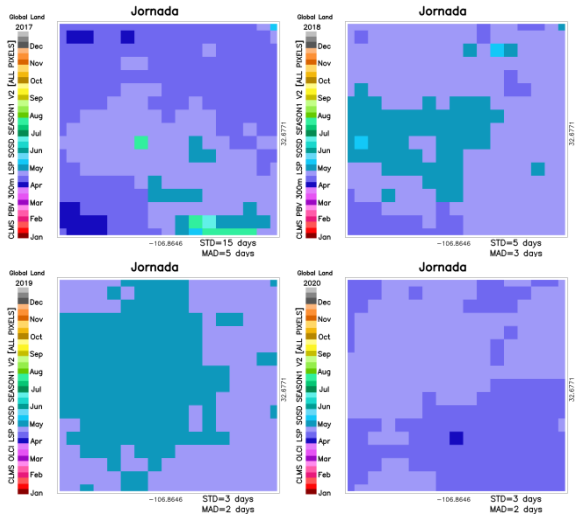


Jornada - SHR (2017-2020)

SOSD



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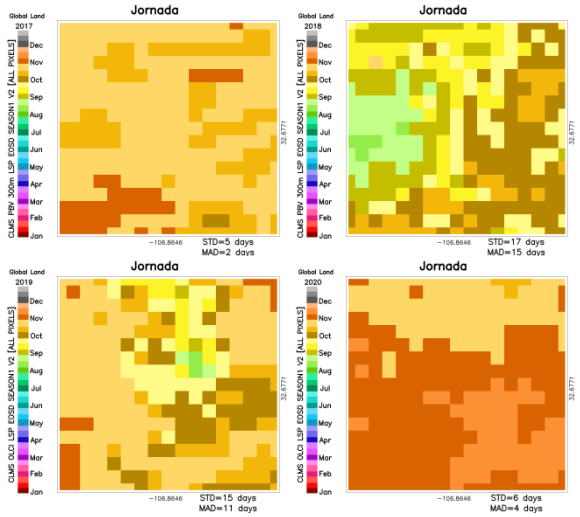
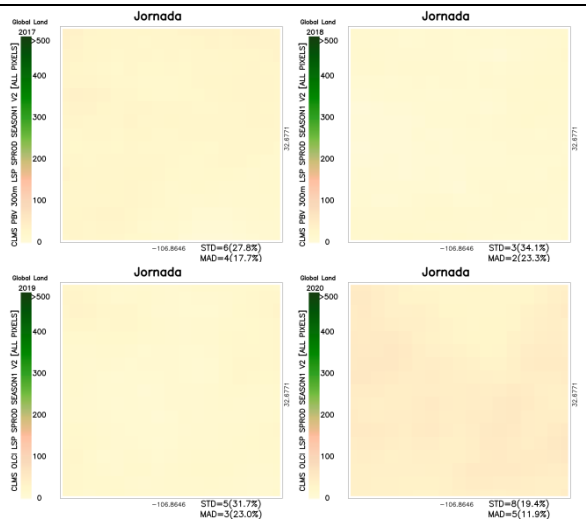


SPROD



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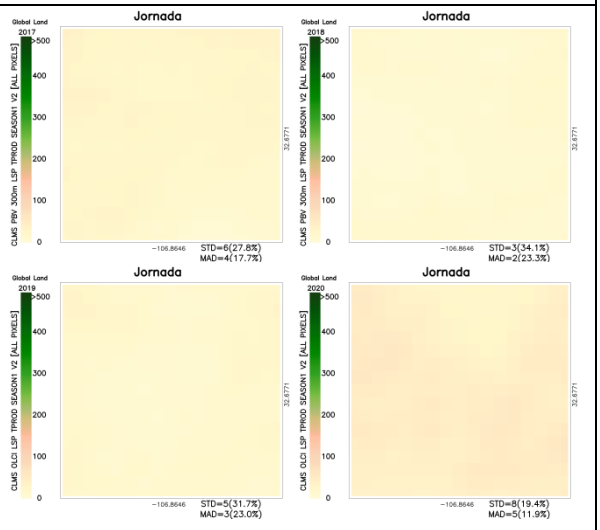
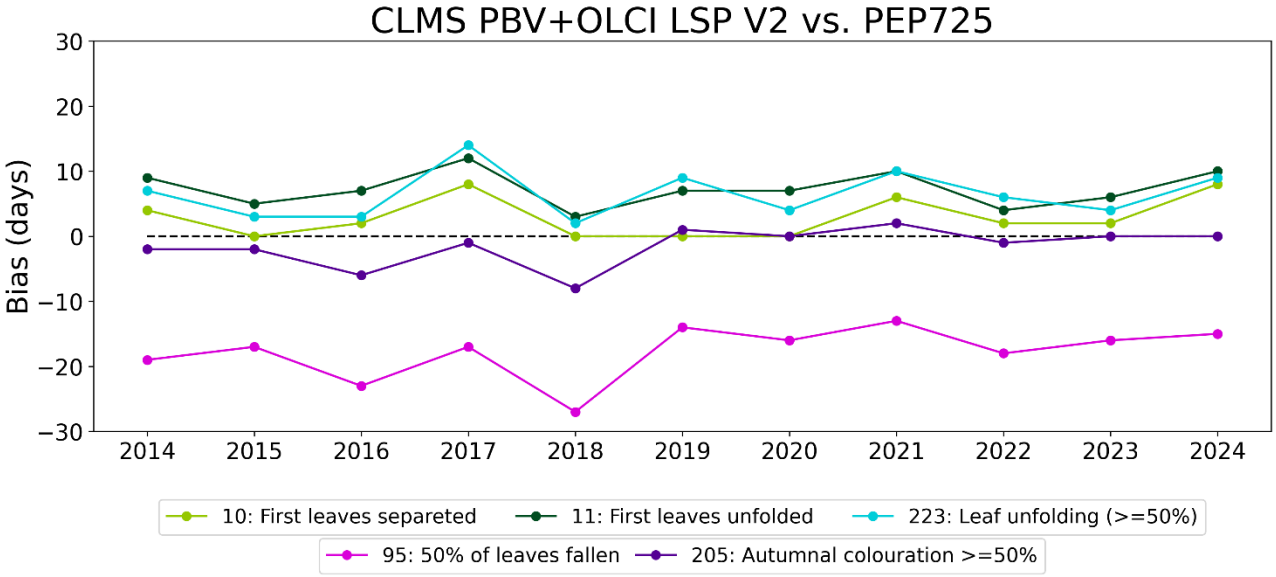
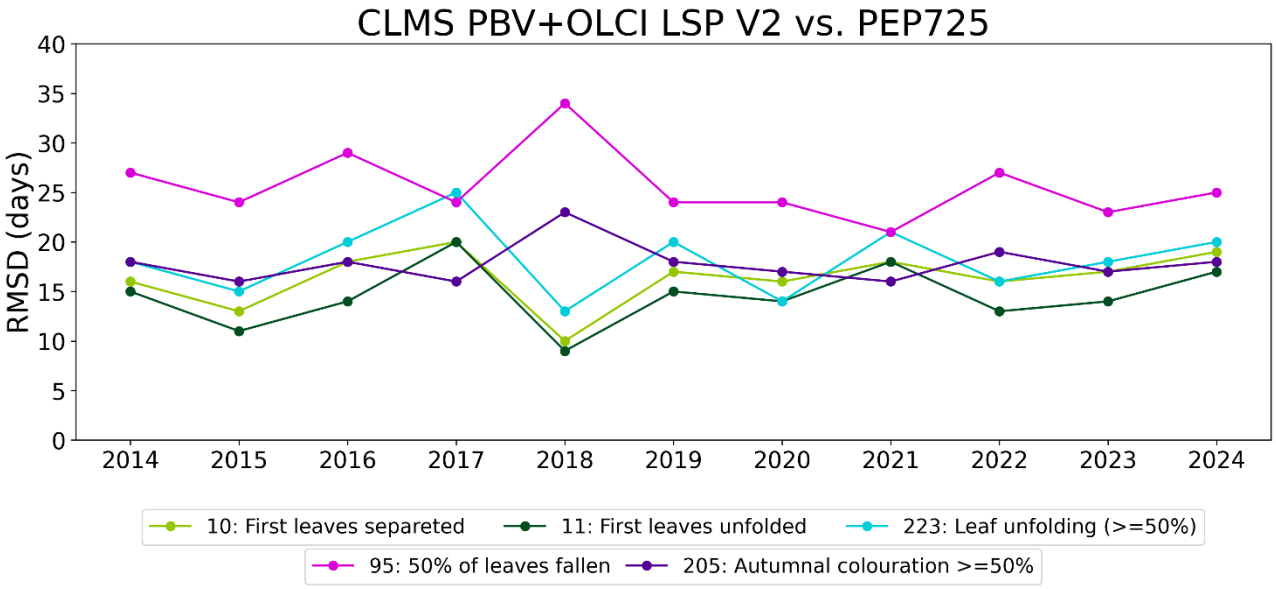


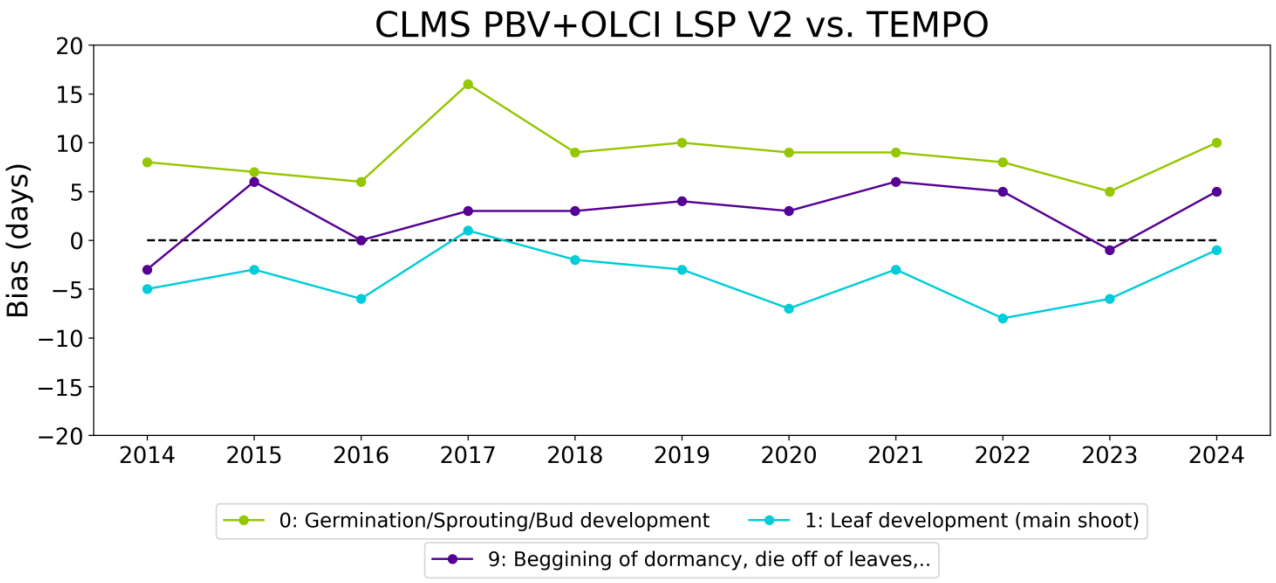
ANNEX III: BIAS AND RMSD OF CLMS LSP V2 VS. GROUND-BASED REFERENCES PER YEAR

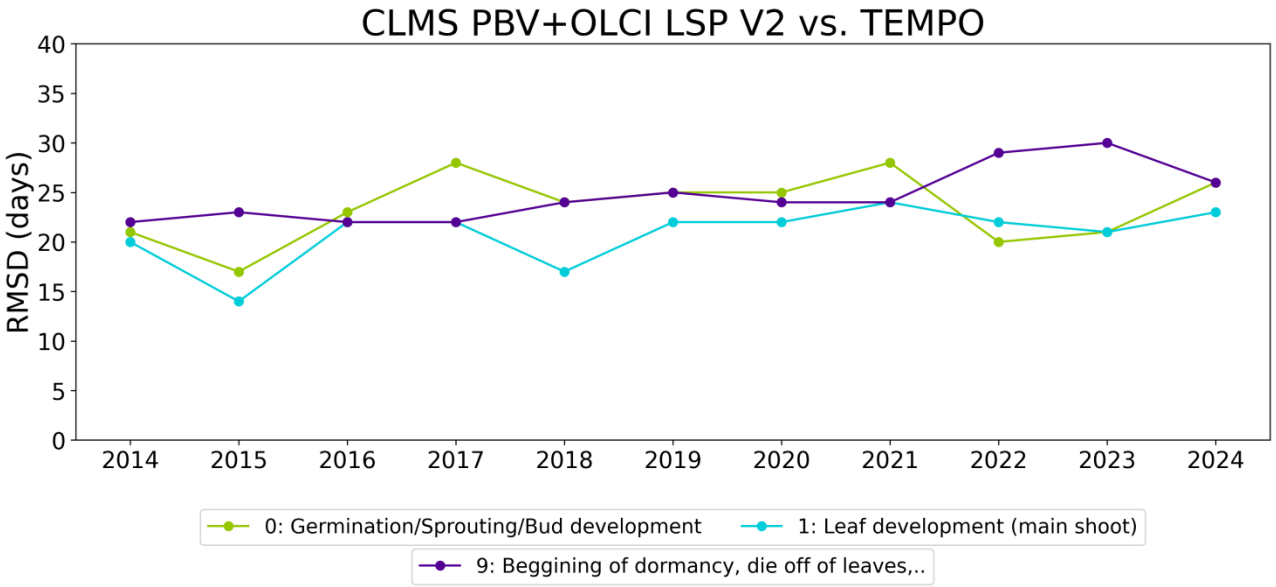
PEP725



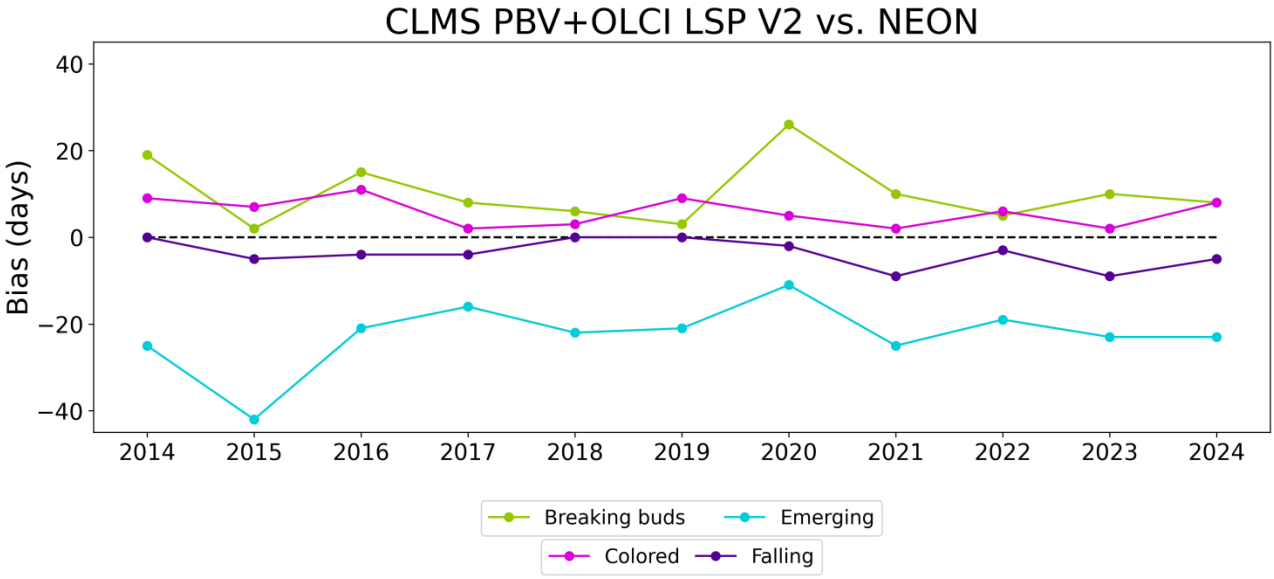


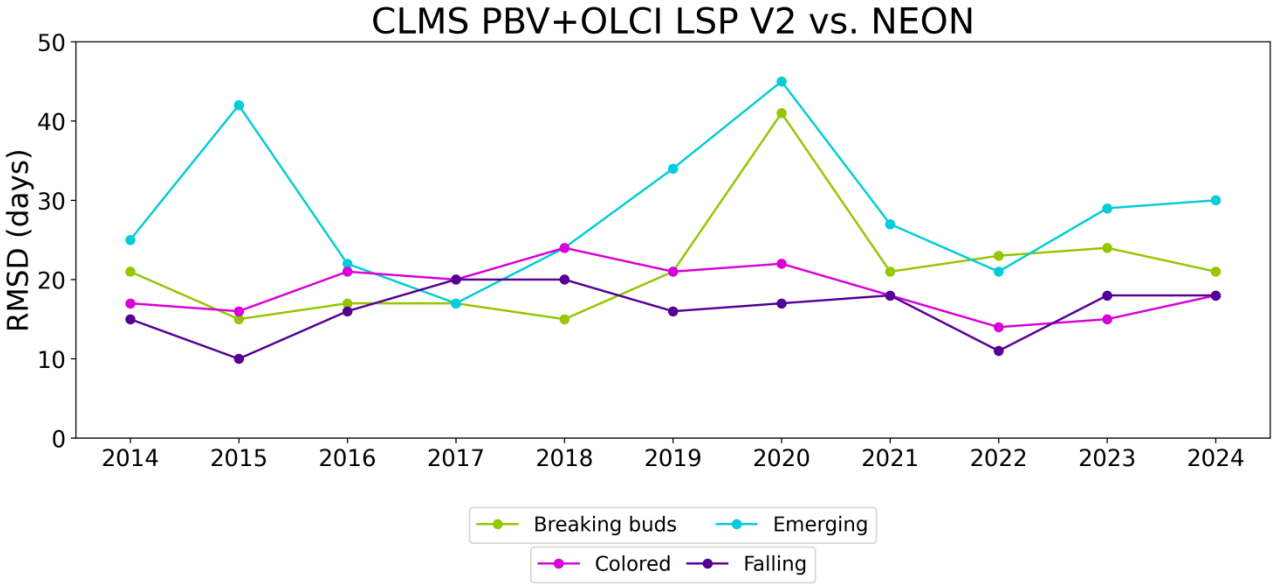
TEMPO


CLMS PBV+OLCI LSP V2 vs. TEMPO
Bias (days)
20
15
10
5
0
−5
−10
−15
−20
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
0: Germination/Sprouting/Bud development
1: Leaf development (main shoot)
9: Beggining of dormancy, die off of leaves,..


CLMS PBV+OLCI LSP V2 vs. TEMPO
RMSD (days)
40
35
30
25
20
15
10
5
0
2014
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
0: Germination/Sprouting/Bud development
1: Leaf development (main shoot)
9: Beggining of dormancy, die off of leaves,..

NEON





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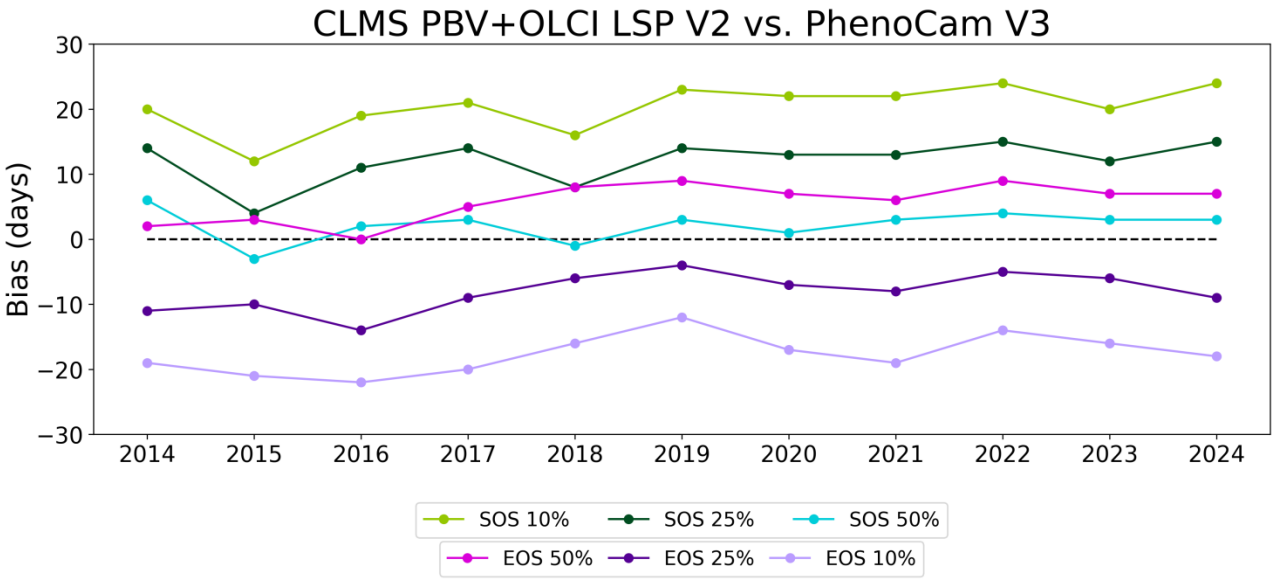
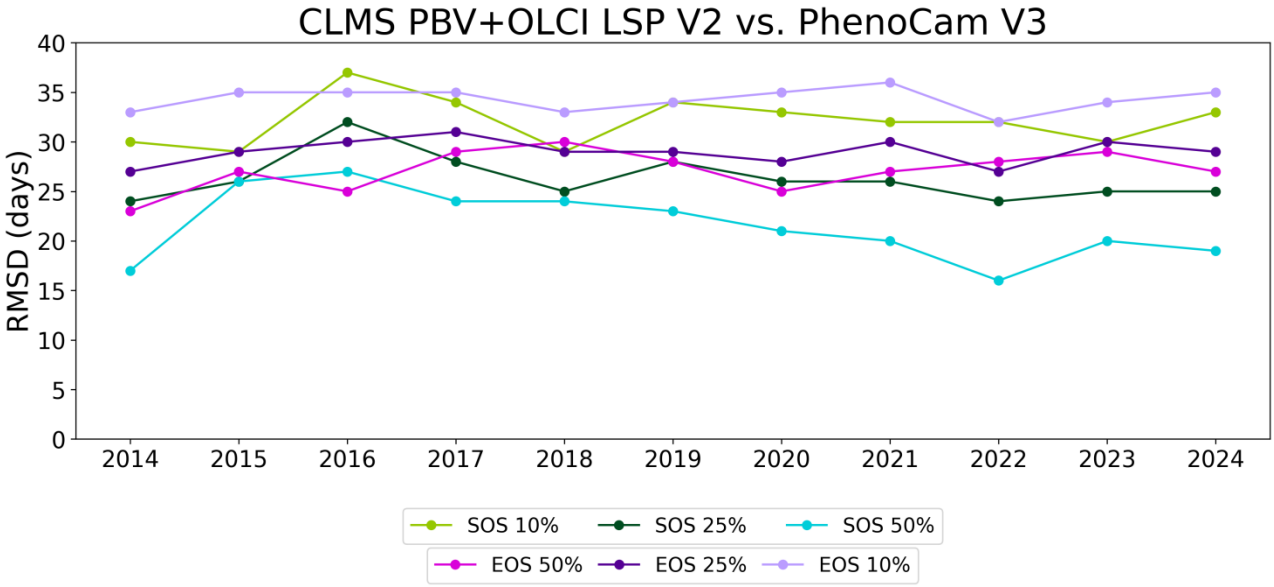
PhenoCam V3





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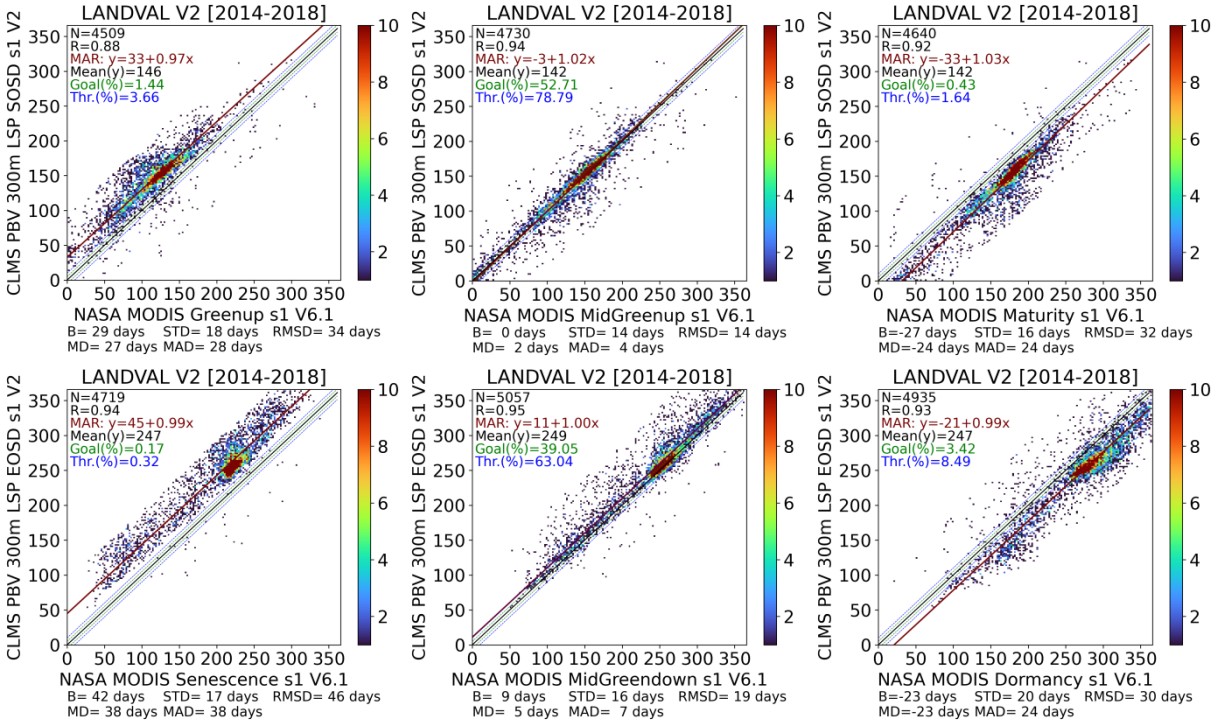
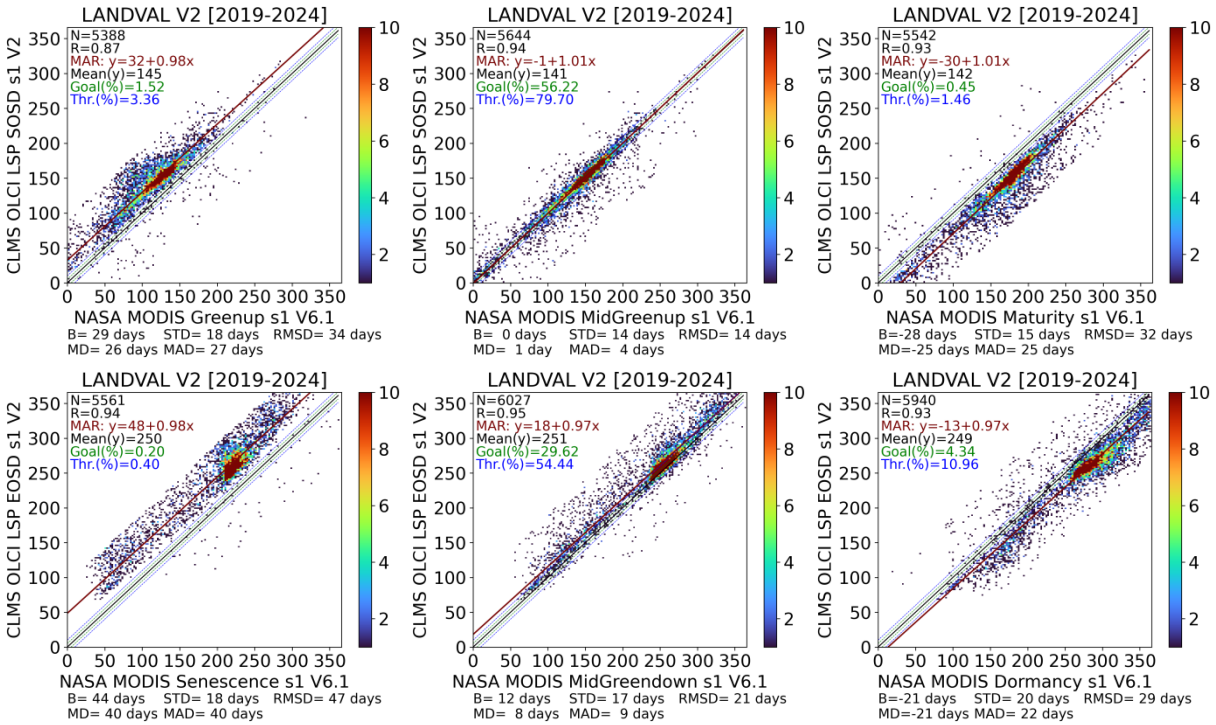
ANNEX IV: CLMS LSP V2 VS. MODIS LCD V6.1 SCATTER-PLOTS FOR 2014-2018 (PBV) AND 2019-2024 (OLCI)

2014-2018

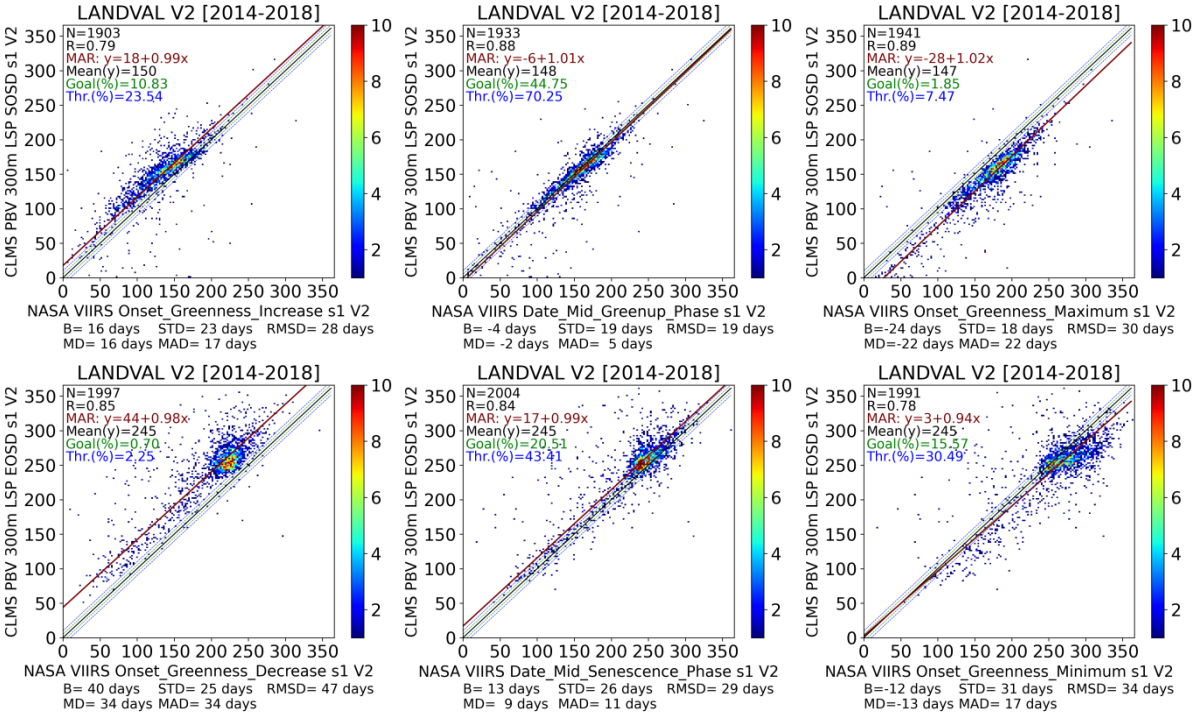


2019-2024



ANNEX V: CLMS LSP V2 VS. VIIRS LCD V2 SCATTER-PLOTS FOR 2014-2018 (PBV) AND 2019-2023 (OLCI)

2014-2018



2019-2023